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OECD ECONOMIC SURVEYS

1999

SPECIAL FEATURES

Origins of the economic boom

Sustaining high growth

IRELAND

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1998-1999

IRELAND

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF IRELAND

THE LAND

Area (thousand sq. km)	70
Agricultural area, 1995, as per cent of total area	57

Population of major cities, 1996[1] census (thousands):	
Dublin (County and Co. Borough)	1 057
Cork, Co. Borough	127
Limerick, Co. Borough	52

THE PEOPLE

Population (April 1998)[1]	3 704 900
Number of inhabitants per sq. km	53
Increase in population: annual average 1993-98	26 160
Natural increase in population: annual average 1993-98	18 320

Net emigration: average 1993-95	2 333
Net immigration: average 1996-98	15 267
Total labour force, April 1997 (thousands)	1 539
Civilian employment in (thousands):	
Agriculture, forestry and fishing	142
Industry and construction	399
Other sectors	840

THE GOVERNMENT

Public current expenditure on goods and services, 1997 (as per cent of GNP)	16
Current government receipts, 1996 (as per cent of GNP)	41
General government debt, 1997 (as per cent of GNP)	77

Composition of Parliament (1997):	Seats
Fianna Fail	76
Fine Gael	54
Labour	21
Progessive Democrats	4
Green	2
Socialist	1
Sinn Fein	1
Others	6

Last general election: June 1997

FOREIGN TRADE

Exports:	
Exports of goods and services, as per cent of GNP (1997)	97
Main exports, 1997 (per cent of total):	
Meat and meat preparation	3
Dairy products	2
Beverages	1
Organic chemicals	11
Medical and pharmaceutical products	6
Textile manufacturing, clothing and footwear	2
Machinery and transport equipment	37
of which:	
Office machines	23
Electrical machinery	8
Main customers, 1997 (per cent of total):	
United Kingdom	25
Other European Union	50
United States	8

Imports:	
Imports of goods and services, as per cent of GNP (1997)	81
Main imports, 1997 (per cent of total):	
Petroleum products	3
Chemicals and related products	12
Textile manufacturing, clothing and footwear	5
Machinery and transport equipment	45
of which:	
Office machines	19
Electrical machinery	10
Main suppliers, 1997 (per cent of total):	
United Kingdom	34
Other European Union	21
United States	15

THE CURRENCY

Monetary unit: Irish pound

Currency unit per US dollar, average of daily figures:	
Year 1998	0.70
March 1999[2]	0.72

1. Preliminary.
2. Based on the fixed rate of the Irish pound against the euro and the euro/dollar rate.

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Ireland by the Economic and Development Review Committee on 8 April 1999.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 23 April 1999.

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The previous Survey of Ireland was issued in May 1997.

Assessment and recommendations

There have been five straight years of impressive economic performance

The Irish economy has notched up five straight years of stunning economic performance. No other OECD Member country has been able to match its outstanding outcomes in a variety of dimensions. Output growth has averaged over 9 per cent per year on a GDP basis in the period 1994-98, bringing GDP per capita in purchasing power parity terms to a higher level than the European Union average. Half of that growth has been reflected in considerable employment gains and the rest in impressive labour productivity growth. Despite substantial increases in the labour force, thanks to Ireland's particularly favourable demographics and to an important reversal in migration flows, the unemployment rate has fallen by nearly 9 percentage points. Even the nagging problem of the long-term unemployed is showing signs of significant improvement. Spending growth has been remarkably well balanced, with export increases in the starring role, ably supported by private investment, and government spending playing only a bit part. Accordingly, the merchandise trade surplus has moved up to fully one-quarter of GDP, and the overall current external account has stayed in healthy surplus. With booming tax receipts, the turnaround in government finances has continued. There is now a sizeable surplus in its accounts (of about 2¼ per cent of GDP), and the level of government debt in relation to GDP has shrunk by nearly half, to well below the key Maastricht threshold of 60 per cent. And even though consumer price inflation is no longer among the lowest in the OECD, as it was a few years ago, it has remained below 2 per cent over the past year, despite the economic boom.

This is in sharp contrast to the first half of the 1980s

This is a far cry from the Ireland of the first half of the 1980s. At that point, output was stagnant; the unemployment rate was surging to the record high level of 17 per cent, despite heavy emigration; real investment was sliding by a cumulative 25 per cent; the current balance was on average in deficit to the tune of over 7 per cent of GDP; annual inflation was typically over 10 per cent per year; and, despite valiant attempts at fiscal consolidation, the public finances were still in a critical condition, with deficits averaging over 12 per cent of GDP and indebtedness soaring to a peak of 118 per cent of GDP in 1987, the second highest in the OECD.

There is no single reason for the improvement

The contrast is therefore sharp. At first glance it is astonishing that a nation could have moved all the way from the back of the pack to a leading position within such a short period, not much more than a decade in fact. It is therefore of great interest to seek the reasons for the turnabout and look for any lessons for other OECD Member countries. Unfortunately, it would seem that there has been no "silver bullet" – no single, overriding policy that could be adopted elsewhere in order to emulate Irish experience. Rather the breaks in trend, first around 1987, when the deterioration ceased and performance improved, and then around 1994, when the boom began, are attributable to the confluence of a series of favourable changes in the environment and other exogenous factors (some of which were specific to Ireland and are unlikely to be replicated elsewhere), as well as prudent planning and a range of policy shifts that lay the foundations for the pickup in growth. Most of the items that have contributed to the improvement are well known to other policy makers, but other countries' situations may not be so propitious as to allow such a strong response, even to fully appropriate incentives and institutional arrangements.

The acceleration in output was associated with a pickup in labour input growth

A decomposition of output growth into its supply-side factors may be instructive. It shows that it is more the pickup in labour inputs than any increase in the pace of what had been already for the most part robust productivity gains that explains the post-1987 upturn in performance. But the impressive magnitude of those productivity gains has been

undeniable. Additional inputs have come from a variety of sources. First, there has been a steady acceleration in the working-age population resulting from Ireland's somewhat unusual baby boom in the 1970s and from an endogenous reversal of traditional net outward migration. Second, a clear uptrend in labour force participation has developed, associated with changing social attitudes to women working and the higher opportunity costs of staying at home for an increasingly educated female population. But work incentives for potential labour force participants were not especially strong until of late, restrained by what was an inimical tax/transfer system. Third, expanding labour inputs were made possible by the drawing down of massive amounts of unemployed labour as well as faster shrinkage in farm labour requirements. Finally, the average quality of labour inputs is rising thanks to an upgrading of human capital, as younger cohorts with higher educational attainment replace their predecessors whose schooling duration was much shorter. This process is a long-term reaction to the introduction of free second-level schooling in 1967 and, to a lesser extent, of third-level education a few years ago, in combination with persistently higher labour-market returns to additional education than in the average OECD country in the form of lower risks of unemployment and higher earnings.

Being part of the European Union has also been of great benefit

Growth has also benefited importantly from Ireland's membership in the European Union. Besides the income received directly and indirectly by the nation's farmers, transfers of structural funds under the Community Support Framework contributed what was initially substantial demand-side stimulus, but whose lasting legacy is likely to be more on the supply side, mainly tangible and intangible capital in the form of improved infrastructure and human resources. This was complemented by Ireland's increasing integration with the economies of its EU partners, primarily through the Single Market Programme. Ironically, this Programme was originally feared likely to put peripheral EU "cohesion" countries at a disadvantage, but the evidence is that it has turned out to be more favourable for Ireland than for most of the rest of the EU. The liberalisation of trade and investment has also elicited static and dynamic gains in output and productivity and, along with the structural funds,

assisted in the nation's transformation into one of the premier host countries for US foreign direct investment in Europe in the 1990s.

But the most significant factor has been inward direct investment

It is the massive inflow of direct investment that has been the major positive shock influencing the economy in the 1990s. Data difficulties abound, but it is clear that there have been substantial cumulative net inflows of foreign direct investment (FDI) in order to take advantage of the favourable environment and set up what are, for the most part, export-platform manufacturing and financial services projects. Ireland's share of OECD total FDI inflows surged in the 1990s, reaching a level out of all proportion with its GDP share; for example, in 1997 it ranked fifth in the world as a destination for US direct investment outflows.

These inflows are attributable to the strategy of industrial support as well as enhanced macroeconomic policy credibility and social dialogue

The reasons for this successful take-off are multiple. Ireland's appeal has been based on the quality, price and availability of its labour, the welcoming attitude to foreign investors, the use of the English language and exploitation of "first-mover advantages": once one producer in a sector establishes production facilities in Ireland, it is generally easier to attract its competitors. But there is no question that the robust and prolonged expansion of the US economy – the source of most of the direct investment inflows – and the attractive policy and institutional setting in Ireland have been the major driving forces. A generally conducive and transparent regulatory framework, an open trade regime and competitive labour costs have generated this outcome. A long-standing favourable tax regime applied to tradable goods sectors, financial incentives and substantial industrial support targeted at a small number of key dynamic sectors with spin-off potential, may also have been important in convincing major multinational corporations to choose Ireland as a production location in order to serve at least the EU market. While the role of fiscal consolidation in the 1980s has been much overplayed in some quarters – with talk of an "expansionary fiscal contraction" – successful fiscal adjustment focused on spending reductions did manage to alleviate earlier concerns for the longer-term sustainability of the low-tax environment. Also, social attitudes coalesced around a consensus view that greater

self-reliance was required, that restructuring was inevitable and that competitiveness had to be safeguarded by moderate wage claims negotiated in a centralised setting through the so-called "partnership" approach. The succession of national wage agreements was crucial in preventing both social discord resulting from needed changes in work practices and excessive wage increases from setting in motion a severe squeeze on traditional, labour-intensive industries.

The economic effects of the investment inflows have been powerful

The macroeconomic and sectoral effects of the foreign capital inflows have been powerful. Foreign-owned firms in manufacturing alone now generate about 30 per cent of GDP. Producing largely on a global scale, they are highly productive and profitable, which gives them the means to pay their workers higher compensation. Even if they still source less of their inputs locally than do indigenous firms, they do have worthwhile linkages with domestic services providers. Furthermore, they provide valuable managerial experience, in some cases (computer software, for example) leading to subsequent related start-ups. In addition, they do a great deal of research and development, albeit slightly less than their Irish-owned counterparts in relation to output. Finally, they open their marketing networks, so that the share of Ireland's exports to the United Kingdom has fallen steadily, thereby providing geographic diversification gains. However, even if policy-makers' attention is no longer focused excessively on creating the right conditions for investment by *foreign* firms to the possible detriment of indigenous enterprises, the long history of activist industrial and tax policies has left Ireland with a possibly distorted industrial structure. Non-government services seem to have an unusually small output share, given the nation's overall development level, thanks to the assistance given to the tradable goods sector. And with full employment within sight, and the cost of scarce inputs bid up, market forces are also more likely to force shrinkage in traditional, labour-intensive sectors within manufacturing.

But over the past year or so signs of overheating have multiplied

In the past year or two, however, the macroeconomic balance in the economy has shifted. The persistent expansionary supply shocks have propagated to factor and domestic goods markets, the non-tradable sectors in particular. Signs

of excess demand and overheating have multiplied, even if headline rates of inflation have eased off since the summer of 1998, mostly because of interest-rate, indirect tax and oil-price declines, all of which are expansionary. But inflation is not as good an indicator of goods market disequilibrium in a country such as Ireland as in others whose economies are less open to trade and factor flows: in effect, Ireland should be seen more as a region in a larger Continental economy, especially now that it is part of Economic and Monetary Union (EMU). With exports and imports each nearly as large as GDP itself and with the labour market so integrated with the UK and, to a lesser extent, the US jobs markets through migration flows, imbalances can exist for quite a lengthy period before they are easily detected or cause much damage.

There are several bottlenecks, labour and housing in particular

There are indeed some short- and longer-run bottlenecks. Labour shortages have moved beyond the realm of specific skills to a more generalised scarcity of even unskilled labour. The most obvious concern is that earnings are now growing more quickly than productivity in a number of sectors, possibly because the unemployment rate has now fallen to below 7 per cent of the labour force, a level that, as recently as April 1998, was set as an official target for the end of the year 2000. And housing prices were rising rapidly at least until the latter part of 1998. This last development has no doubt limited the supply of immigrant labour and helped to push up wage demands. The form of adjustment to supply constraints that is occurring is that higher wage rates are running ahead of productivity gains and pushing up costs, eroding competitiveness and braking the growth in exports. The risk is that this process could be too rapid, with unrealistic expectations leading to an overshooting of sustainable wage levels and a sharp cost-based slowdown in growth to well below potential rates.

The central projection is that slowing export increases will moderate growth, but there are risks that wages will overshoot

Fortunately, there have been some signs of cooling off since the middle of 1998. Most of these result from the external side: export market growth has apparently begun to moderate in line with the slowdown in the EU, along with continued weakness in certain Asian markets. The central projections call for that easing in foreign demand to persist. There are few reasons to expect any such deceleration in domestic demand, however. Household income growth is expected to remain robust, confidence is high, wealth gains have probably not yet been fully taken into account, taxes have been cut and interest rates are lower than they have been in a generation. Nevertheless, with the softening in demand for exports, real GDP growth may ease somewhat and by next year could be below 7 per cent. But that is still a rate which will increase the pressures on available resources and slice further into the stock of the unemployed. This could result in a further acceleration in labour compensation and costs, cutting into competitiveness and squeezing the foreign balance. As mentioned above, there are additional risks that adjustment will proceed through an even sharper slowdown in direct investment inflows and exports as a result of deteriorating competitiveness or weakness in trading partner economies. At the same time, with buoyant expectations and rising housing-related wealth, household saving rates could fall even in the face of robust disposable income gains, pushing up consumption and imports and leading to an even more rapid erosion of the current account surplus.

Monetary policy was already constrained in 1997-98, and the appreciation of the exchange-rate parity was half-hearted

The major policy question is how to manage the pressures arising from the rapid expansion in such a way to ensure that growth is sustained at a high rate. That challenge would always have been difficult, but in Ireland's specific circumstances of late it has been formidable. Monetary policymakers would normally have wished to react not only to the pickup in price and wage inflation, but also to the steady acceleration in money and credit aggregates. However, with Ireland now being one of the eleven founding members of EMU, the formation of monetary policy has passed from Dublin to Frankfurt. And even with the imminence of membership, it was widely recognised in 1997-98 that interest rates would have to drop to much lower levels, thereby limiting the impact of official rates on market rates at any

other than fairly short and steadily shortening maturities. Thus, the only leverage the monetary authorities had was over the speed of the convergence of interest rates and the level at which the currency would ultimately be irrevocably exchanged into euros. It was, therefore, appropriate for the Central Bank to put off the convergence for as long as possible, even if the effects were limited. But, while some advantage was taken to exploit the final chance to set the exchange rate in order to try to stabilise the balance of payments, the actual increase of 3 per cent was rather modest. The chosen entry rate was still over 7½ per cent weaker than the central rate that prevailed prior to the 1993 devaluation, since which time cost competitiveness against trading partners has soared. Trying to protect traditional sectors by helping them retain capital and labour resources was ultimately pointless: the equilibrium real exchange rate has almost certainly appreciated, and these sectors will have to shrink, whether it be the result of higher costs or lower prices.

Only fiscal and incomes policies remain to control demand

Thus, there remain only fiscal and incomes policies with which to control demand and stabilise the macroeconomy. In theory, joining EMU should render fiscal policy more effective, as both domestic interest and exchange rates have become essentially exogenous, and therefore leakages resulting from budgetary changes will be limited to the trade and savings channels. But even if taxpayers are not perfectly far-sighted, the trade channel is so strong in Ireland's case, and likely to get even stronger as a result of the common currency area, that crowding-in effects following budgetary moves toward restriction are likely to prove to be significant. The role of budgetary policy as a demand management tool is therefore more limited than in other countries, although it may nevertheless be needed should serious overheating occur.

An even larger budget surplus would be appropriate

The current situation is one in which fiscal tightening is difficult, as the budget was already in surplus in 1998, and the outlook for 1999 is for an even larger excess of revenues over outlays – though this is entirely due to the decline in interest rates and the effects of the above-trend growth. On the other hand, government debt remains quite high in

relation to GDP. Furthermore, much of the present good fortune is attributable to the unusually favourable demographic situation, with low spending pressures arising from a relatively small share of the aged in the total population, something which will begin to reverse in less than a decade. Thus, on economic grounds, an even larger surplus would be appropriate. However, in addition, the budget has been largely embedded in the Partnership 2000 incomes policy process, and certain groups had claimed that sizeable tax cuts skewed towards the poor were required to sustain the partnership approach to centralised wage negotiations. These demands were acceded to. In the future, however, the authorities should at least avoid fuelling demand in full consciousness of the increasing severity of supply constraints. Thus, cuts in tax rates should be offset by base-broadening moves and other revenue-raising measures.

Incomes policy may prove no longer as successful in moderating wage and price pressures as in the past

While incomes policy has probably contributed to the containment of wage and price inflation over the past decade, its effectiveness may be more limited in the current environment of strong labour-market pressures. Nevertheless, the partnership process still deserves support. In particular, during the forthcoming crucial negotiations among the social partners the government should impress upon them the importance of the various linkages between affordable earnings and profitability, the prices of other inputs and foreign costs. At the same time it should itself commit to ensuring that public-sector wage settlements take full account of the impact on national competitiveness, so that the benchmark for private-sector wage-setting remains market conditions rather than public-sector wages.

Labour supply can be enhanced by improving work incentives for all and, if desired, by facilitating greater female participation

Budget and other economic policies may be more effective in stimulating the supply side to respond efficiently to growing demand than in restraining that demand, so that competitiveness effects are minimised and the growth period is prolonged. Regarding labour supply, recent budgets have appropriately focused on improving work incentives by means of cutting marginal tax rates and net replacement ratios, but the latter remain high for married people with children. The effects of housing-related benefits on incentives still look significant for some limited groups. One

demographic group that could potentially participate to a greater extent in the labour market – again if the social and political will is there – is adult women. But to achieve a faster increase in their participation rate than heretofore will require the government to encourage more actively the private provision of childcare and pre-school facilities. Firms too will have to implement more family-friendly policies. Those currently unemployed represent a further source of additional labour supply. A shift towards a so-called "preventative" approach to long-term unemployment for young people, as promised in last year's Employment Action Plan, and, to a lesser extent, a strengthening in the conditionality of unemployment benefit payments have apparently revealed substantial numbers of claimants who were not truly unemployed. This is being appropriately extended to other groups. Further sustainable declines in the unemployment rate as a source of under-utilised labour after this year, when a 6 per cent rate looks set to be recorded, will be more difficult to achieve unless labour market flexibility is enhanced. The substantial spending on active labour market policies may make no more than a marginal contribution to prospective labour supply. In that regard, there remains a need to rationalise the large number of schemes that focus mainly on the long-term unemployed.

The housing constraint has become more severe, even if some recent measures may have eased the situation slightly

The shortage of housing has become an important bottleneck. Alleviating the housing constraint is important not only in order to attract more immigrants but also to meet the growth in demand resulting from both the increasing numbers of citizens in the age groups in which first-time buyers are located and the probable decline in average household size. The government acted quickly last year to implement many changes designed to stabilise house prices, mainly by facilitating an increase in supply. It is also moving in response to those higher prices to try to improve the access to housing by low-income households. However, such moves, by themselves, only worsen the imbalance in the short run by raising demand without stimulating any corresponding supply. Having made available enough serviced land for a few years, the authorities should now give priority to implementing a strategic physical plan for the Dublin area, expediting planning procedures and increasing

urban densities. The private rental market should eventually be deepened in order to allow greater labour mobility, even if short-term considerations persuaded the government to discourage the development of this market in the measures it took to cool the housing market last spring. Other desirable longer-term reforms would be either to phase out mortgage interest tax relief or tax the imputed rents on owner-occupied housing and to expose owners to higher holding costs through either the re-introduction of a residential property tax or through a restructured capital gains tax.

Public infrastructure is also in urgent need of enhancement, especially in the domains of transport and environmental services

Other eventual bottlenecks are in a variety of forms of public infrastructure, most prominently in transport and environmental services, although the authorities believe that there is also a externalities-based argument for public funding in telecoms as well. With housing affordability concerns coming to the fore, it is clear that commuting time considerations will be prominent in determining where and whether many people are willing to take available jobs. Thus, unless both rail and road infrastructures are quickly improved, labour supply will start to suffer, and the efficiency of job matching will deteriorate. There are several deficiencies in environmental outcomes, which, even if the danger they pose to sustainable development is sometimes more distant, could eventually harm the tourism and fisheries industries. These include the large albeit rapidly declining share of untreated waste water generated by households, agriculture and industry and the over-reliance on landfill for dealing with solid waste, which will have to end in view of the depletion of existing sites and the lack of obvious replacements. In addition, without further measures greenhouse gas (GHG) emissions will grow too rapidly for the nation to meet its international commitments. In view of diminished EU funding, rectifying all these shortcomings will have potentially heavy budgetary implications. To minimise them, the authorities need to make greater use of market mechanisms to ration access to these infrastructures and to look seriously at private-sector involvement through, for example, the Public-Private Partnership idea. Any new motorways should be financed in part by toll charges. Water should no longer be provided free at the margin to

residential users. Recycling should be encouraged with appropriate incentives. But, most importantly, it seems unlikely that administrative decisions, even with the possible contribution from emissions trading, will achieve the target in GHG emissions, without recourse to some sort of tax/subsidy scheme.

Skills development must remain a high priority

The final major bottleneck is in the area of skills. But here laudable progress has been made with substantial funding increases in the area of educational technology and research. Nevertheless, the system of human capital development is still in need of further upgrading, as was attested to by the recent international comparison of adult literacy levels, which showed the Irish population in a relatively poor light. While average initial educational attainment is still improving, there is too little adult education and training, especially for the unemployed. Unfortunately, available tools to encourage employers to provide greater worthwhile on-the-job training are few. Demand for such training by employees would be greater if the skills mastered were more widely acknowledged; to this end a planned national qualifications framework is a worthy endeavour. The initial education system, too, has room for improvement. At the tertiary level, with a view to the alleviation of overcrowding and possible favourable effects on equity, the restoration of fees combined with a programme of income-contingent student loans should be seriously considered. At lower levels, standards still need to be raised further. The school-leaving age remains unusually low, and present plans call for it to be increased only to the minimal international norm. The curriculum will benefit from adjustment towards a greater emphasis on natural sciences, foreign languages and vocational and technical programmes, and adoption of a continuous system of assessment might also make sense.

Industrial policy needs to place still greater weight on the scarcity of labour

Industrial policy has very recently begun to adapt to changing economic circumstances, but sustainability considerations point to the need to further shift its focus. In an era of near-full employment, the weight accorded to calculations of cost per job "created" should be further reduced. Labour has become a scarce resource. The cost-benefit methodology that the authorities use should be further updated to

take fuller account of that scarcity as well as of the rise in the shadow price of public funds that results from declining EU transfers for enterprise support. Also, the assessment process needs to include, to the maximum extent possible, job quality considerations and the regional dimensions involved. The government has gone to great lengths to emphasise that some of the nation's regions have not progressed to the same level as others, and regional factors are appropriately being taken explicitly into account in the implementation of the industrial support schemes.

The efficiency of markets will assume greater importance in the post-EMU era

There are also other areas of structural policy whose importance for national welfare has increased in the post-EMU era in which Ireland has given up important instruments of national adjustment. Markets must be made to function as well as possible in order to transmit price signals clearly to producers and consumers alike. Also, greater competition in the markets for business services will keep their prices low, thereby maintaining competitiveness. Issues of competition and regulatory policies, financial supervision and public ownership come to the fore. Utilities are the primary case. Full deregulation of the telecoms sector has now been achieved, and producers have responded with major price cuts that have probably eliminated the previous gap between Irish and foreign tariffs. Electricity prices are set to be freed for the largest users next year; current cross-subsidies from industrial to household users should then be abolished. Indeed, Ireland should not be satisfied with the minimum amount of regulatory reform imposed by EU directives but should proceed with utilities deregulation on its own. Various privatisations are also planned, and all look set to return to the private sector activities that are no longer appropriately in the public domain. The competition authorities have moved to strengthen enforcement of the relevant statutes, thereby gaining credibility for themselves as well as the law. But officially imposed entry controls on taxis and public houses would best be phased out in order to eliminate the resulting rents.

Labour-market flexibility will also be crucial, and a minimum wage may not do much to reduce poverty

Labour markets will also have a greater responsibility in adjustment to shocks, especially those that are peculiar to Ireland. This raises the question of the proposed national minimum wage, which was promised by the current government in order to address the issues of exclusion, marginalisation and poverty. There is no doubt that these are important issues in Ireland, as elsewhere, even if recent increases in real incomes have probably moderated their severity somewhat. However, it is unlikely that intervening directly in labour markets outcomes is the best way to deal with them. A minimum wage is generally a weak anti-poverty device, and is certainly so in Ireland, because most of the poor are jobless: only 3.3 per cent of low-wage workers live in households that would conventionally be described as subject to poverty, even in relative terms; and only 1.7 per cent of all adults living in poor households are low-paid employees. Concerns about income distribution are best addressed by budgetary and social policies that concentrate on these issues alone, while labour markets are left solely with their primary function of efficiently allocating individuals to jobs.

An excessively high minimum wage could destroy jobs, especially if applied inappropriately to youths, the unskilled and those in outlying areas

But assuming that the government will not reconsider its decision to implement a minimum wage, a number of points should be kept in mind. First, the level at which it is introduced is crucial in determining its overall economic impact. The rate recommended by last year's Commission – two-thirds of median wages of full-time workers – would be the highest relative rate in the OECD. The risk would therefore be strong that it would lead to a fall in employment, especially among the young and the unskilled who would be priced out of the market and might therefore be worse off owing to their exclusion. Even the nominal amount considered, 4.40 pounds an hour effective as from April 2000, would pose important risks in this direction, thereby making any further uprating inappropriate. Any minimum wage at all for those under 18 might lower the average effective school-leaving age by encouraging such teenagers to look for work rather than staying on longer in the school system to equip themselves with valuable human capital. For similar reasons the suggested reduced rate of 70 per cent for such youths would best be applied to those in the 18 to

21 age group whose combination of skills, experience and anti-poverty needs are unlikely to justify eligibility for the full rate. Finally, despite its rejection by the Commission, a regionally differentiated minimum wage would be worthy of the added complexity involved. The cost of living outside of Dublin is clearly lower, implying that less income is required to avoid poverty. Furthermore, with an equal rate it would be much more difficult to attract investment into such outlying areas, an objective that is widely supported, not only for regional development considerations but also for easing the strains on the capital's infrastructure.

The long-term stability of the pension system is being appropriately re-examined before the ageing process gets underway

The authorities are also rightly considering longer-term issues, especially the pensions question. Ireland has the luxury of an ageing process that will get underway later than elsewhere, and it needs to have in place systems that will prove robust with respect to this eventual demographic shift. An important report published last year recommended the creation of a reserve fund to smooth the burden of rising costs over time and an increase in the generosity of the first pillar of the pension system – the social welfare pension. Advance funding is also being considered for public-service pensions. The report also urged the introduction of Personal Retirement Savings Accounts (PRSAs) to increase the share of the population benefiting from such supplementary coverage. The government is pursuing these ideas. Advance funding should alleviate political pressures to increase current spending that are likely to arise because of the budget surplus. Privatisation proceeds should be used to minimise the budgetary effects of establishing the fund. However, fund management and portfolio selection will need serious thought in order to avoid having the fund gain effective control of any private-sector firms, for example. It is difficult to oppose the idea of boosting the generosity of public pensions, but indexation to real wage increases should be granted only if actuarial review of the fund's financial outlook confirms affordability. Finally, PRSAs would increase the range of the population with a stake in the economy's performance, might increase saving rates and, by eliminating portability problems, would improve labour market flexibility.

Ireland can potentially continue to grow at superior rates

To recapitulate, Ireland has enjoyed yet another year of very rapid growth, low inflation, declining unemployment and external and budgetary surplus. Such outcomes are the envy of countries around the world. They have resulted from a variety of sources. Most will continue to contribute to performance in the future, although many at declining rates. How much longer output can grow at these superior rates will to a large extent be determined by how quickly the well-known bottlenecks are overcome and to what extent income and spending restraint occur. But it would be prudent to lay the groundwork for the day when growth slows – as it inevitably will. The nation no longer has many instruments of macroeconomic policy at its disposal, and product and input markets will, thus, have to function that much more flexibly to handle unexpected developments.

I. The origins of the recent economic boom

It is little more than a decade since Ireland was labelled the "sick man" of Western Europe. Despite fairly rapid population increases, real GNP had stagnated in the six years ending in 1986, and employment fell steadily through the period, leaving in its wake an unemployment rate of around 17 per cent, the second highest in the OECD, despite well below-average labour force participation and sizeable outward migration. Notwithstanding serious attempts at fiscal consolidation, the public finances were still in a critical condition, with the general government deficit at nearly 12 per cent of GNP, most of which was judged to be not amenable to cyclical improvement. As a result government debt was en route to nearly 130 per cent of GNP, again second worst among Member countries. Finally, although the current external deficit had been brought down sharply, it remained at a still-worrisome level of over 3½ per cent of GNP.

Today Ireland is a world leader in a number of aspects of economic performance. For a number of years its economy has been termed the "Celtic tiger", even though, following the financial crisis in Asia which began in 1997, its outcomes have clearly surpassed those of such etymological rivals in the 1990s. Output has been growing fairly steadily at near-double-digit rates for some five years; the unemployment rate has fallen around 9 percentage points in that period; the external accounts are in a healthy surplus; and, with the public finances the envy of most of its EU partners, government debt has halved in relation to GNP. The worry has shifted from how to make fuller use of under-utilised resources to one of how to manage congestion and shortages.

This amazing turnaround was hardly predicted,[1] and its causes have only recently begun to receive widespread examination. This chapter reviews the sources of this improvement. It focuses in particular on: the role of the pickup in productivity growth as compared with the more intensive use of available factors of production; the shift of those factors from traditional low-productivity sectors to more high-technology industries; and the importance of macroeconomic and structural policies both in catalysing the growth process and now prolonging its lifetime.

Setting the scene: accounting for growth performance

The Irish economy has generated clearly superior outcomes for over a decade now. But the period since the mid-1980s should be split into two, the periods prior to and after 1993. In the first of the two, output growth resumed, but only at levels which were the norm prior to the earlier stagnation, and while there was renewed job creation, it was insufficient to make much of an impact on the level of unemployment (Table 1). However, substantial progress was made in resolving a number of the economy's other fundamental problems. The prevalent slack was sufficient to continue the disinflation process, with moderate wage gains and consumer price increases regularly below those of its trading partners. Accordingly, unit labour cost rises were also fairly limited, because of improved productivity growth. Since the exchange rate of the Irish pound was little changed in trade-weighted terms, Ireland's competitive position was steadily enhanced (Figure 1). Exporters – to a large extent foreign multinational corporations – were able to gain market share without having to translate lower costs into lower prices, resulting in both massive increases in profitability[2] and a sharp turnaround on the external accounts: the current account surplus surged to around 3½ per cent of GNP. Furthermore, spending restraint in the public sector finally managed to

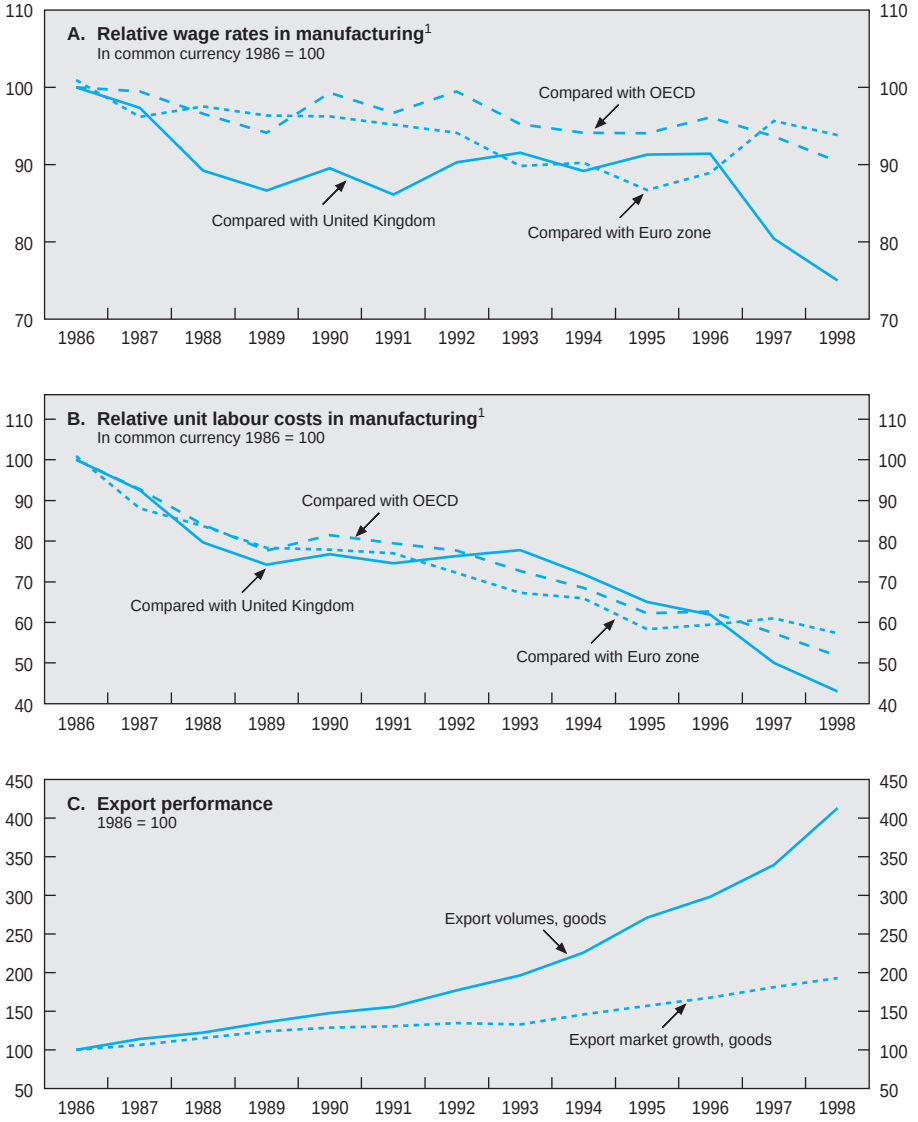
Table 1. **Historical performance measures**

Average annual percentage change, except as noted

	1961-70	1971-80	1981-86	1987-93	1994-98
Real GDP	4.2	4.7	2.1	4.8	9.2
Real GNP	4.2	3.9	0.1	4.1	8.3
Consumer prices	4.8	13.6	10.8	2.9	2.1
Employment	–0.0	0.9	–1.3	1.1	4.9
Unemployment rate					
(level)	4.8	6.8	13.8	15.2	11.4
(end of period)	5.6	7.0	17.1	15.7	7.7
Net lending of general government as a ratio to GNP					
(level)	n.a.	–10.3[1]	–12.5	–3.9	–0.2
(end of period)	n.a.	–12.7	–11.9	–2.7	2.4
Current balance as a ratio to GNP					
(level)	–3.3[2]	–6.1	–7.5	0.5	2.9
(end of period)	–4.7	–11.1	–3.6	4.4	2.2

1. 1977 to 1980.
2. 1962 to 1970.
Source: OECD.

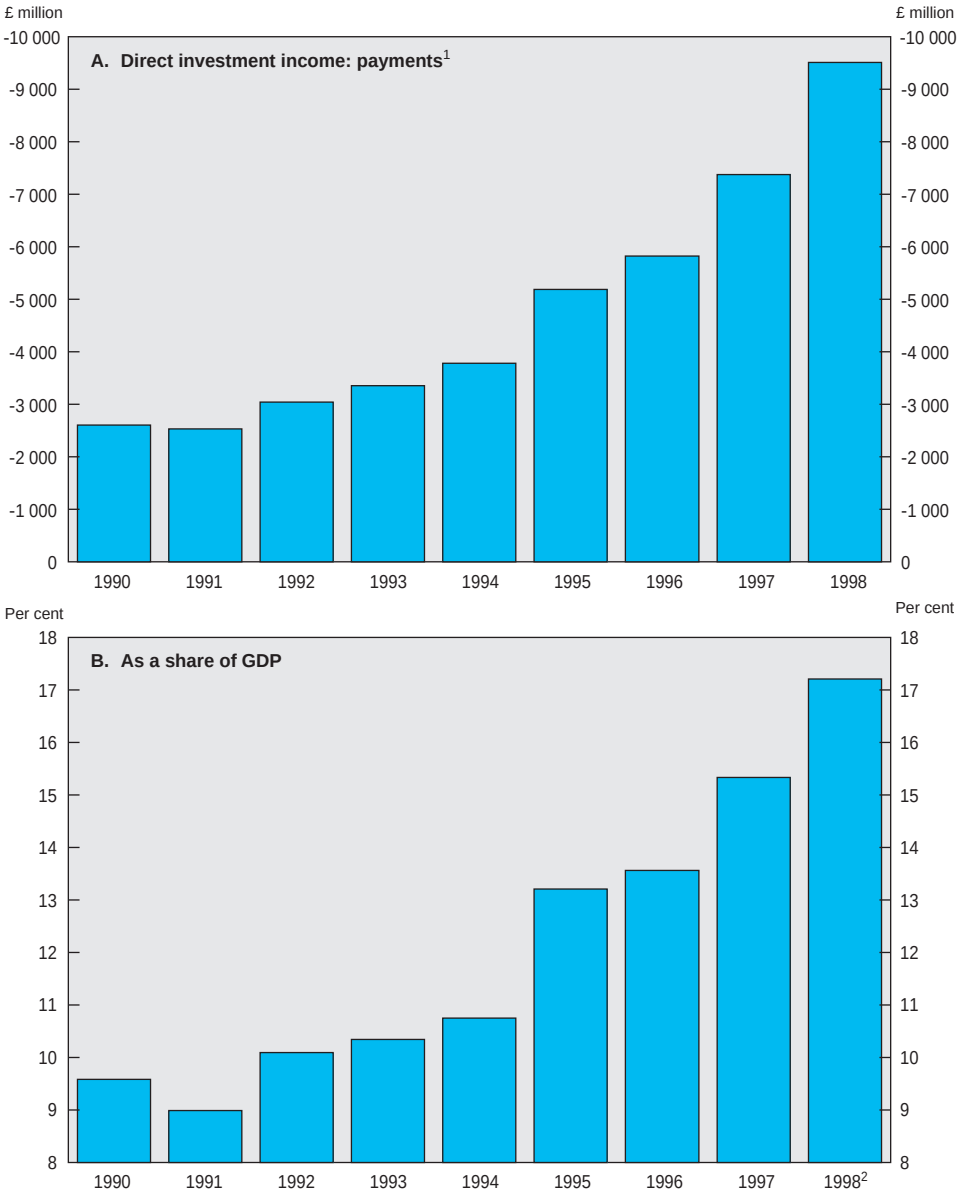
Figure 1. **Steadily improving cost competitiveness and market-share gains**



1. Relative to trade-weighted average of trading partners excluding the Czech Republic, Hungary, Iceland, Korea, Mexico, Norway, Poland, Switzerland and Turkey. Fixed 1997 weights.

Source: OECD.

Figure 2. **Direct investment income**



1. Dividends, distributions of branch profits, inter-affiliate interest and reinvested earnings.
2. Based on OECD estimate for GDP.

Source: Central Statistics Office.

break the trend to ever-increasing budget deficits and government debt, thereby earning significant benefits in terms of confidence in macroeconomic policies (see below).

It has really only been in the past five years that the Irish economy has demonstrated peerless performance. Average output growth essentially doubled. The most impressive figure is based on real GDP, but, as is often pointed out, this measure is not the most appropriate for Ireland. It includes all the profits earned by its large foreign-owned manufacturing and internationally traded services sectors,[3] which are not available to domestic residents. The significance of these profits has been rising in line with growing foreign ownership and unusually and persistently large margins in a number of sectors (see below). By 1998, they reached 17½ per cent of GDP, up from 9½ per cent in 1990 (Figure 2). Thus, GNP is almost 15 per cent smaller than GDP, and the gap is widening (it was 10.7 per cent in 1990). But it is also often argued that even GNP misstates Ireland's economic welfare, since it excludes net current transfers received from abroad.[4] Inflation-adjusted growth in output has exceeded that in disposable income by an average of around 2½ percentage points in the 1990s.

It is also of interest to look at the evolution of Ireland's growth performance in relation to that of its EU neighbours and of the OECD more generally. International capital mobility might be expected to lead to convergence of GDP per capita across countries, but it would have to be supplemented by labour mobility to allow the prediction of GNP per capita convergence as well. Figures converted by means of purchasing power parities show a substantial amount of convergence for per capita GDP over this period (Table 2), much greater than the average 2 per cent per year observed internationally (Barro and Sala-i-Martin, 1995). Indeed, convergence could be judged to be essentially complete by 1998: European Commission figures released in February show Ireland's GDP per capita to be 105 per cent of the EU average, while the corresponding GNP-based figure

Table 2. **Convergence of real GDP per capita, 1985-96**

OECD = 100

	1985[1]	1990	1993	1996
Ireland	**50**	**61**	**73**	**92**
EU 12	78	91	87	n.a.
EU 15	n.a.	n.a.	90	99

1. The source provides figures only for United States = 100. It has been assumed that US GDP per capita was 122 in OECD = 100 terms (the figure observed in 1990).

Source: OECD.

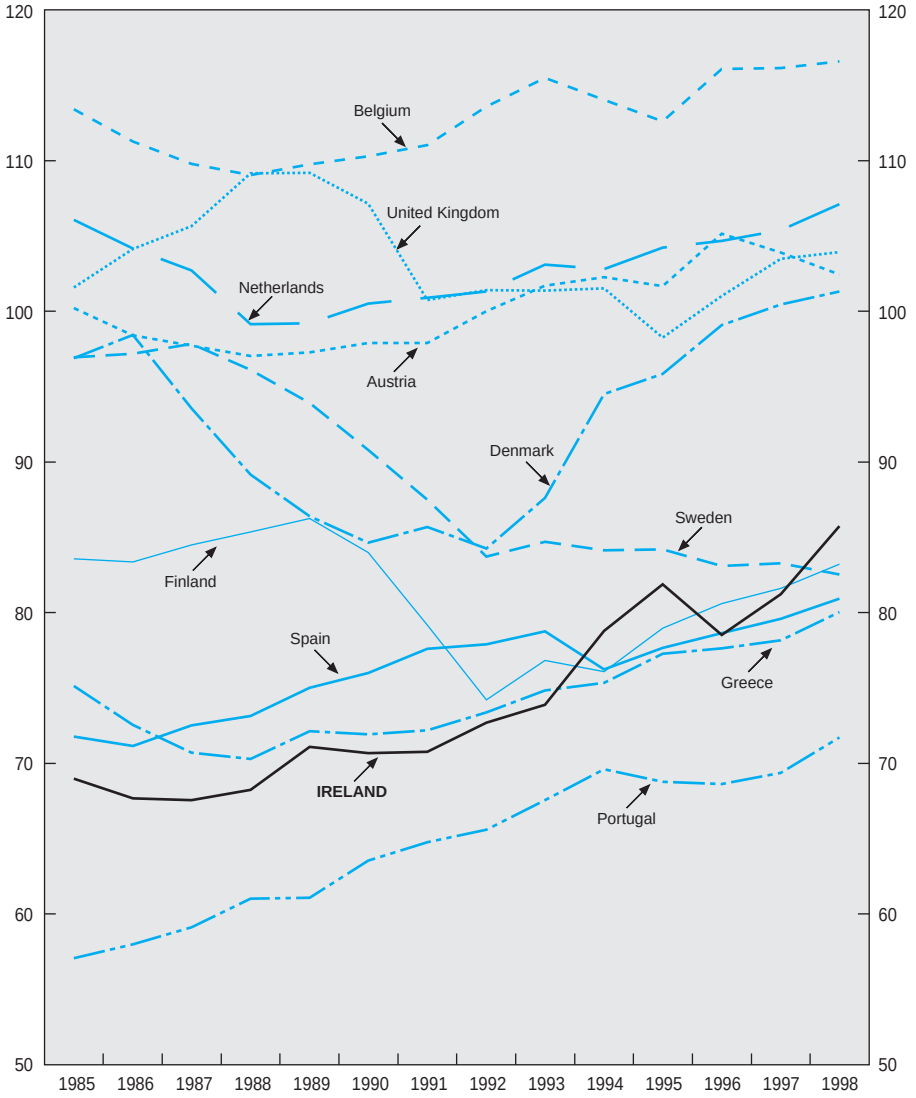
was 87.[5] However, it must be remembered that GNP convergence would have been much more gradual and that other figures show that the gap with the EU had previously widened slightly in the quarter-century to 1985 (Barry, 1996).

In any case, the ultimate measure of economic success for a nation must be its ability to support the private consumption of its residents. Since so much of Ireland's growth has been concentrated on exports, with the associated value added going increasingly to foreign capital, and since it is widely accepted that there remain substantial productivity gaps between indigenous Irish producers, mainly small and medium-sized enterprises, and their foreign counterparts,[6] it is perhaps not surprising that such a comparison is somewhat less flattering: again using purchasing power parity exchange rates, Ireland's position was the second lowest in the EU in the mid-1980s and right up to the growth takeoff five years ago (Figure 3). Since then its lag over the EU average has shrunk, and it has overtaken four more EU partners. But it remains well behind the leaders, and even with another year of strong growth in private consumption expected in 1999, there will still remain a shortfall of about 10 percentage points. Furthermore, it is noteworthy that Denmark has done at least as well as Ireland on this measure over the recent past.

As noted, real GDP growth in the past five years has picked up substantially from historical trends. A simple growth accounting exercise reveals that there has been an amazing confluence of supply-side sources of this acceleration (Table 3), even though it is premature to claim that the process has been mainly supply driven. As was discussed, the GDP/GNP gap has continued to widen in recent years, and this has occurred at a slightly faster pace than previously, even if the contribution of foreign factors of production to output growth has not been as marked as it was in the crisis period of the early 1980s. Faster labour productivity growth has also been recorded, owing in part to its cyclical nature, but the average annual gain has not been out of line with historical averages. Similarly, population growth has been somewhat more rapid, but again there has been no clear break in trend. All of the pickup is attributable to migration developments (see below). The age structure of the population continues to be propitious, as Ireland's youthful demographic composition continues to generate more potential labour-market entrants than retirees because of a baby boom which was late by international standards, with a peak in 1980.

However, two phenomena are truly new and unique in the past five years. First is the extent to which those potential entrants are indeed joining the labour force. Trend labour force participation rates[7] had been falling in Ireland for an entire generation up to the late 1980s. There followed a period of stabilisation, and in recent years a clear uptrend has developed. This is no doubt because of the ease of finding employment, the fall in the birth rate and changes in social attitudes towards women working outside the home and simultaneously raising

Figure 3. **Private consumption levels in EU countries**
Per capita PPP's, EU = 100



Note: Care should be taken to avoid reading too much into year-to-year developments since they may result from methodological changes. For example, a new rent survey in Ireland in 1996 resulted in a much lower PPP for the Irish pound that year.
Source: OECD.

Table 3. **Supply-side factors in output growth**[1]

Average annual percentage change

	1961-70	1971-80	1981-86	1987-93	1994-98
Real GDP	4.2	4.7	2.1	4.8	9.2
Excess of real GDP over real GNP	–0.0	0.8	1.9	0.6	0.9
Real GNP per employee	4.3	3.0	1.4	3.1	3.2
Employment/labour force	–0.0	–0.2	–1.9	0.3	1.8
Labour force/working age population	–0.2	–0.5	–0.4	0.1	1.7
Working age population/population	–0.2	0.2	0.4	0.7	0.9
Population	0.4	1.4	0.7	0.1	0.8

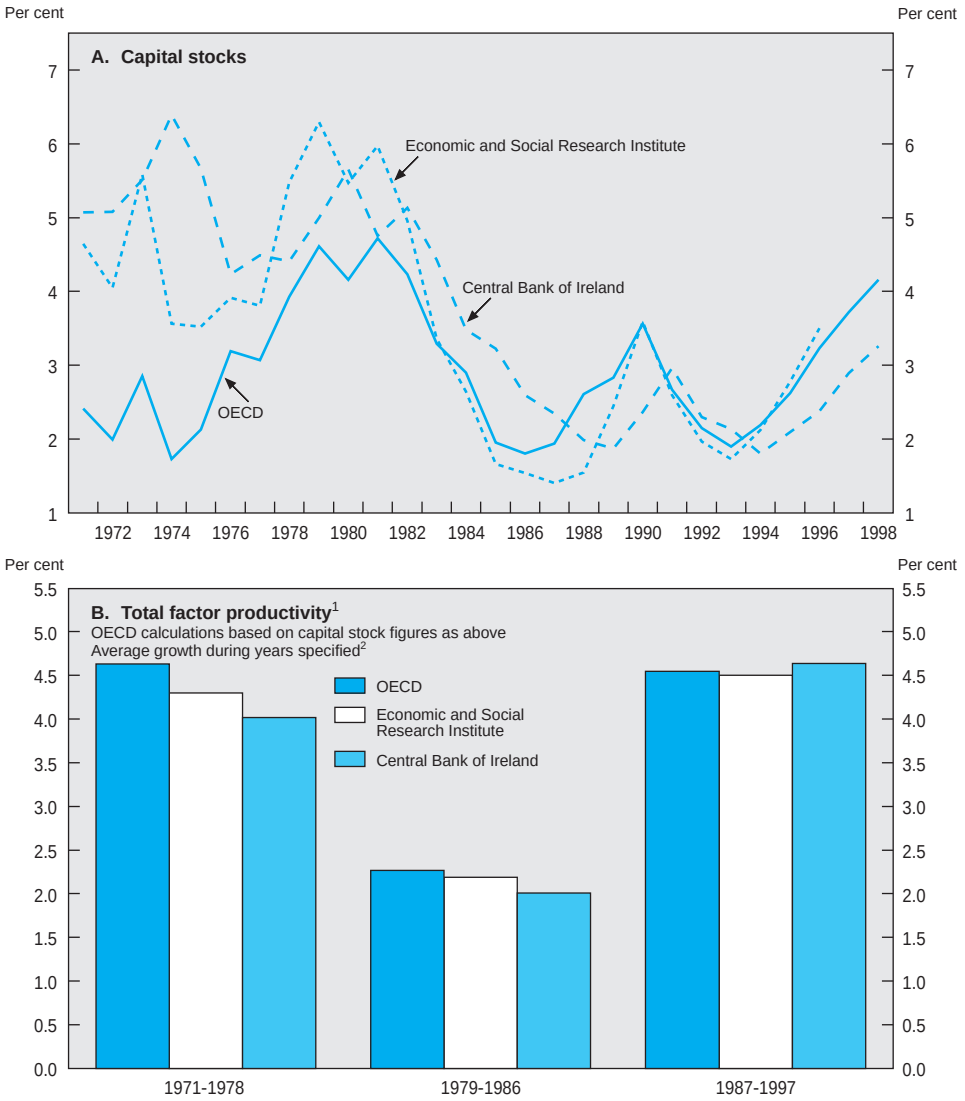
1. In principle, real GDP growth should be the sum of the factors listed below it. Any residual differences are due to rounding and the fact that growth rates are calculated exactly, rather than as logarithmic differences.

Source: OECD.

children, and the rising opportunity cost of staying at home for an increasingly educated population. The other phenomenon is the drawing down of the previously massive stock of the unemployed. As mentioned above, the unemployment rate has been cut by 9 percentage points in the last six years, providing a tremendous boost to overall available labour inputs. This drop is clearly the result of robust demand growth in the economy, which has served not only to boost the derived demand for labour, but also to raise participation and encourage a reversal of net migration flows, especially for skilled workers.

The other major factor input is fixed capital. Its contribution is difficult to assess, because of a lack of any official data on the aggregate capital stock. Internal Central Bank of Ireland estimates (Kenny, 1996), figures from the Economic and Research Institute and those in the OECD's Interlink model all provide evidence of a downshift in the rate of capital accumulation as from around 1980 (Figure 4, Panel A). With real business fixed investment falling by a cumulative 32.5 per cent in the five years to 1986, capital stock increases waned over most of the rest of the decade and have averaged less than 3 per cent in the 1990s, although not surprisingly there has been some pickup of late.[8] Nonetheless, with such a modest growth rate of capital input it is no surprise that measured total factor productivity (TFP) growth has strengthened considerably since the first half of the 1980s. TFP has increased at an average annual rate of more than 4½ per cent since 1987, similar to the rates achieved prior to 1979 (Panel B). No doubt some of that strengthening is attributable to the improvement of input quality, of labour in particular. Human capital is more difficult to measure than its physical counterpart, but a wide variety of measures indicate that a substantial upgrading has taken place over this period (see below). Adjusting for enhanced human capital this results in average TFP increases of around 2 per cent in the decade to

Figure 4. **Different estimates of business capital and TFP growth**



1. Real business-sector GDP growth less a weighted average of capital and employment growth in that sector. The weights are fixed at period-average income shares, with the self-employed assumed to earn the average private-sector compensation per employee.
2. For the Economic and Social Research Institute, the third time period is for 1987 to 1996 only.

Source: Central Bank of Ireland, Economic and Social Research Institute and OECD.

1996, with only a slight acceleration in the second half, possibly ascribable to cyclical developments (Durkan *et al.*, 1999). Nevertheless, it is clear that while factor input increases have played an important part in fuelling the growth process, total factor productivity gains have played at least as large a role, in contrast with what may have been the case in the Asian tigers, according to Paul Krugman and Alwyn Young for example. Furthermore, despite the increased use of both labour and capital, shortages, in some cases severe, have developed, as is discussed in Chapter II with regard to labour and below for infrastructure capital.

Looking beneath the surface at the causes of Ireland's success

There is a surprising degree of consensus as to the array of factors that should be included in a comprehensive list of the reasons for the two-part acceleration in growth that began around 1987. It is only as to emphasis that the experts begin to disagree. In this section each of these factors will be enumerated and assessed. The order in which they will be treated is from those which are least under the control of the Irish authorities to those where their influence has been most telling. Based on this approach, the section first looks at labour supply influences, followed by the role of human capital improvements. It then turns to the use of EU structural funds and the economy's increased openness in the context of the EU Single Market. Finally, it examines the important macroeconomic and structural policy changes that helped transform the potential growth into realised outcomes. It closes with an assessment of the strategy before a brief summing-up.

Increased labour supply

Unusually favourable demographics

As was mentioned above, Ireland has had an unusual demographic history in recent decades, with a uniquely late baby boom that got underway in the 1970s and peaked only in 1980. The result is a hump-shaped distribution of the population with a peak in recent years in the 15-19 age group. The natural increase in the population (that is, the number of births in excess of the number of deaths), though remaining positive, had been steadily sliding until recently. A trough of some 17 000 (0.5 per cent of the population) was reached in 1994-96, with a sizeable pick-up in 1997 and again in 1998 (to 21 400), as a greater share of the female population has moved into peak child-bearing years, and the total fertility rate shows some signs of unexpected stabilisation.[9] Such a fertility rate is, however, just over half what it was in 1979 to 1981. But in any case the dependency ratio (the ratio of the non-working-age population to the working-age

population) has continued to move down, and that trend is projected to persist as the population structure adjusts to lower total fertility rates (Table 4). Following a period in the 1980s and most of this decade when Ireland had a decidedly unfavourable dependency ratio by international standards, the decline in the ratio will soon bring it below the OECD average. And the old-age dependency ratio is still edging down at a time when most other countries are experiencing rising trends, some of them quite steep. The latest figures (a ratio of 17.2 per cent in 1998) would place Ireland in fourth lowest position among OECD countries by this measure.

Table 4. **Dependency ratios for OECD countries, 1985-2005**

	Total			Old-age dependency		
	1985	1995	2005	1985	1995	2005
Australia	50.9	49.9	48.3	15.3	17.6	17.9
Austria	48.2	48.1	45.4	21.2	21.8	22.1
Belgium	48.3	50.5	50.2	20.7	23.8	24.9
Canada	46.0	48.0	45.3	14.9	17.7	18.9
Czech Republic	54.2	46.1	40.6	17.8	18.4	17.3
Denmark	50.4	48.3	50.9	22.7	22.3	22.2
Finland	46.8	49.6	49.3	18.3	21.1	22.6
France	51.9	52.9	51.4	19.7	23.2	25.3
Germany	44.0	45.6	46.8	21.0	22.2	26.2
Greece	52.9	48.7	51.9	20.3	23.7	29.2
Hungary	51.2	47.2	44.5	18.7	20.6	21.0
Iceland	58.2	54.6	51.5	16.3	17.8	17.0
Ireland	**67.4**	**55.1**	**45.9**	**17.7**	**17.5**	**17.1**
Italy	47.7	45.0	48.7	18.8	23.3	28.5
Japan	46.7	43.6	50.6	15.1	20.4	28.0
Korea	52.1	40.7	40.0	6.5	7.9	11.1
Luxembourg	43.5	46.4	49.2	18.8	20.4	22.1
Mexico	85.2	66.1	56.4	6.9	7.1	8.2
Netherlands	46.0	46.0	46.0	17.6	19.2	20.7
New Zealand	53.5	52.9	51.2	16.1	17.4	17.4
Norway	55.5	54.3	51.7	24.5	24.5	21.9
Poland	53.7	51.1	43.7	14.5	16.6	17.3
Portugal	54.7	48.2	49.2	18.7	21.9	24.3
Spain	54.1	46.2	46.1	18.5	21.9	25.1
Sweden	54.8	56.4	54.0	27.7	27.0	25.6
Switzerland	46.3	46.5	46.3	20.7	21.0	22.3
Turkey	68.4	57.0	51.1	7.1	8.1	9.6
United Kingdom	52.3	54.1	51.2	23.0	24.3	24.0
USA	50.5	53.1	48.6	17.8	19.2	18.5
Mean	52.9	50.1	48.5	17.8	19.6	20.9

Source: United Nations.

A tradition of emigration

In principle, immigration is another source of labour input. Ireland's longer-term trend has of course been toward emigration: the cumulative total of net emigration from Ireland in the 110 years up to 1991 was 2.8 million persons. At peak times, such as in the 1950s, around 1½ per cent of the population was departing every year in net terms, and in 1958 there were as many emigrants as births; with few immigrants the total population fell. The 1990s, on the other hand, look set to register modest net immigration, totalling perhaps 100 000 (nearly 3 per cent of the total population), with gross emigration below 1 per cent of the population per year and gross immigration slightly above it. The turnaround has contributed up to more than 1¼ percentage points per year of population. But this is not unique: the 1970s also witnessed net immigration of about this order of magnitude. Nevertheless, what is unique is that net immigration has very recently provided a greater contribution to population increase than have natural sources (the estimated 1998 figures are 22 800 versus 21 400). A scissors phenomenon has developed in which gross immigration has risen and gross emigration has shrunk (Table 5); both trends are attributable to the strength of the local labour market. By now migration balances with all major destination/origin countries are in favour of Ireland. The surplus is largest with respect to the United Kingdom, where the largest stock of previous Irish emigrants are known to reside.[10] Emigrants are heavily concentrated in the 15 to 24 age group, but immigrants are more likely to be slightly older (nearly three-quarters are 15 to 44) or younger (their children, aged 0 to 14). These migrants have been heavily recruited in particular sectors, especially where linguistic skills are important such as telemarketing and the hotel and restaurant trade, or where shortages have been the most acute (construction). Immigrants are also more skilled than in the past (see below), and since skilled labour is likely to be complementary with capital, at least in the manufacturing sector, the availability of immigrants has allowed faster growth in this sector and, indirectly, increased the demand for services, which are then provided in many cases by unskilled workers (Fitz Gerald and Kearny, 1998).

There have recently been some attempts to assess the factors that have played a role in these migration trends. The stock of previous emigrants (the diaspora), especially those who have not been out of the country too long and who are high-skilled,[11] is one of the most important factors. One approach has been to proxy this with a long-term moving average of net emigration. Using a 15-year term, this gives a stock of 150 000 to 200 000 in recent years (Fitz Gerald and Kearny, 1998). However, a fair number of these are nearing the age when return is seen to be less likely (around 40). But the fact that they are more skilled than in the past raises the probability of return, thereby boosting the elasticity of labour supply with respect to domestic labour market conditions. Another vital factor has been the relative strengths of the demand for labour in Ireland and in potential destination/origin countries. Recent research has shown a

Table 5. **Migration by destination/origin and age group, 1992-98**

Thousands

	1992	1993	1994	1995	1996	1997[1]	1998[1]
A. Migration by destination/origin							
Emigrants							
United Kingdom	16.9	16.4	14.8	13.3	14.1	12.9	8.5
Rest of EU	7.5	7.3	5.5	5.1	5.1	4.1	4.3
United States	3.5	5.6	9.6	8.2	5.2	4.1	4.3
Rest of world	5.5	5.8	4.9	6.6	6.8	7.9	4.1
Total	33.4	35.1	34.8	33.1	31.2	29.0	21.2
Immigrants							
United Kingdom	22.7	17.5	15.2	15.6	17.6	20.0	21.1
Rest of EU	6.5	6.6	5.8	6.3	7.2	8.1	8.7
United States	4.6	5.0	4.3	3.8	6.4	6.6	4.9
Rest of world	6.9	5.7	4.8	5.5	8.0	9.3	9.3
Total	40.7	34.7	30.1	31.2	39.2	44.0	44.0
Net migration	7.4	–0.4	–4.7	–1.9	8.0	15.0	22.8
B. Migration by age group							
Emigrants							
0-14	0.7	1.1	1.4	1.2	0.9	0.7	1.4
15-24	22.5	23.6	24.6	22.6	21.4	17.9	11.8
25-44	8.8	9.1	8.2	8.5	8.1	9.6	7.0
45-64	1.2	1.2	0.6	0.8	0.7	0.9	1.0
65 and over	0.2	0.2	0.0	0.0	0.0	0.1	0.0
Total	33.4	35.1	34.8	33.1	31.2	29.0	21.2
Immigrants							
0-14	6.2	5.6	4.4	5.3	6.6	6.4	7.2
15-24	12.5	10.3	9.7	8.0	10.9	13.8	12.0
25-44	16.5	14.5	12.1	14.6	16.9	18.2	19.1
45-64	4.1	3.6	3.1	2.6	3.6	4.4	4.2
65 and over	1.4	0.8	0.9	0.7	1.2	1.3	1.6
Total	40.7	34.7	30.1	31.2	39.2	44.0	44.0
Net migration	7.4	–0.4	–4.7	–1.9	8.0	15.0	22.8

1. Preliminary.
Source: Central Statistics Office.

semi-elasticity of net emigration as a share of the labour force with respect to the domestic unemployment rate of around 0.6 (Kearney, 1998). The major reference country has traditionally been the United Kingdom, and it has become widely accepted that the Irish and UK labour markets are very closely linked.[12] Third, the explanation of migration trends would have to comprise relative real wage developments; these do not appear to have followed any particular trend in recent years. Fourth, with improved transportation and communications, the perceived costs of migration may have been reduced, because it is more easily reversible.[13] Finally, for less-skilled potential migrants the relative generosity of the social

welfare systems is also likely to be relevant. Ireland's system has become steadily more generous over time[14] and the long-standing gap with the United Kingdom, for example, in this domain was reversed in the 1980s.

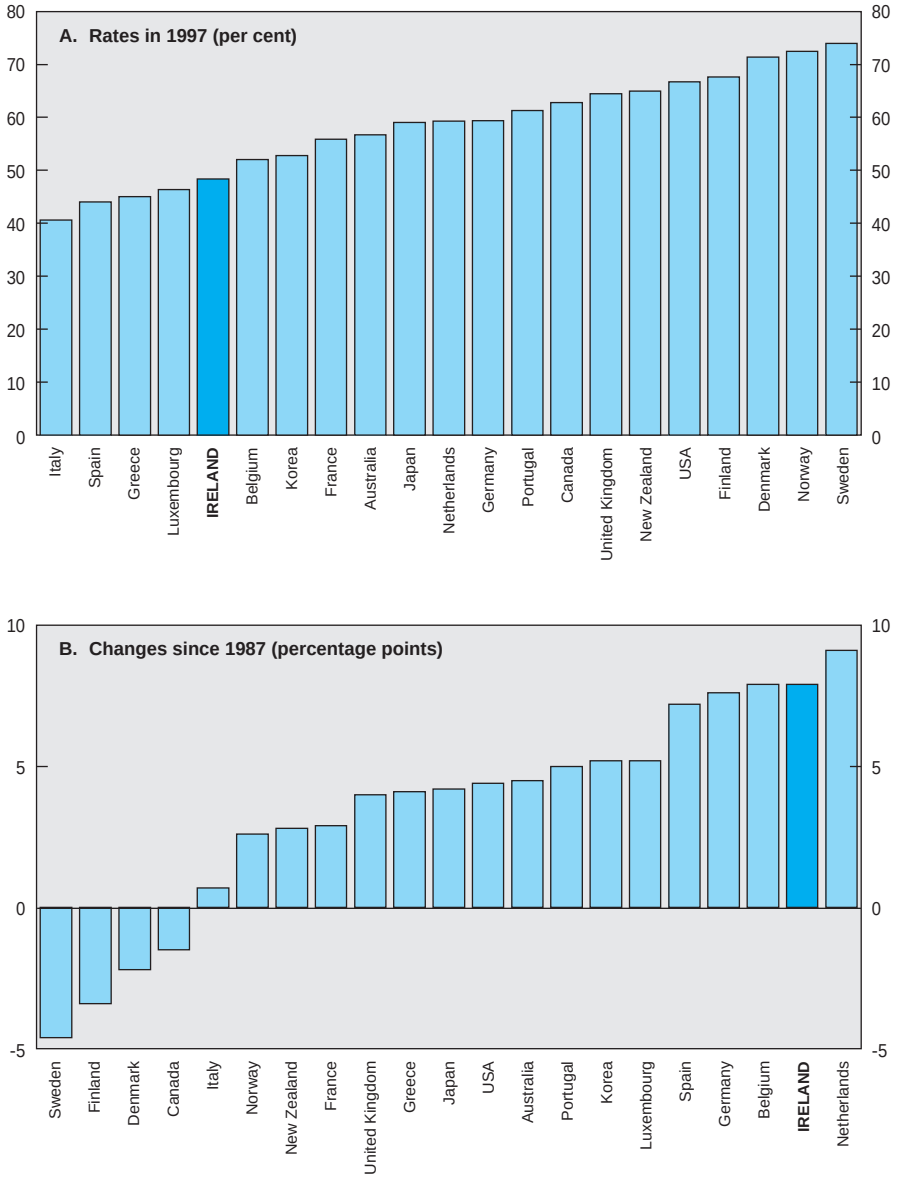
Rising but still-low participation

The final source of increased labour input to the economy as a whole is rising labour force participation. In Ireland's case, participation has long been low by international standards, at least for women. Most other OECD countries experienced rapidly rising female participation rates in the 1970s, but Ireland's trend was much shallower (Figure 5). Since then increases have been more rapid in line with improvements in educational attainment,[15] but a significant gap remains nonetheless. For example, an Irish woman of working age is nearly 20 per cent less likely to be in the labour force than her UK counterpart. The gap with respect to the Nordic countries is around 25 percentage points.

Irish women are less likely to be in the labour market for a number of reasons. First, work incentives are restrained by the tax and transfer system (OECD, 1997*a*). While the average total tax wedge does not appear high by international standards, marginal rates facing low-income workers are fairly high, and the wedge is very high for some categories of single workers. However, it is well known that it is married women whose participation is the most sensitive to economic incentives, and Ireland's tax system allows personal tax allowances to be transferred between spouses, boosting secondary earners' marginal tax rates to over 50 per cent. As mentioned above, social welfare benefits have become increasingly generous over the past 20 years or so[16] (Figure 6), thereby offsetting much of the favourable participation effect which would have otherwise resulted from higher opportunity costs of not working following steady rises in educational attainment (see below).[17] But some progress has been made in recent years in reducing the tax wedge and work disincentives more generally (see Chapter III). Personal tax rates have been cut; allowances have been raised; the standard rate bands have been widened; and social security contribution rates have been lowered. Social welfare reforms in recent years (to housing, family income and medical benefits) have also sought to mitigate disincentives by integrating the tax and transfer systems to a greater extent and by reducing the preponderance of means testing in benefit entitlement determination (O'Connell, 1998; Walsh, 1999).

Second, the very rise in educational attainment just referred to has meant that fewer workers of both sexes, especially in the 15-24 age group, are available for work: in recent years the number of people over 15 identifying their principal economic status as students has risen by nearly ten thousand per year, and the 15-24 year-old participation rate has fallen about a percentage point per year. But this factor has been more important for women, with an average increase double that of men. A third limiting factor is the extent to which the demand for

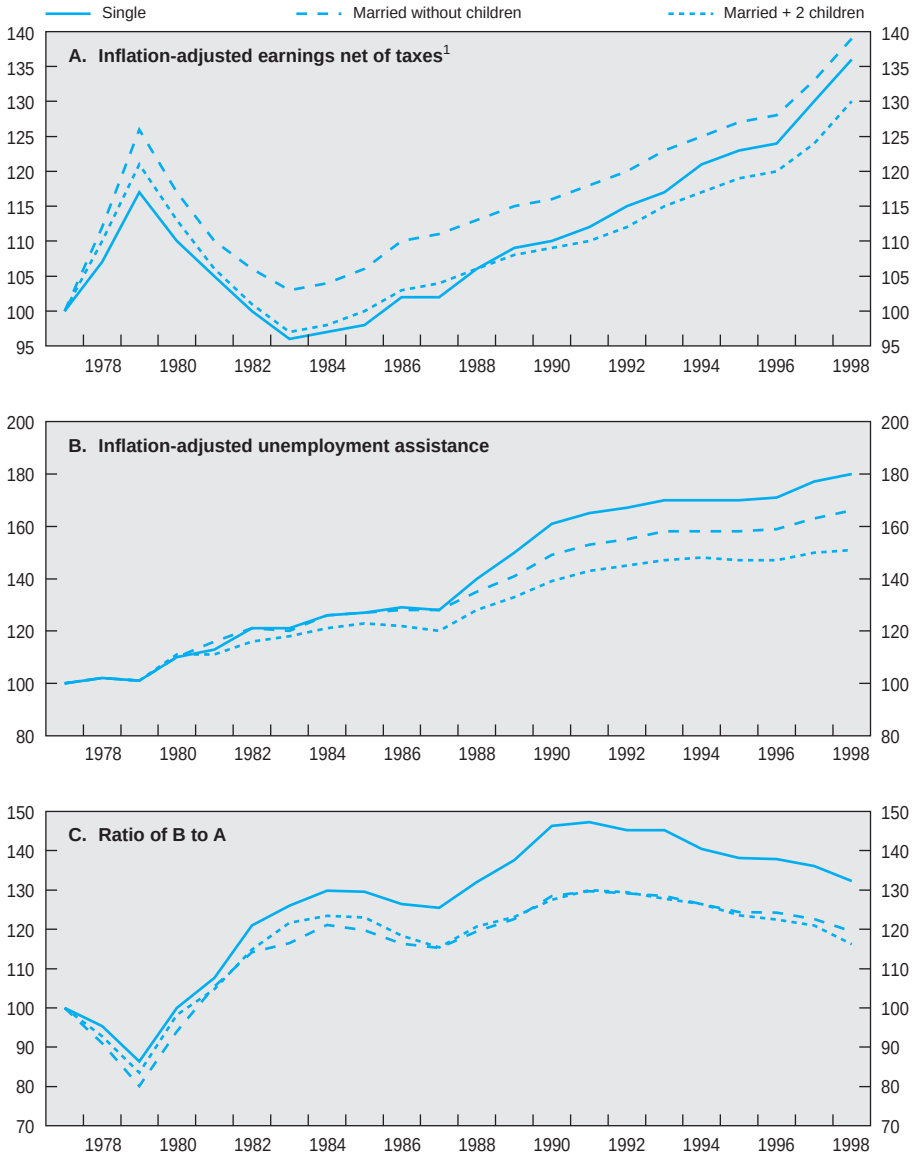
Figure 5. **Female participation rates in OECD countries**



Source: OECD.

Figure 6. **The generosity of social welfare benefits, 1977-98**

1977 = 100



1. At average wage in the manufacturing sector.

Source : Department of Finance.

affordable childcare places now exceeds supply. This is a controversial area of long-standing concern.[18] While in previous generations it had been marriage that put an end to many women's careers (the so-called "marriage bar"), it is now most often child-bearing, particularly for those who are single. But even this has been less of an influence in the past decade than in the preceding ten years, as the fall in participation for women from their early 20s to their early 30s is only half as large as it was previously (FAS, 1998). And the return to work after raising a family has become more pronounced as well. Closely related to this discussion is the historically limited share of part-time jobs for women in the economy (although this seems now to have risen to international norms),[19] a form of employment especially suited to working mothers' needs, and insufficient working-time flexibility. Finally, employment has become increasingly costly for women as well as men in recent years in non-financial terms due to the increase in average commuting distances and times[20] resulting from growing congestion not only in Dublin, but nation-wide.

Freed-up labour from shrinking agriculture

The needs of the expanding part of the economy can also be met by the shrinkage in declining industries, the primary sector in particular. In employment terms, agriculture, forestry and fishing were together responsible for nearly 40 per cent of total employment as recently as 1960. The amount of freed-up labour was about 10 per cent of total employment in each of the following two decades, but fell to around 4 per cent in the 1980s. Since then it appears to have picked up again to a decennial rate of around 8 per cent. But given that the employment share is now only around 9 per cent, such a pace cannot be maintained for much longer.[21]

Improved education and skills

The preceding discussion shows that increased labour supply has been a significant part of the story of rapid economic growth. But the effective supply of labour includes a quality component attributable to human capital accumulation.[22] Recent estimates[23] vary from around 0.6 percentage points of labour input per year in the 1980s and 1990s (Fahey *et al.*, 1998) to as much as 1 per cent in the period 1986-96 and even 1.5 per cent in 1991-96 (Durkan *et al.*, 1999), as Ireland's population structure has shifted from generations born prior to 1930 to those born in the 1960s and 1970s who have a much higher level of educational attainment.[24] While only 30 per cent of the population in the 55 to 64 age group had completed at least upper secondary education in 1996, the corresponding figure for the 25 to 34 age group was 66 per cent (Table 6, Panel B). That gap, an indicator of changing human capital, was well above the OECD country mean. The same holds for university-level education attainment.

Table 6. **Educational attainment and incentives to invest in human capital**

	Total population		Labour force	
	Ireland	OECD country mean	Ireland	OECD country mean
A. Educational attainment for adult (25-64) population, 1996 (per cent)				
Below upper secondary	50	40	43	34
Upper secondary	28	40	29	43
Non-university tertiary	12	10	14	11
University level	11	13	14	15
	At least upper secondary		At least university level	
	Ireland	OECD country mean	Ireland	OECD country mean
B. Percentage of the population with at least a certain level of attainment by age group, 1996				
25-34	66	72	14	15
35-44	54	65	11	14
45-54	38	55	9	12
55-64	30	42	6	8
25-64	50	60	11	13
	Age 25-64		Age 30-44	
	Ireland	OECD country mean	Ireland	OECD country mean
C. Percentage point differences in unemployment rates by educational attainment and gender, 1996				
Below upper secondary minus university: men	13	7	15	8
women	15	7	19	9
Upper secondary minus university: men	3	3	3	4
women	4	4	6	4
	Age 25-64		Age 30-44	
	Ireland (1994)	OECD country mean (latest year)	Ireland (1994)	OECD country mean (latest year)
D. Relative earnings for persons with income from employment by educational attainment and gender (upper secondary = 100)				
Below upper secondary: men	77	77	78	81
women	62	75	61	75
University level: men	171	162	169	160
women	187	157	197	159

Source: OECD calculations based on OECD (1998*a*).

According to the most recent OECD figures (for 1996), a greater share of Ireland's labour force has completed tertiary education than for the average Member country (28 versus 26 per cent), but a much higher share has less than upper secondary education (43 compared to 34 per cent) (Table 6, Panel A). This process is not complete: the replacement of older cohorts of workers having modest levels of formal education with younger, more schooled cohorts will continue for about another 20 years, as it takes up to 50 years for early school leavers to exit the labour force, and it was only in 1967 that free second-level education was introduced.[25] According to Duffy *et al.* (1997), human capital upgrading will still be contributing an average of 0.4 percentage point per year to effective labour supply over the next decade; Durkan *et al.* (1999) claim the benefit will be even larger, some ¾ percentage point per year.

Despite the steadily increasing supply of well-educated new entrants into the labour force, the demand has risen apace. In fact, average earnings figures for 1987 and 1994 by level of education show that the earnings penalty to less than upper secondary education rose over this period, as did the marginal returns to university education for men, although not for women. Overall, the returns to education rose (Barrett, Callan and Nolan, 1997). By the end of that period the return to education in Ireland was higher for men than on average in the OECD area, mainly at the post-secondary level, and much higher for women at both pre- and post-secondary levels (Table 6, Panel D). Besides the return to education in the form of earnings, unemployment rates also vary according to educational attainment, and the differentials were much larger in Ireland than in the average OECD Member country, at least between those not having completed upper secondary education and those who have (Table 6, Panel C).

The result is that the numbers staying on in upper secondary and tertiary education have grown strongly over the last generation,[26] and claims were recently made that the level of capital spending on education has not kept up,[27] although very recent budgetary allocations have alleviated some of the shortfall at the primary level at least. Currently about 96 per cent of any cohort complete the Junior Certificate and 82 per cent upper secondary Leaving Certificate, but the goal is to boost that latter figure to 90 per cent by 2001, since the current level of school-leaving is regarded as excessive[28] in view of the enormous implications for future income prospects. The compulsory schooling age is still 15, although it is widely expected to be raised to 16 in the near future. However, even that level might be considered inadequate.[29] Overall school expectancy is low by OECD standards, and that is mainly at the upper secondary level, although much of that is attributable to a frequent early transition to university-level education. The so-called "transfer rate" (the share of those eligible who do go on to tertiary education) has risen from about 45 per cent in 1990 to around 60 per cent today, putting tremendous strains on tertiary capacity.[30] These pressures are still building, but demographic trends should begin to bring about an easing in the near future.

EU structural funds

Even though around 65 per cent of EU transfers to Ireland are still farm-related, it is clear that structural funds under the Community Support Framework (CSF) have made a significant contribution to the economy since its inception in 1989. Several evaluations of the CSF have been conducted (Barry *et al.* (1997) and Honohan (1997), for example), and their assessment is that growth was boosted as the transfers geared up at the outset. After four years payments had reached 2.8 per cent of GNP and nearly 3½ per cent if one includes other structural funds not strictly part of the CSF; this implies substantial demand effects even without any multiplier considerations. But in the following three years payments under the second CSF programme were actually lower than those made in 1993. It was only in 1997 and 1998 that there was a renewed direct demand-side stimulus of around ¼ percentage point per year, bringing the *level* of CSF funding back to 2¼ per cent of GNP (around 2½ per cent for all structural funds).

But it is the longer-term supply-side effects that are likely to be the most significant. CSF funding is used primarily to pay for new capital stock both in tangible and intangible forms (road construction, higher education, job retraining, etc.),[31] many of which can be expected to yield positive externalities, and as these investments are implemented the economy's productive potential is progressively enhanced. Even if there were to be no further structural funding following the expiry of the current CSF at the end of this year (see below), the long-run boost to the level of potential supply is estimated to be 1 to 2 per cent (Barry *et al.*, 1999).[32] Furthermore, there have been other less direct benefits as well. The programme has prevented under-funding of infrastructure and higher education which might have occurred in the late 1980s drive to cut all forms of public spending (Bradley, 1998). This might well have led to a diminished ability to capture such a high share of global direct investment flows in the 1990s (Barry, 1999). And it has raised the quality of public investment outlays by forcing the introduction of longer-term project planning, so that short-term budgetary pressures have not led to stopping an undertaking with the extra cost of subsequently re-starting it. Thus, summing up, average growth in the most recent decade has benefited from structural funds to the extent of ¼ percentage point per year plus any multiplier, supply-side and other indirect static and dynamic effects, a total which might easily attain ½ percentage point per year or more.

Increased openness and EU market integration

Besides the foreign linkages in the labour market through migration flows already discussed and those in the capital market through direct investment discussed below, productivity and output in Ireland have profited from increased openness and integration in the goods market. Ireland has been liberalising its trade regime ever since the late 1950s. It has had free trade with the

United Kingdom since 1965 and joined what is now the EU in 1973. A simple shorthand measure of openness to trade is the ratio of the sum of exports and imports of goods and services to total GDP. By this proxy Ireland has the most open goods market in the OECD (other than Luxembourg), with total trade amounting to 163 per cent of GDP in 1998. But what is relevant here is the indication that this fraction has surged from only 114 per cent in 1990, a far faster rise than elsewhere (Figure 7) and one which is reflected in both export and import components. The principal reason for this jump is the export-platform inward investment that has been underway for many years but which accelerated over the past decade (see below). But one of the attractions for that investment has no doubt been the Single European Market (SEM) programme. The reforms that have flowed from the SEM have been much more important than the related increase in structural funds that ironically were intended to offset the disadvantages that were thought to exist for the peripheral EU Member countries including Ireland – the so-called cohesion countries. But the impact of the SEM programme has turned out to be more favourable for some of these countries, including Ireland, than for the core EU Members.

The SEM has operated through a number of different channels. First, it has brought about a slight reduction in government employment as a result of the abolition of customs barriers. Second, there have been the usual static gains in productivity and output from liberalisation (through the exploitation of scale economies and the elimination of X-inefficiencies),[33] all of which have put downward pressure on costs and prices. Third, there have no doubt been dynamic gains as well from faster growth, as described in the endogenous growth literature, operating through learning-by-doing, human capital accumulation and R&D channels (Barry, 1999). Fourth, it has led to rationalisation in a number of sectors, most prominently associated with distribution. Model simulations of these effects were carried out by the Economic and Social Research Institute (Barry *et al.*, 1997); the results (Table 7) show larger effects than were even predicted by the original Cecchini analysis for the core countries. The economy is estimated to have grown about a percentage point faster per year since 1992 than it would otherwise have done, with most of the lift experienced by the manufacturing sector. The stimulus to output and incomes is thought to have pushed down the unemployment rate, but mainly to have boosted productivity. Because so much of the demand increase has been focused on exports, the impact on the trade balance has probably been favourable.

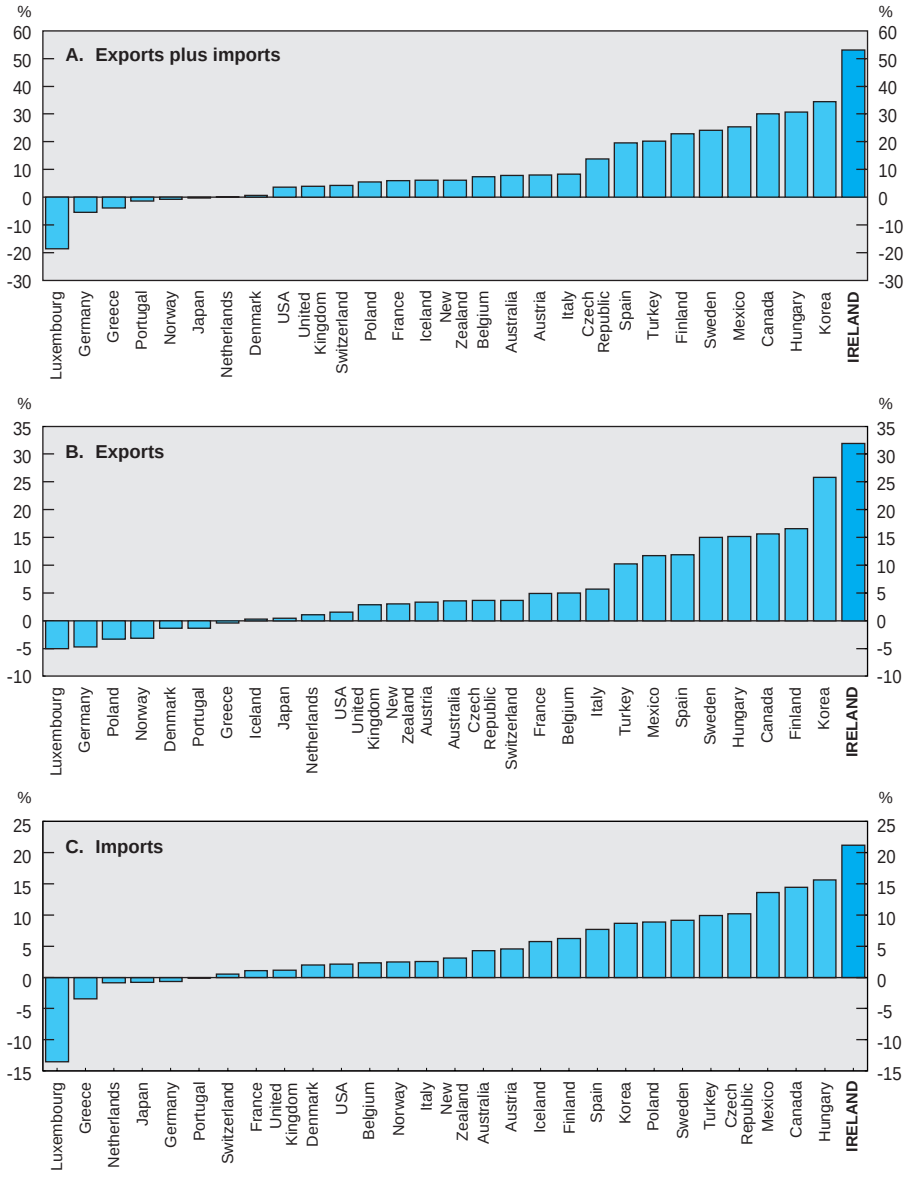
The domestic policy environment

The part played by fiscal policy

As mentioned at the outset one domain that has shown remarkable progress since the mid-1980s is the public finances. The fact that this

Figure 7. **The change in trade in relation to GDP in the 1990's**

Percentage points, 1990-1998



Source: OECD.

Table 7. **The total effect of the Single European Market programme on Ireland**

	1995	2000	2010
Deviations from a no-SEM baseline (in per cent)			
GDP at factor cost	5.0	7.2	9.2
Manufacturing GDP	11.1	15.1	17.1
Market services GDP	2.4	3.6	4.8
Deviations from a no-SEM baseline (in percentage points)			
Unemployment rate	–1.4	–1.6	–1.2
Net trade surplus (per cent of GDP)	0.9	1.3	1.6
National debt (per cent of GDP)	–5.5	–11.1	–21.6

Source: Barry *et al.* (1997).

improvement coincided with the pickup in output growth has prompted the view that the increase in growth might have resulted from the fiscal consolidation. In theory an "expansionary fiscal contraction" might result from an increase in spending by far-sighted private-sector agents, who correctly discount the implied reduction in future tax rates to meet the government's inter-temporal budget constraint, that is sufficient to more than offset the direct contractionary effects of the budgetary tightening. However, this seems highly unlikely to have actually occurred (Bradley and Whelan, 1997). Indeed, Irish fiscal policy has most often been accused of having been pro-cyclical (Lane, 1998; Bradley, 1998). Rather it seems likely that Ireland was merely fortunate in undertaking much of its fiscal correction at a propitious moment with respect to world economic developments. The combination of declining interest rates at home and abroad, buoyant export market growth, improving cost competitiveness and terms of trade gains provide a satisfactory alternative explanation for what occurred in the last half of the 1980s. Nevertheless the fiscal improvement did remove those concerns for all private agents and thereby made for a more supportive environment. Furthermore, Ireland's case does provide support for the hypothesis (Alesina and Perotti, 1997) that fiscal adjustments based on expenditure reductions are more likely to succeed than those relying on tax increases. Much of the adjustment was focussed on health spending and public service salary costs. Finally, assessing the impact of fiscal policy in the last few years is difficult because of the lack of a clear measure of the effect of the business cycle on budgetary outlays and tax revenues. But it should be pointed out that discretionary changes in fiscal policy in the last five years have indeed been stimulative (Bradley, 1998): the automatic stabilisers have not been allowed to function fully to brake the economy's underlying accelerating trend.

A change in attitudes and institutions: the role of incomes policy

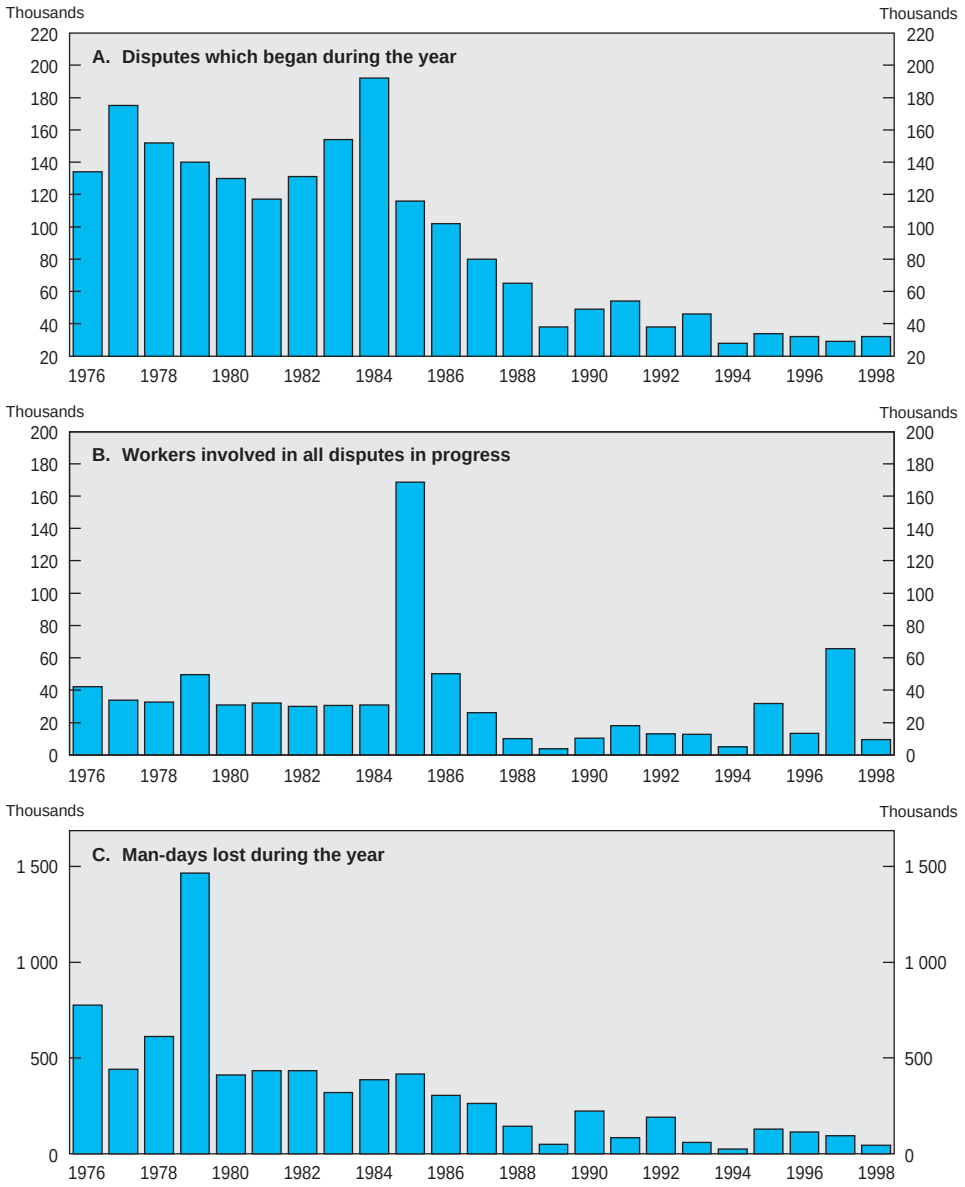
The economy has been supported by the coalescing and maintenance of a national political and social consensus since 1987. Since the situation had become so desperate, a spirit of self-reliance took hold and radical policy changes became politically feasible. Besides the aforementioned expenditure cuts, a tax amnesty in 1988 raised revenue in the short term and permanently reduced the size of the underground economy. Overstaffing, restrictive practices and cross-subsidisation became the focus of major restructuring, both in the private sector and among state-owned enterprises. Trade union leadership took a far longer-term perspective, with the result that pay claims became more reasonable and industrial actions fewer and less severe (Figure 8): the average number of working days lost due to strikes dropped by around two-thirds following the institution of centralised bargaining (Taylor, 1996).

The clearest manifestation of the change in the policy environment was in the series of four three-year national pay agreements (the most recent of which is P 2000, expiring at the end of March 2000 in many parts of the private sector and at the end of September 2000 in the public service) and in the broadening of the so-called "partnership" approach to policy setting. Centralised accords have assisted management in its introduction of flexible working practices and of innovative forms of technology (Taylor, 1996). These agreements may have prevented productivity gains that resulted from new foreign investments from propagating to wages more generally. This would have quickly led to the development of a "Dutch disease" phenomenon in which traditional, lower-productivity sectors would have been squeezed out of production by high real wage increases emanating from the modern high-growth sectors in spite of very high levels of joblessness. Similarly, hysteresis was avoided, as the agreements bound labour-market insiders to take account of the plight of their outsider counterparts. The pacts may have also limited the widening in the wage dispersion, thereby helping to build widespread support, despite the possible longer-term negative supply implications. Finally, they interacted favourably with budgetary policy by pushing towards tax cuts and especially spending moderation rather than tax increases as the means to deficit reduction.[34] The result of what ultimately amounted to a national strategy of wage moderation[35] has been a sharp drop in labour's share of value added since 1987, which has enabled a dramatic recovery in profitability following an earlier deterioration. Restoring profitability to adequate levels has no doubt contributed importantly to the much improved employment outcomes and the resurgence of investment witnessed recently.

The important effect of industrial policy

But possibly the most important policy channel that has underlain Ireland's economic success is its "outward-oriented interventionist strategy"

Figure 8. **The declining frequency and severity of industrial work stoppages**



Source: Central Statistics Office.

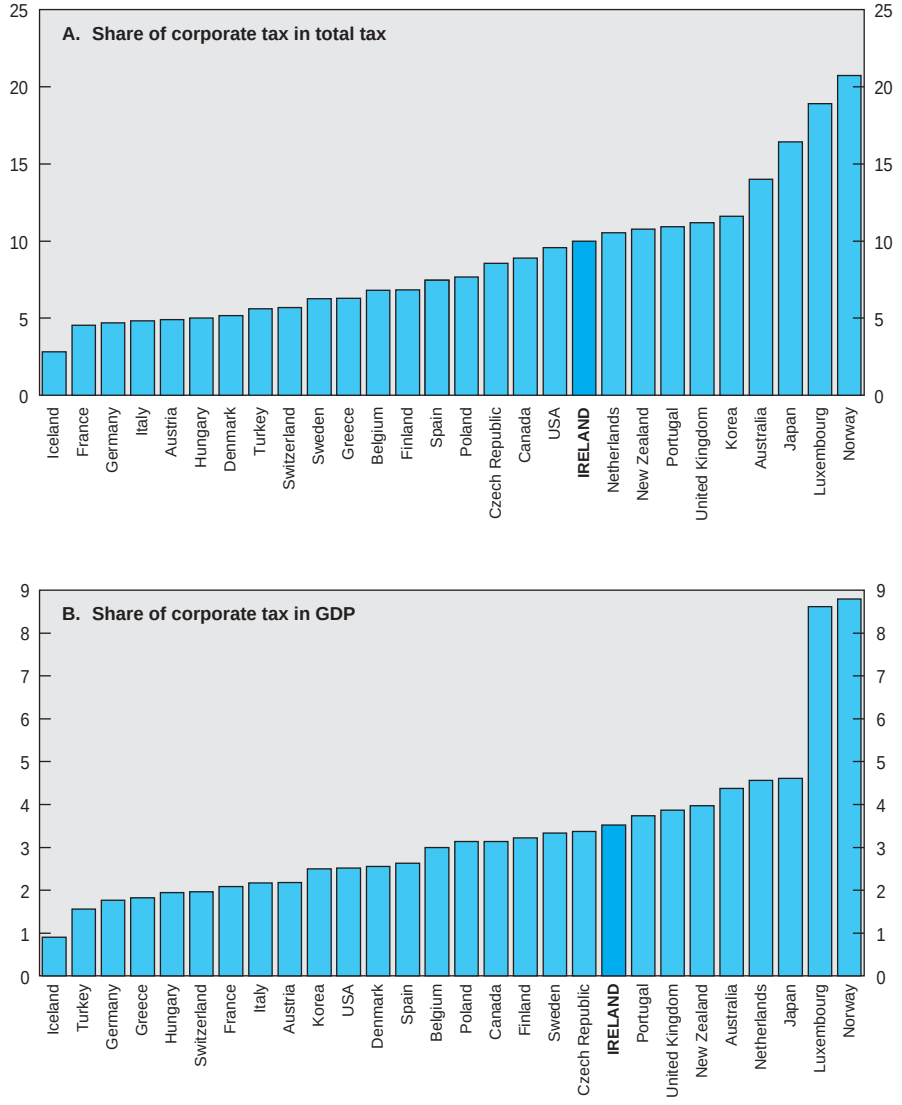
(Barry, 1999): the consistent use of tax and financial incentives to attract inward foreign investment, mainly in manufacturing, since the late 1950s.[36] While other countries also made financial incentives available, their offers have not generally been as complete. Tax treaties and global averaging rules combined with the low corporate tax rate applied to manufacturing and internationally traded services[37] make the tax regime especially attractive to US firms. However, while statutory rates have long been low, the base has been comparatively broad, with the result that corporate tax does not represent an unusally small share of either total tax or of GDP (Figure 9).

Industrial policy advice has been provided by Forfas[38] since 1994. Current and potential foreign-owned firms are the target of the Industrial Development Authority (IDA), which has 280 employees and 15 overseas offices. Over time it is said to have put an increasing focus on the services sector. Originally, that was mainly the computer software sector and, as from 1987, the International Financial Services Centre (IFSC) in Dublin. Then in the early 1990s Ireland began successfully to penetrate the "teleservices" industry (call centres), and in recent years it has concentrated on so-called "shared services" – corporate overhead services such as personnel, accountancy and sales support. Electronic commerce-related activity is expected to expand in the future. However, in order not to disadvantage indigenous industry, grants have long been available to Irish-owned firms as well. Most recently, there has been an amalgamation of several of the State agencies dedicated to the development of indigenous firms[39] to form Enterprise Ireland, with employment of around 1 300 and an annual budget of 220 million pounds (about ½ per cent of GDP).

Industrial support is provided to subsidise fixed investment,[40] to provide employment/training grants, to take an equity position in Irish start-ups and to assist in R&D activity, management development and market research. The justification is that there are a number of externalities that private agents will not internalise and that will provide favourable spillovers that are worth paying for – most prominently the reduction in unemployment[41] but also, increasingly, the undertaking of innovative activity in frontier industries by small firms and start-ups.[42] Cost-benefit analysis is used to help select the individual fixed investment projects that will receive assistance. Final decisions do, however, rely on judgements that take a variety of sectoral and regional factors into account. However, there has been some criticism of some of the methodology used in this analysis, and its use is being reviewed (see Chapter II).

Besides the indirect benefits from EU structural funds, the private sector also enjoys further direct aid amounting to several hundred million pounds a year out of the government's budget (Table 8). Besides agriculture, fisheries and railways, it is of course largely the manufacturing sector that benefits from such assistance, primarily for regional objectives. At both the manufacturing and the

Figure 9. **Ireland's corporate tax position**
Per cent in 1997[1]



1. 1996 for Japan, Italy, Belgium, Netherlands, Poland and Turkey.
Source: OECD.

Table 8. **State aid to the private sector, 1994-96**

	Ireland	EU 15
A. Total aid to manufacturing (million ECU)	215	38 318
ECU per person employed	838	1 238
Per cent of value added	1.5	3.0
Objectives (in per cent)		
Horizontal	37	30
of which:		
R&D	6	9
SMEs	17	7
Sectoral	7	13
Regional	56	56
Total	100	100
B. Fisheries – per cent of gross value added	8.4	4.9
C. Railways – per cent of gross value added	6.7	29.4
D. Overall national aid (exc. agriculture)		
Per cent of GDP excluding agriculture	0.8	1.4
ECU per person employed	312	573
Per cent of total government expenditure	2.1	2.6

Source: European Commission (1998), "Sixth Survey on State Aid in the European Union in the Manufacturing Sector and Certain Other Sectors", COM(98)417.

overall levels, direct aid was less generous than the EU average in the period 1994-96, but more recent figures show a reversal for manufacturing.[43] In the 1990s there has been a fairly steady acceleration in the number of jobs "created" by government grants. However, the grant cost per job *approved* rose from around Ir£ 10 400 in 1989-91 to Ir£ 17 100 in 1994. The authorities use the grant cost per job *sustained* for a period of seven years (since a substantial share of grant-aided jobs had earlier been shown to be ultimately "lost", especially for indigenous firms) following the initial grant as a measure of policy efficiency. This has fallen some 42 per cent in real terms since 1989 to around 10 900 pounds in 1998.[44]

Successful attraction of foreign investors

Data problems are particularly severe for Ireland, but it appears that foreign direct investment (FDI) inflows slackened off in the second half of the 1980s, when lack of investor confidence in European Community institutions fundamentals reached its peak (Table 9). But with the development of the Single European Market Programme and the more favourable host environment, it took off strongly again in the early 1990s and again more moderately in the mid-1990s,[45] thereby gaining a greater share of OECD host country *inflows*. Over

Table 9. **Ireland's receipt of OECD direct investment flows**

Average shares in per cent[1] except where noted

	1981-85	1986-90	1991-97
Average annual level ($ millions)	178	96	1 124
Share of OECD total inflows	0.48	0.06	0.66
Share of OECD total outflows	0.48[2]	0.17	0.17
of which outflows from:			
United States	1.58[2]	0.48	0.45
United Kingdom	0.28[2]	0.19	0.11
Germany	0.26[2]	0.20	0.09
Japan	0.27[2]	0.07	0.07
Other EU	0.29[2]	0.08	0.02
Other OECD	0.10[2]	0.04	0.09

1. Irish figures are for net inflows. Thus, all figures are biased towards zero to the extent that Ireland is also a source of outward direct investment flows. US figures show that Ireland's direct investment position in the United States in 1997 was $10.5 billion in historical-cost terms, as compared to $14.5 billion in the opposite direction.
2. 1983-85 only.

Source: OECD (1998), *International Direct Investment Statistics Yearbook.*

the past fifteen years a cumulative Ir£ 6 billion (an average of 2 per cent of GDP per year) have been invested in Ireland on direct investment account, about two-thirds of which from the United States (over 80 per cent in more recent years).[46] With virtually all of it being in the form of greenfield investments and expansions rather than mergers and acquisitions, the first-round impact must be considerable. Nevertheless, Ireland's share of total origin-country *outflows* may have been slightly lower in the 1990s than it had been in the 1980s.

The key questions here are what have been the reasons for Ireland's success in attracting this kind of foreign investment and then what has been its total impact? Besides the tax and other financial incentives offered and the barrier-free access to the EU market, especially following the SEM programme, the Irish industrial authorities have managed to capitalise on a number of other trump cards at their disposal. First, there has never been any antipathy to foreign investors in either the government or the wider public, in possible contrast with a number of other potential host countries. Second, the ready supply of both highly skilled and less skilled labour at a reasonable cost has until recently been an important advantage. Third, the regulatory environment has never been overly burdensome, even if the infrastructure still leaves some important gaps with leading competitor nations (see Chapter II). Fourth, given the English-speaking environment, there is no language barrier for US investors, who also often have a historical/cultural connection in any case. Fifth, Irish banks have been willing to provide loans to foreign investors who repay them out of cash flow before repatriation. Finally, there are significant first-mover advantages in this field. The IDA

had the chance to develop the specialised skills needed to market the nation as an investment location. Once one or more large firms decide to locate in Ireland, other competitor companies often make the same decision (a sort of "band-wagon" effect). Furthermore, once a critical mass is established, so-called "agglomeration economies" may begin to set in (Barry *et al.*, 1999), and, if conditions are appropriate, "clusters" may eventually begin to form, as they seem to have done in the software industry (O'Gorman *et al.*, 1997).[47]

The upshot of all this inward investment has been the creation of an export-oriented, skilled-labour-intensive sector, primarily concentrated in high-technology areas, notably electronics (including software), health care and pharmaceuticals, but also in internationally traded services such as software, telemarketing, financial services and shared corporate services (accounting, for example). By 1996, foreign-owned firms were responsible for 47.0 per cent of manufacturing employment, 66 per cent of the sector's gross and 77 per cent of its net output and 84 per cent of industrial exports (Table 10), all much higher shares than in earlier years. The foreign-owned share in this sector alone represented about 26 per cent of GDP already in 1996 and probably more than 30 per cent today. About half the employment and more than two-thirds of the net output attributable to foreign-owned firms are from companies originating in the United States, with the United Kingdom and Germany being other important inward investors. Foreign-owned firms have far larger plants;[48] they have much higher labour productivity levels,[49] only part of which is reflected in higher wages and salaries per head,[50] resulting in much higher profitability; they import much more of their material inputs and export nearly all of their output (except those firms from the United Kingdom). The destinations of their exports differ from those of Irish-owned firms in that they are directed less to the United Kingdom and more to the rest of the EU. US figures allow some useful disaggregation of direct investment *stocks* and *profitability* by sector (Table 11).[51] Manufacturing is of course the main recipient of US direct investments, but the importance of the non-bank financial sector has been growing rapidly in recent years. It appears that Ireland's share of the stock of US direct investment capital held abroad has jumped since 1995, both overall and in most sectors except the residual services sector. Rates of return on those investments in Ireland have consistently been nearly double those earned on average (Panel B), with persistently large differentials recorded for the chemicals and electric and electronic equipment sub-sectors.[52]

Favourable linkages and spin-offs

The presence of so many foreign enterprises has no doubt had a substantial influence on the rest of the domestic economy. Even though these subsidiaries tend to import a great deal from abroad, they nevertheless spend a lot on

Table 10. **Manufacturing plants: characteristics by ownership, 1996**

Characteristics	Nationality of ownership							
	Irish	Other EU			Non-EU		Total foreign	Total
		Total	United Kingdom	Germany	Total	United States		
Number of plants	3 871	344	117	98	384	286	728	4 599
Total persons engaged	120 224	37 114	12 283	10 684	69 296	54 167	106 410	226 634
Gross output (Ir£ millions)	12 188	4 765	1 960	855	19 343	15 814	24 108	36 296
Purchased materials (per cent imported)	25.1	65.2	52.9	80.3	65.7	61.6	65.6	47
Output exported (per cent)	34.0	70.5	53.4	93.2	93.9	95.3	89.3	70.7
Net output per person engaged (Ir£ thousands)	34.6	65.3	87.4	35.9	166.7	177.9	131.3	80
Wages and salaries per person engaged (Ir£)	14 324	17 716	19 672	15 679	17 467	17 473	17 554	15 841
"Profit"[1] per person engaged (Ir£ thousands)	20.3	47.5	67.7	20.2	149.2	160.4	113.8	64.2
Destination of exports (per cent of total)								
United Kingdom	42.2	36.8	74.3	12.3	19.9	20.1	22.6	25.7
Other EU	32.2	46.8	10.9	72.7	50.6	52.5	50	47.1
United States	8.2	6.8	5.4	6.4	11.4	9.5	10.7	10.3
Rest of the world	17.3	9.6	9.3	8.6	18.1	17.9	16.8	16.9

1. Net output less wages and salaries.
Source: Census of Industrial Production.

domestically produced goods and services. The government instituted a National Linkage Programme in 1985 to try to encourage domestic raw materials purchases, but it is unclear how well this has worked, as foreign firms still source significantly less of their inputs locally than indigenous firms.[53] Nonetheless, by 1996 foreign enterprises in the manufacturing sector were responsible for nearly as much Irish economy expenditure as their Irish-owned counterparts (Table 12). A much greater share of these linkages was in the form of purchased services, since the foreign firms tend to be located in sectors with a low labour component and with few materials requirements. An input-output approach shows that already by 1991 there were 62 vertically-related jobs for every 100 direct manufacturing jobs in foreign-owned firms and only 51 in Irish-owned firms (Barry *et al.*, 1999).

The presence of leading multinational corporations in Ireland has also provided a number of other benefits. For example, they expose their Irish employees to best practices in a number of domains (technology, marketing, etc.) and provide a useful training ground for managers and potential entrepreneurs.

Table 11. **Recent trends in US foreign direct investment stocks and rates of return**

Per cent except as noted

	1994			1995			1996			1997		
	Ireland $ billion	Irish % share in Europe	Irish % share in Total	Ireland $ billion	Irish % share in Europe	Irish % share in Total	Ireland $ billion	Irish % share in Europe	Irish % share in Total	Ireland $ billion	Irish % share in Europe	Irish % share in Total
A. Investment position (historical cost basis)												
All industries	7.24	2.44	1.18	8.00	2.32	1.14	10.20	2.67	1.31	14.48	3.44	1.68
Manufacturing	3.83	3.86	1.91	4.57	3.73	1.87	6.01	4.35	2.21	8.46	5.94	2.94
Food and kindred products	0.26	2.30	1.06	0.32	2.32	1.11	0.40	2.61	1.20	0.64	3.53	1.65
Chemicals and allied products	1.26	4.55	2.63	1.59	4.30	2.59	1.94	4.25	2.69	2.77	6.16	3.77
Primary and fabricated metals	0.14	2.88	1.41	0.14	2.68	1.22	0.16	2.40	1.12	0.16	2.29	1.07
Industrial machinery and equipment	0.03	0.23	0.13	0.01	0.07	0.04	0.22	1.19	0.70	0.56	2.79	1.67
Electric and electronic equipment	0.35	5.08	1.81	0.53	4.94	1.92	0.99	7.57	3.13	1.75	12.80	5.17
Transportation equipment	0.03	0.24	0.10	0.02	0.18	0.07	–0.01	–0.01	–0.01	0.01	0.04	0.02
Other manufacturing	1.75	7.74	3.84	1.96	8.10	3.85	2.31	9.07	4.14	2.59	10.21	4.47
Wholesale trade	0.25	0.83	0.42	0.22	0.65	0.33	0.36	1.02	0.51	0.35	1.02	0.51
Financial sector[1]	2.82	2.66	1.44	2.58	2.18	1.18	3.10	2.40	1.29	5.11	3.33	1.82
Services	0.29	1.86	1.07	0.52	2.72	1.75	0.60	2.77	1.68	0.32	1.29	0.79
Other industries	0.03	0.25	0.08	0.02	0.11	0.04	0.02	0.11	0.04	0.02	0.12	0.04
	Ireland	Europe	Total	Ireland	Europe	Total	Ireland	Europe	Total	Ireland	Europe	Total
B. Rate of return[2]												
All industries	21.3	10.3	11.3	26.9	11.9	12.5	22.5	11.3	11.9	22.1	11.4	11.7
Manufacturing	27.9	10.7	12.9	29.7	12.9	14.1	26.3	11.8	12.6	27.2	11.9	13.0
Food and kindred products	14.1	14.0	16.6	15.9	12.6	15.5	20.2	13.3	14.6	n.a.	11.6	13.3
Chemicals and allied products	35.9	13.3	13.9	37.3	13.5	14.0	32.4	12.2	13.2	35.4	12.6	12.8
Primary and fabricated metals	16.5	5.8	8.1	9.9	10.1	11.9	3.8	10.3	9.5	11.5	11.0	10.4
Industrial machinery and equipment	484.8	3.7	8.5	61.5	13.7	14.3	5.0	12.8	14.4	18.4	11.5	15.1
Electric and electronic equipment	50.0	15.0	16.8	59.0	16.0	16.2	51.9	14.4	13.3	33.5	15.6	14.4
Transportation equipment	n.a.	9.6	12.2	n.a.	6.5	10.9	300.0	4.4	9.4	66.7	10.1	13.3
Other manufacturing	12.4	10.6	11.8	19.5	14.3	14.6	15.6	12.6	12.0	n.a.	10.6	11.5
Wholesale trade	54.0	15.3	14.2	18.4	15.0	13.4	13.4	12.5	12.2	23.3	13.0	13.1
Financial sector[1]	11.3	10.0	10.0	26.1	11.7	11.3	19.3	12.6	11.5	17.9	11.7	10.6
Services	–18.6	8.8	10.6	3.3	12.8	13.9	1.7	8.3	10.0	–45.8	11.5	12.9
Other industries	207.4	0.7	4.8	n.a.	–0.6	7.1	11.1	1.6	6.1	n.a.	2.2	6.5

1. Excluding depositary institutions.
2. Income net of withholding taxes and without a current-cost adjustment as a percentage of the investment position on a historical cost basis.

Source: US Department of Commerce (1998), *Survey of Current Business,* 78, 10, October.

Table 12. **Irish economy expenditures of manufacturing firms**

Shares in per cent, except as noted

	1983		1996	
	Irish-owned	Foreign-owned	Irish-owned	Foreign-owned
Wages and salaries	23.6	35.4	18.4	23.9
Materials	57.4	35.8	59.1	28.1
Services	17.7	25.3	15.9	43.5
Profits (tax)[1]	1.3	3.5	6.5	4.5
Total	**100.0**	**100.0**	**100.0**	**100.0**
In Ir£ million at 1996 prices	8 717	3 650	12 381	10 434

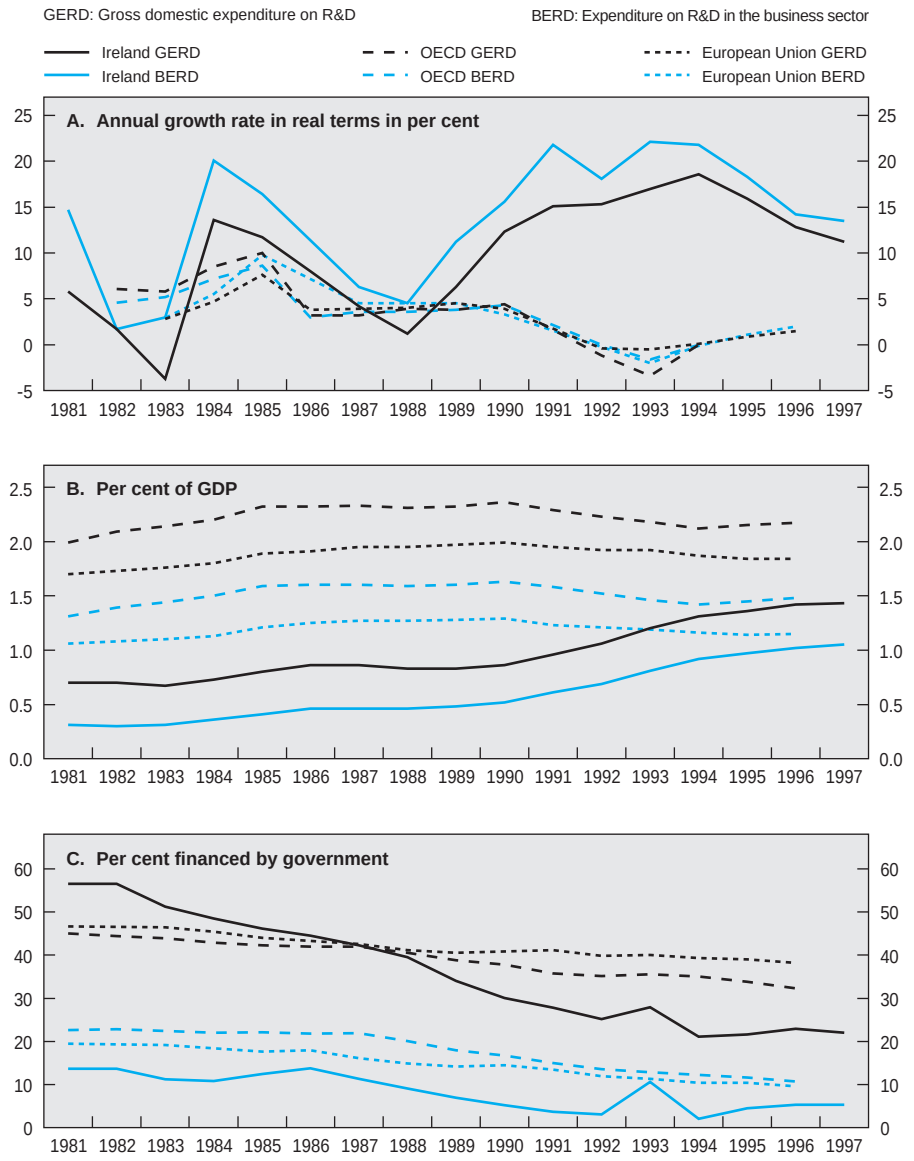
1. All profits for Irish-owned firms but only corporation tax paid by foreign-owned firms.
Source: Barry, Bradley and O'Malley (1999) for 1983 figures; Forfas (1998) for 1996 figures.

One widely cited case is that of software, where one-third of all indigenous software firms were started up by people who had just left foreign firms and a further third by someone with similar previous working experience (Barry *et al.*, 1999). Access to leading multinational corporations' marketing networks has also allowed the decoupling of the Irish economy from almost total dependence on the United Kingdom as an export destination[54] (Barry *et al.*, 1999).

Furthermore, although Ireland was originally a laggard in terms of research and development financed by all sectors, foreign firms have helped to raise the intensity of such business expenditures at a very fast rate – an average of nearly 20 per cent per year in real terms in the first half of the 1990s – bringing it to near the level achieved by leading small countries in the EU (Figure 10). They are responsible for two-thirds of all R&D performed by business, easily the highest share in the OECD (OECD, 1998*f*). However, such activity is concentrated in only one-half of all foreign firms, and the growth rate since 1986 has been smaller than in indigenous firms.[55] The authorities are anxious to get more firms to undertake research in Ireland, as they have observed that R&D performers had superior employment outcomes over the period 1986-95 than non-performers. However, causality could run the other way. One of the important areas of current policy focus is to improve the collaboration between business and the academic research community by easing the formation of "campus companies". Ireland's government spending on R&D is, however, very low – only 22 per cent of all domestic outlays on R&D, far less than the 34 per cent OECD average (Panel C). This was despite the availability of EU structural (CSF) funds, which amounted to Ir£ 133 million in 1997, one-sixth of the public-sector total (Forfas, 1997). In part, this reflects the lack of a set of well-funded public research organisations.

OECD 1999

Figure 10. **Rapid increases in Ireland's research and development outlays**[1]



1. Some missing observations have been filled in through interpolation.
Source: OECD.

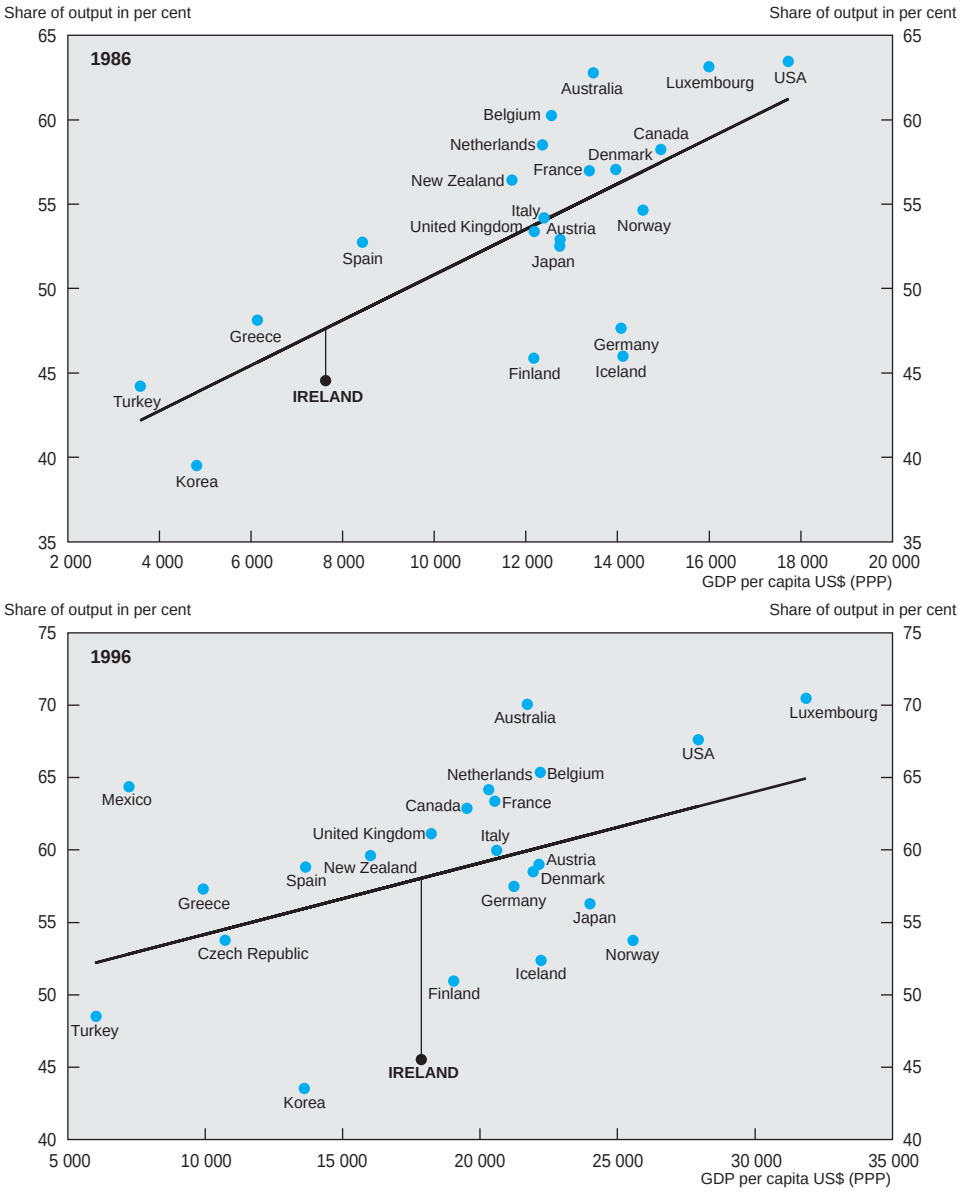
Possible costs of the strategy

But there have been costs to pay from such a heavy emphasis on attracting foreign investment as well. Some critics have charged that the 1990s boom is almost entirely attributable to the exports made by foreign multinationals, especially in a narrow range of high-technology fields (Murphy, 1998), exports which could easily disappear if those producers decide to shift production elsewhere or if there is a significant downturn in one of Ireland's key export sectors (Leddin and Walsh, 1997).[56] In earlier years there was clearly an excessive focus of policy-makers' attention on the creation of favourable conditions for the subsidiaries of multinational corporations, but policy has become more balanced since the publication of the so-called Culliton Report (Industrial Policy Review Group, 1992). Nevertheless, some observers believe that the industrial policy regime has failed to encourage innovation and has fostered a dependency culture (Maillat, 1997). It remains an open question whether clusters will emerge from an industrial structure based originally on foreign branch plants.

But the most obvious side effect of such a long history of support for the manufacturing sector is an atypical industrial structure. Ireland's share of non-government services in output was already fairly low in 1986, given its level of GDP per capita in purchasing-power parity terms at that point (Figure 11).[57] In the following decade Ireland became increasingly an outlier in this respect.[58] By 1996, Ireland had the smallest output shares in the OECD for both wholesale and retail trade, restaurants and hotels; and transport, storage and communication. It also had the third lowest share in finance, insurance, real estate and business services, despite the support given to internationally traded services,[59] all of which would probably be found in this sector.[60] It is only the final component of the non-government services sector whose output share is not small in Ireland, quite the contrary; but these community, social and personal services include a number which are in the broad public sector (such as health and education) and which are clearly not subject to the same kind of market forces as are the others. If one were to exclude these services, however, the significance of the relationship with per capita GDP would break down. Nevertheless, Ireland's output share of this narrowly defined non-government services sector, around 27 per cent in the mid-1990s, can be compared with that of the United Kingdom, for example, at nearly 52 per cent.[61]

This evolution of the overall industrial structure is separate from the ongoing shift from traditional, largely Irish-owned manufacturing[62] to modern, largely foreign-owned manufacturing through ongoing direct investment inflows.[63] Since the FDI inflows are largely not in the same sectors as indigenous firms, there is no resulting loss of market share and profits/rents of domestic producers. But the FDI supply shock does have certain similarities to the phenomenon known as the "Dutch disease" (Barry and Hannan, 1995), which usually refers to

Figure 11. **The share of services in output, 1986 and 1996**[1]



1. Both services and output figures exclude government services but include community, social and personal services. Where 1996 figures are not yet available, the latest data are used.

Source: OECD.

the displacement of activity by some development such as a resource discovery – the Dutch discovered natural gas reserves (Corden and Neary, 1982). Such a process might be expected to operate because the newer high-technology sector can compete globally at the higher real exchange rate that would normally ensue and can afford to pay higher wages to attract additional workers, which the traditional sector cannot afford to match in the longer term. This phenomenon has not been very powerful in Ireland (Walsh, 1996; Barry *et al.*, 1999). This may have been either because there has been chronic excess supply in the labour market, because the partnership approach to pay determination may have helped to prevent its emergence, or because the expanding sector is less labour-intensive than the contracting sector, restraining real wage gains and therefore negating the real effective exchange rate appreciation (see above), at least for the moment. Nonetheless, with wage growth virtually identical in the two sectors from 1980 to 1992, employment growth was robust in the modern sector and fell sharply in the traditional sector, which resulted in productivity growth in excess of wage growth (Barry, 1996). On the other hand, employment at all Irish-owned firms in manufacturing and internationally traded services began to grow in 1993, even if their growth rate has been only half as fast as that of foreign-owned firms – 3.3 per cent per year through 1997 on a full-time-equivalent basis (Forfas, 1998*a*).

More recently, with labour-market tightening and accelerating wage gains, the rate of shrinkage and closure of firms in traditional, low-productivity sectors has picked up, even though the growth rate of the modern sector has slowed slightly. For example, in October 1998 the year-on-year change in the volume of manufacturing production was 20 per cent, the modern sector (comprising pharmaceuticals, office and data processing machinery, electrical engineering and instrument engineering) having grown by nearly 33 per cent, but a set of traditional sectors representing nearly 38 per cent of the index (determined by net output weights in 1987) having shrunk.[64] By way of comparison, in January 1996 the overall index was up 22 per cent, the modern sector was growing at a 43 per cent rate and the share of the index representing shrinking industries was only 33 per cent.

Summing up

Thus, the improvement in economic performance that began around 1987 and shifted to an even higher gear in 1994 is attributable to a variety of factors. Increased labour input has been a key component. This has resulted from favourable demographic and migration developments, higher labour force participation and a drawdown of the stock of jobless and agricultural workers, as well as a continuing upgrading of the nation's resources of human capital. Also considered and found to have played important supporting roles have been EU membership,

including the increased openness and market access this has brought and the associated receipts of structural funds. Domestic policies have also been supportive. Fiscal consolidation may not have been as powerful as some have claimed, but has been helpful nonetheless. Similarly, incomes policy has allowed the achievement of a substantial degree of social consensus. But possibly the most important causal influence has come from the success of the strategy of attracting inflows of foreign direct investment, especially from the United States. These flows have generated substantial direct economic stimulus and significant spin-offs to domestic suppliers and potential entrepreneurs. However, the assistance to the exporting sectors has left the industrial structure atypical by international standards, with an unusually small share of output in the form of non-government services.

II. Prospects for sustained high growth and the role of policies

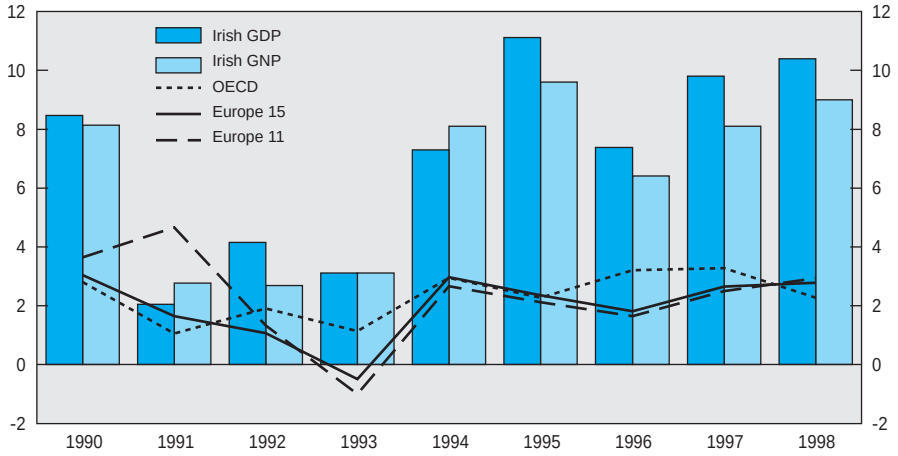
The recent economic boom, whose origins are described in the preceding chapter, seems to have reached a stage where there are real risks of overheating. Signs of bottlenecks to further growth have emerged. The prices of immobile factors, notably houses, have risen, boosting private consumption through wealth effects and adding to wage demands. The problem of housing shortages has restrained an increase in labour supply from immigration, which in turn may be contributing to wage acceleration. This chapter first discusses recent economic trends and the diminished role of macroeconomic policy as a demand management tool with the economy's increased openness and the advent of EMU. Short-term prospects, overheating risks, as well as underlying adjustment mechanisms are then assessed. The chapter ends with a discussion of longer-term sustainability issues. It concludes with an ultimately upbeat assessment: with several prudent policy adjustments, Ireland can continue to grow much faster than virtually all of its trading partners, even if substantially less rapidly than in recent years. But the extent to which such rapid growth does in fact occur involves important social choices regarding traditional lifestyles (childcare options, for example).

Recent economic trends

Continuing strong growth in a virtuous circle

The Irish economy is still growing very fast as the benefits of inward direct investment propagate through the rest of the economy. Real GDP, having risen more than 11 per cent in 1995, moderated only slightly in 1996, before bouncing back to about a 10 per cent rate in 1997 and 1998. This growth record has been the best in the OECD for four consecutive years (Figure 12). Although GNP is a more appropriate measure of the performance of the economy in such a small, open economy as Ireland's (Box 1), the overall picture remains unchanged, because its growth rate has been only 1½ percentage points lower than its GDP counterpart in

Figure 12. **Outstanding Irish growth performance**
Annual percentage change



Source: OECD.

Box 1. A small open economy as a production base

Ireland's is clearly a small open economy: it is the third smallest in the OECD, ahead of only Iceland and Luxembourg, with a quarter per cent of OECD total GDP and around one per cent of the EU total. Openness measured by the ratio of the sum of export and import values to GDP was nearly 1.7 in 1998, the second highest in the OECD, behind Luxembourg. Most high-tech multinational companies, many of which started to operate in Ireland in the 1990s, export the bulk of their production to overseas markets. Therefore, Ireland may be likened to a region with a strong industrial base in a large country, such as the Detroit area in the United States, for example (Krugman, 1997). This approximation is reinforced by Ireland's participation in Economic and Monetary Union, much like a region in the larger Continental economy. The fastest growing industries in Ireland are chemicals, computers and instrument and electrical engineering whose output shares together accounted for about 22½ per cent of real GDP at factor cost in 1997. They are greatly affected by external developments because about 90 per cent of their sales come from exports.

this period. Net exports accounted for about 4 percentage points on average of the strong real GDP growth in 1997 and 1998. The rest, about 6 percentage points, is attributable to domestic demand. The same pattern was observed in the 1994 to 1996 period, in contrast to the early 1990s when net exports accounted for the larger part of output growth. This indicates that a virtuous cycle continues to work: initial favourable supply shocks, in the form of start-ups of high-productivity operations of multinational companies, lead to increased income and thus domestic demand, which then brings about an increase in production in indigenous industries, especially in services.

Thanks to rapid increases in production, the growth rate of aggregate employee compensation gradually accelerated to above 10 per cent in 1997, which, together with implemented income tax cuts, ensured a pick-up in disposable income growth from about the 6 per cent level in the previous years to about 9 per cent in 1997. Falling long-term interest rates further boosted consumers' confidence, which, along with wealth effects from rising house prices (see below), resulted in a fall in the saving rate from about 13 per cent in the early 1990s to about 10¾ per cent in 1997 and 1998 on average. Private consumption growth gradually sped up (Table 13), with a double-digit growth rate in expenditures on durable and semi-durable goods such as automobiles and clothing. Rapid income growth as well as demographic factors and substantial immigration have boosted housing demand, causing house prices to soar and residential investment to surge.

This buoyancy of domestic demand, in turn, has benefited indigenous sectors. For example, those industries not dominated by foreign companies registered about 5 per cent output growth in 1996 and 1997, about 2 percentage points higher than in the first half of the 1990s. Another piece of evidence supporting the hypothesis of spreading strength is the growth rate of profits in the indigenous sector, which is calculated by subtracting remitted investment income from aggregate non-agricultural sector profits on a national accounts basis: 5.3 per cent per annum from 1991 to 1995, followed by 8.5 per cent from 1995 to 1997. The significant profit growth from 1993 to 1998 raised the capital income share in the business sector by 7¾ percentage points to 37¾ per cent; this was the largest increase among OECD Member countries, but it merely eliminated most of the gap with the other EU nations. The restoration of profitability partly resulted from incomes policy in Ireland: income tax cuts have had a damping effect on wage growth, though this moderating effect might be now waning in the current tight labour market situation. Improved profitability has encouraged business investment, which has been growing at an annual rate of more than 10 per cent since 1995. In particular, investment in commercial and industrial buildings increased at a very rapid pace in 1996 and 1997, another demand pressure on the construction sector, while some slowdown was observed in commercial building in 1998 (see Table 26). Disaggregated data on capital asset acquisitions in the manufacturing

Table 13. **Demand and output**

1990 prices

	1995	1995	1996	1997	1998 estimate
	Per cent of GDP in constant prices	Percentage change over previous year			
Private consumption	54.4	4.1	6.1	6.3	8.0
Public consumption	13.1	2.6	1.5	4.8	5.0
Gross fixed investment	15.9	10.9	13.7	10.9	12.0
Building and construction	9.2	12.7	18.6	13.3	13.0
Machinery and equipment	6.7	8.6	6.8	7.2	10.5
Final domestic demand	83.4	5.1	6.8	7.0	8.4
Stockbuilding[1]	1.0	1.8	0.2	0.4	–0.6
Total domestic demand	**84.4**	**7.2**	**6.9**	**7.4**	**7.5**
Exports of goods and services	81.5	19.6	11.7	17.0	20.9
Imports of goods and services	65.9	15.8	12.2	15.6	20.2
Foreign balance[1]	15.6	4.9	1.5	3.7	4.3
Gross domestic product[2]	100.0	11.1	7.4	9.8	10.4
Net factor income[1]	–11.9	–2.5	–1.7	–2.7	–2.7
Gross national product[2]	88.1	9.6	6.4	8.1	9.0
Gross national disposable income		5.8	6.7	7.1	n.a.
Memorandum item:					
Household saving rate		10.8	9.6	10.5	4.2
Output by sector					
Agriculture, forestry, fishing	7.7	1.2	2.3	1.4	n.a.
Chemicals, computers, electrical engineering	18.6	20.9	12.6	27.3	n.a.
Other industries	23.1	12.7	4.9	4.8	n.a.
Distribution, transport and communication	17.7	8.0	15.6	8.8	n.a.
Other services	32.9	3.7	3.7	3.7	n.a.
GDP (factor cost)	100.0	9.2	7.6	9.3	n.a.

1. Contribution to GDP.
2. Expenditure base estimates.
Source: Central Statistics Office and OECD estimates.

sector (which have been available only since 1996) show that the pick-up in 1997 was manifest not only by the leading industries such as chemical products (17 per cent) and office machinery and computers (28 per cent) but also by indigenous industries: for example, investment in the basic metals sector more than doubled,

while those in other non-metallic mineral products rose 17 per cent. However, in 1998 it was an outstanding increase by the office machinery and computers industry (more than 90 per cent) that raised the overall investment growth rate in manufacturing to 19 per cent in the first three quarters of the year.

Volume growth of goods and services exports has continued to pick up, to around 20 per cent in 1998. This is sensitive to developments in EU countries (Figure 12 and Table 13), with a slowdown in 1996 and a rebound thereafter, while exports to the United States have rapidly accelerated since 1995 (Table 14). The

Table 14. **External trade**

Annual growth rate in per cent

	Share in 1997	1994	1995	1996	1997	1998
Customs trade						
Exports						
Volumes		14.8	20.1	9.9	14.6	23.8 [1]
Prices		–0.1	1.8	–0.6	1.1	3.8 [1]
Values	100.0	14.7	22.3	9.3	15.9	29.1
Agricultural exports	7.2	2.9	12.6	–15.1	0.8	6.9 [2]
Industrial exports	90.6	19.2	24.5	13.0	17.7	30.0 [2]
of which:						
Chemicals	25.2	24.4	10.9	27.5	31.8	58.8 [1]
Machinery and transport equipment	37.3	21.2	37.7	10.7	23.5	26.9 [1]
United Kingdom	24.6	13.5	11.1	5.3	16.3	14.9 [1]
Germany	12.5	22.9	22.4	–1.3	11.0	53.9 [1]
Other European countries	30.5	18.3	49.7	2.7	14.9	29.0 [1]
EU 15	67.6	17.2	28.5	2.8	14.7	29.0 [1]
United States	11.3	6.4	19.2	24.0	41.3	54.0 [1]
Rest of the world	21.2	10.9	4.4	27.5	9.8	12.4 [1]
Imports						
Volumes		13.2	14.4	10.0	15.0	17.6 [1]
Prices		2.6	4.3	–1.2	0.5	2.3 [1]
Values	100.0	16.1	19.3	8.8	15.2	22.4
Capital goods	11.7	22.7	20.5	16.5	25.9	37.3 [2]
Consumer goods	21.4	11.2	7.7	13.9	15.9	17.9 [2]
Raw materials	62.1	17.9	24.5	6.6	15.0	20.5 [2]
United Kingdom	34.6	18.1	15.6	6.9	14.7	16.5 [1]
Germany	6.0	10.5	15.5	4.7	2.1	23.9 [1]
Other European countries	15.6	13.6	38.3	14.6	18.9	10.0 [1]
EU 15	56.2	16.1	20.6	8.6	14.3	14.5 [1]
United States	15.0	24.7	12.9	–4.6	12.0	31.4 [1]
Rest of the world	28.8	10.7	20.9	18.5	19.5	26.0 [1]
Tourism						
Overseas visitors		10.4	15.6	11.3	9.0	10.4 [2]

1. Up to November.
2. Up to September.

Source: Central Statistics Office.

negative impact on overall developments from the crisis in emerging markets since mid-1997 seems to have been limited because of the small share of exports to the affected countries and the lack of any pronounced indirect effects on Ireland's major markets, at least until mid-1998. Business survey evidence does, however, point to the likelihood of some slowing in export growth around the end of 1998.

Growth in exports has caused an increase in imports of raw materials for production, with a share of over 60 per cent of the total, pushing up overall imports in 1998. On balance, the trade account has continued to register a huge surplus, nearly one-third of GNP in 1998 (Table 15). However, the other side of the

Table 15. **The current account of the balance of payments**

		1993	1994	1995	1996	1997	1998
		Ir£ million					
Exports		19 459	22 424	27 698	30 723	36 506	45 966
Imports		14 633	17 028	20 239	21 967	25 422	30 570
Trade balance		4 826	5 396	7 459	8 756	11 084	15 396
Services	Net	–1 366	–1 978	–2 991	–3 782	–4 690	–7 711
Royalties	Net	–897	–1 218	–1 617	–2 071	–2 655	–4 166
Other services	Net	–469	–760	–1 374	–1 711	–2 035	–3 545
Factor incomes	Net	–3 521	–3 575	–4 508	–5 151	–6 322	–7 676
Company earnings	Credit	151	296	556	478	674	806
	Debit	3 355	3 786	5 193	5 826	7 382	9 519
of which:							
Reinvested earnings	Credit	149	292	512	455	670	782
	Debit	664	577	1 248	1 372	1 377	2 025
National debt	Debit	1 020	1 081	1 015	915	765	663
Other	Net	703	996	1 144	1 112	1 152	1 700
Current transfers	Net	1 309	1 156	1 110	1 354	1 290	1 038
Current account balance		1 248	998	1 070	1 176	1 362	1 047
		Per cent of GNP[1]					
Memorandum items:							
Trade balance		16.7	17.0	21.4	23.2	26.4	32.3
Royalties		–3.1	–3.8	–4.6	–5.5	–6.3	–8.7
Other services		–1.6	–2.4	–3.9	–4.5	–4.9	–7.4
Company earnings		–11.1	–11.0	–13.3	–14.2	–16.0	–18.3
National debt interest		–3.5	–3.4	–2.9	–2.4	–1.8	–1.4
Other factor incomes		2.4	3.1	3.3	2.9	2.7	3.6
Current transfers		4.5	3.7	3.2	3.6	3.1	2.2
Current balance		4.4	3.1	3.1	3.1	3.2	2.2

1. 1998 figures are OECD estimates.
Source: Central Statistics Office.

same coin is a sizeable deficit on both services and factor income accounts: multinational companies have made a substantial and increasing amount of royalty payments and profit repatriation,[65] 9 per cent and 18 per cent of GNP in 1998, respectively. Current transfers, which are mainly from the EU and are declining, have provided a partial offset. The current account has continued to register a surplus that had been stable in relation to GNP at around 3¼ per cent until 1998, when preliminary figures indicate that it fell to about 2¼ per cent, as services imports surged.

These external trade developments also reflect a continuing improvement in cost competiveness. A competitiveness measure based on relative wage rates shows a loss against Euro zone countries, but a gain against the United Kingdom, because of significant appreciation in sterling from 1996 to 1998 (Figure 1, panel A). However, this relatively high wage growth has been offset by a surge in productivity. As a result, competitiveness measured by relative unit labour costs is still improving globally (Figure 1, panel B). Because there seems to be a large difference in productivity growth between the modern and traditional sectors, the overall picture is not reflected throughout the economy. The higher wage inflation is slowly eroding the competitiveness of indigenous industry. For example, according to IDA Ireland, a reason for the job destruction in 1998-99 by IDA-assisted companies is that several are moving their labour-intensive operations to lower-cost locations.

Tightened labour market

Because of the strong output growth and buoyant expectations, employment increased by 4 per cent per annum from April 1994 to April 1997, and then its growth rate shot up to an estimated 6.9 per cent in the year to spring 1998 (Table 16), even after allowing for methodological changes in the data.[66] The increase in the number of part-time workers has been highly variable, reaching about 35 per cent in the year to spring 1998[67] (after adjusting for methodological changes), while the number of full-time workers has shown a steady increase of 3 to 4 per cent per annum in recent years. With regard to sectoral composition, sharp increases in construction (23.5 per cent) and market services (14.6 per cent) pushed up employment growth in 1997-98, while significant albeit slowing growth has also been observed in manufacturing. Public-sector employment had been flat until the spring of 1998 when a moderate jump occurred. The buoyancy in construction and market services has been reflected in substantial employment expansion in Dublin and the surrounding area, which is in contrast with developments in the first half of the 1990s when Dublin was an under-performing area (Figure 13, Panel A). Strong labour demand pushed the unemployment rate down, even though it encouraged new entrants into the market. About three-quarters of the increase in employment from 1995 to 1998 came from increases in the labour force, with the rest from a decline in unemployment (Figure 13).

Table 16. **Labour market**

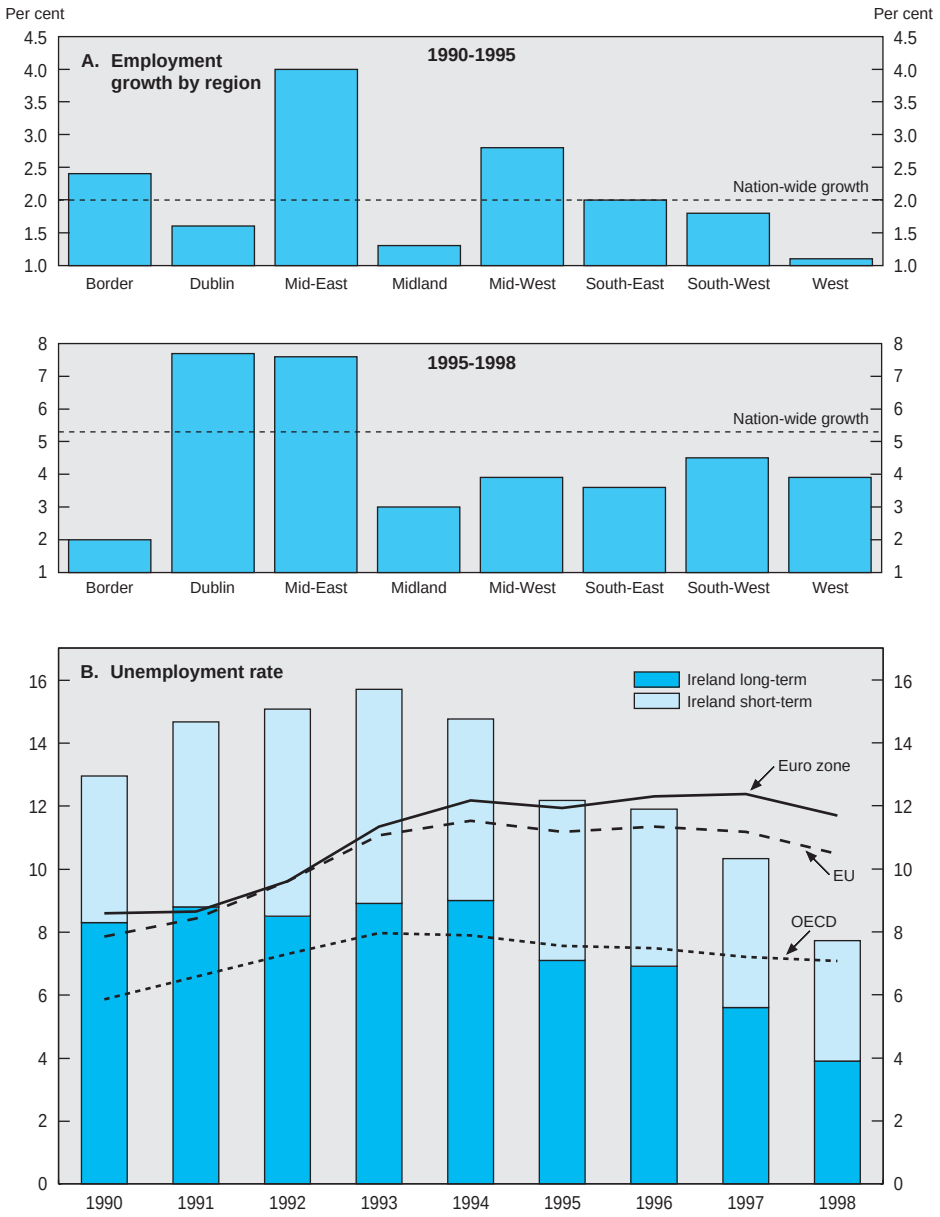
Year-on-year growth rates, per cent

	April 1993	April 1994	April 1995	April 1996	April 1997	March-May 1998[1]	Adjusted[1] basis
Population	0.6	0.3	0.4	0.7	1.0	1.2	
Natural increase (contribution)	0.6	0.5	0.5	0.5	0.5	0.6	
Net immigration (contribution)	0.0	–0.1	–0.1	0.2	0.4	0.6	
of which:							
United Kingdom	0.0	0.0	0.1	0.1	0.2	0.3	
Rest of EU	0.0	0.0	0.0	0.1	0.1	0.1	
United States	0.0	–0.1	–0.1	0.0	0.1	0.0	
Population above 15	1.3	1.1	1.3	1.6	1.7	1.9	
Labour force	2.3	2.0	1.9	3.3	2.1	5.3	4.0
Male participation rate (per cent)	67.9	68.0	68.0	68.1	67.8	69.4	
Female participation rate (per cent)	38.2	39.0	39.7	41.4	42.0	44.0	
Employment	1.5	3.3	5.0	3.7	3.9	8.3	6.9
Full-time	–0.5	2.6	4.1	4.3	2.9	2.9	
Part-time	22.0	7.8	12.0	–1.2	11.7	46.9	35.1
Agriculture, forestry and fishing	–4.9	–1.9	1.0	–5.2	0.1	–3.9	
Non-agricultural	2.5	3.9	5.7	4.8	4.2	9.7	
of which:							
Manufacturing and other production	n.a.	n.a.	4.7	1.0	8.3	4.4	
Construction	n.a.	n.a.	5.6	4.3	9.5	23.5	
Market services[2]	n.a.	n.a.	5.8	4.9	3.8	14.6	
Public sector	0.3	0.7	1.0	2.4	–1.3	0.9	

1. Some figures produced by direct comparison of the 1998 and 1997 surveys are misleading because of methodological differences. Therefore, adjusted figures are also given here. See footnote 66 in the text. Even when no adjustment figure is provided, the 1998 figures, especially for the sectoral disaggregation, should be treated with caution.
2. This is the sum of "wholesale and retail trade", "hotels and restaurants", "transport, storage and communication", "financial and other services" and "other services".

Source: Central Statistics Office.

Figure 13. **Employment and unemployment**



Source: Central Statistics Office and OECD.

Labour force growth accelerated from about 2 per cent per annum in the 1993-95 period to more than 3 per cent since then.[68] While domestic demographic factors accounted for a stable portion of about 1¼ percentage points, the pick-up in the growth rate came from other sources. An increase in the labour force participation rate explains about 1 percentage point on average, though its movement has been erratic. The recent increase in the female participation rate is significant, although room for a further rise seems to exist, given that it is still below the average level in EU countries. Net immigration started to contribute positively in 1996, and its contribution increased to 0.6 percentage point in 1998. Although this unusually high elasticity of labour supply, a particular privilege for Ireland, has alleviated the nascent labour shortage, it does not mean that Ireland faces a horizontal labour supply curve. Two factors make elasticity less than infinite: a limited supply of houses, around Dublin in particular, and depletion of the stock of would-be immigrants to Ireland (see Chapter I).

The unemployment rate has continued to decline, reaching 7.8 per cent in the spring of 1998,[69] and a provisionally estimated 6.7 per cent in March 1999, which is less than half of the most recent peak of 15.7 per cent registered in 1993 and a record low since 1980. This has resulted in the complete erosion of the difference from the OECD average and a far lower level than the EU average. The long-term unemployment rate (those unemployed for one year or more) also declined significantly to 3.9 per cent in the spring of 1998. The incidence of long-term unemployment at 50 per cent, however, remained above the OECD average of 32.7 per cent in 1997, although very near the EU average.[70] Other alternative measures of under-utilisation of labour (adding back discouraged, underemployed or passive job seekers) show even more rapid declines than the official unemployment rate over the year to the spring of 1998. With the unemployment rate this low, labour shortages have increasingly become a bottleneck to further growth.

Anecdotal evidence indicates that labour shortages have spread across the skills spectrum and are especially severe at the high and low ends, for example software workers and those working in hotels and restaurants. Within the manufacturing sector, the problem has also grown more severe, even if only 5 to 10 per cent of all respondents to the IBEC/ESRI Monthly Industry Survey report production constraints resulting from either insufficient skilled or unskilled labour.[71] The official wage data seem to support the hypothesised significance of shortages (Table 17). Aggregate data are available only with a long lag, but the OECD estimates that employee compensation per head accelerated to 6 per cent in 1998 from a subdued growth rate of about 2 per cent in 1994 and 1995. Wage increases in the construction sector are high, and, in particular, growth in skilled workers' wages in that sector sped up to double-digit rates in 1997, with little sign of moderation since then.[72] Hourly earnings gains in industry have been moderate relative to registered productivity growth but started to increase more rapidly

Table 17. **Prices, wages and profits**

Percentage change from previous period

	1993	1994	1995	1996	1997	1998
Consumer prices						
All items	1.4	2.4	2.5	1.7	1.4	2.4
Except mortgage interest	2.0	3.0	2.3	1.9	1.1	2.4
Food	0.3	3.5	3.0	1.9	1.7	4.2
Energy	0.2	0.7	0.5	3.9	3.0	–0.6
Services	3.4	3.6	2.9	1.1	1.6	3.5
Wholesale prices						
All items	4.8	1.0	2.1	0.5	–0.4	2.3[1]
Manufacturing	4.6	1.1	2.5	0.7	–0.6	0.8
Agriculture	6.5	1.5	2.3	–4.9	–6.5	–0.6
Price deflators						
GDP	4.4	1.2	0.4	1.7	2.3	4.0[2]
Private consumption	1.9	2.8	2.1	1.4	0.9	2.6[2]
Total domestic demand	3.3	2.8	2.2	1.9	2.7	4.1[2]
Exports of goods and services	6.2	0.7	1.9	–0.3	1.1	3.0[2]
Imports of goods and services	4.3	2.7	4.0	–0.7	0.8	2.2[2]
Wages and profits						
Compensation of employees	8.1	6.3	7.7	8.5	10.2[2]	14.5[2]
Agricultural profits	2.2	5.0	7.5	1.4	–4.9	n.a.
Non-agricultural profits	13.2	6.8	20.3	11.1	15.5	n.a.
Average weekly earnings						
Skilled workers in construction sector	4.2	5.5	5.8	5.3	13.0	14.6[1]
Unskilled workers in construction sector	1.7	9.4	4.0	5.4	5.6	5.0[1]
Manufacturing	5.4	2.9	2.2	2.5	2.8	5.0[1]
Banking, insurance and building societies	5.8	2.6	1.1	3.4	2.8	3.3[1]
Public sector	6.0	2.8	2.9	4.1	4.8	4.5[3]
Memorandum items:						
Compensation per employee, private sector	5.3	1.8	2.3	3.7	5.4[2]	6.0[2]
Terms of trade, goods and services	1.9	–1.9	–2.1	0.3	0.3	0.6[2]
Average hourly earnings in manufacturing	5.8	1.8	2.2	2.6	3.0	5.6[1]

1. January to September.
2. OECD estimates.
3. January to June.

Source: Central Statistics Office, *Economic Series*, and OECD.

in 1998, reaching 5.8 per cent in the year to September. Wages in the financial and public sectors have shown a modest pick-up in recent years: weekly earnings in the insurance sector have accelerated to about 5½ per cent in the year to September 1998, while those in banking and building societies have risen more moderately; in the public sector (including the semi-state sector but excluding health) earnings have shown a pick-up to 5 per cent in the year to September 1998.

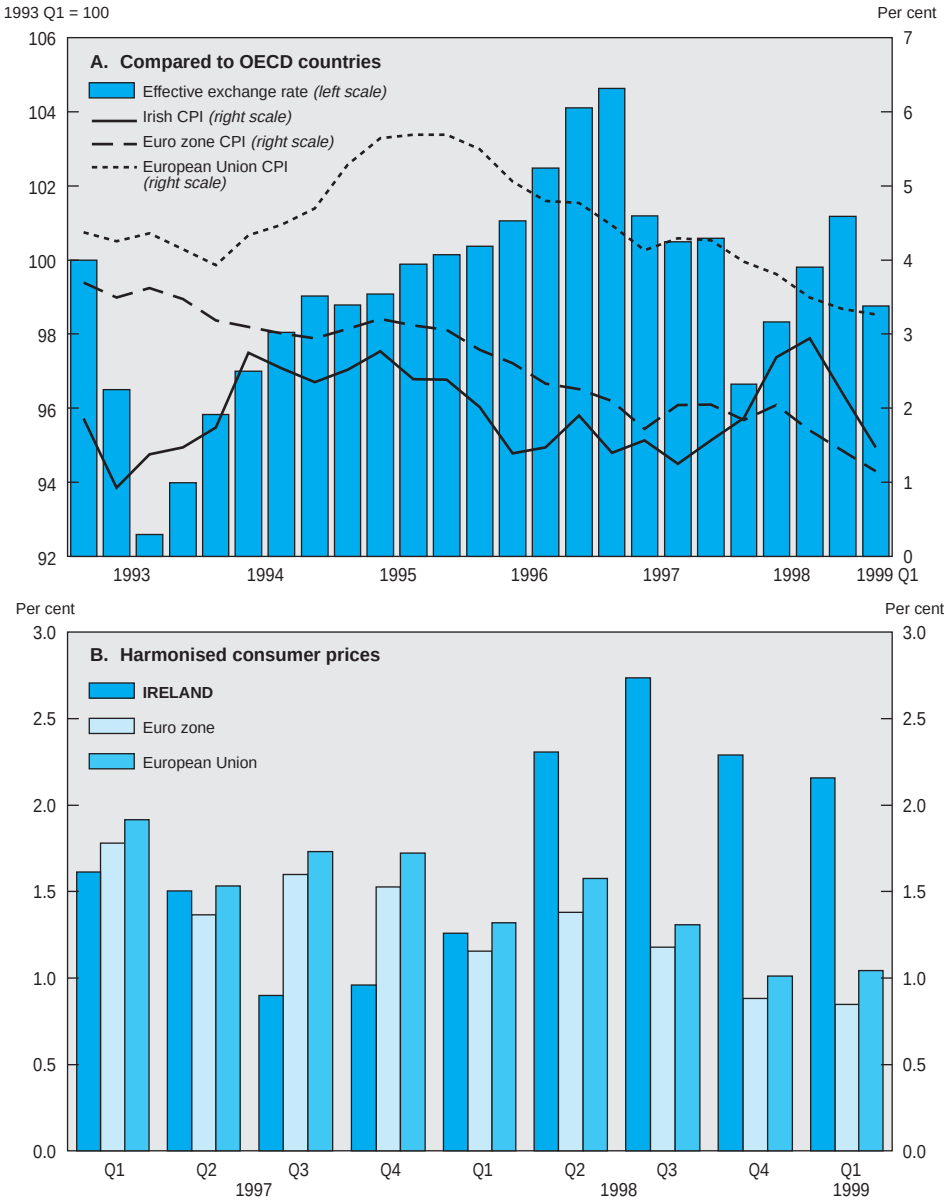
Although these wage data are not directly comparable with the terms agreed in the Partnership 2000, because of compositional changes, for example (see below), they can be taken as indirect evidence that pressures are causing substantial wage drift beyond the increases implied by the wage agreement.

The inflation rate as measured by the consumer price index (CPI) is subdued on the whole. Demand pressures do not necessarily cause faster consumer price rises for such a small open economy because the share of imports and import-competing goods in the consumption basket is substantial. Although the prices of non-tradable goods, with less than a 40 per cent weight in the Irish CPI, reflect domestic conditions, the rest are determined by the evolution of the effective exchange rate. Therefore, in the tradable goods sector, the increase in wage inflation has been easily absorbed by persistently robust productivity gains and by falling non-labour costs (such as raw materials and financial costs). The stable and lower inflation rate than in other EU countries until 1997 was mainly due to an appreciation of the effective exchange rate from mid-1993 to early 1997 of 13 per cent in cumulative terms (Figure 14, Panel A). The decline in mortgage interest rates also contributed to relative inflation reduction in 1997 and again more recently. But with a reversal of this exchange-rate trend and a cumulative depreciation of 7½ per cent over the year to early 1998, CPI growth picked up, reaching over 3 per cent in the summer of 1998, about 1½ percentage points higher than EU average on a harmonised basis (Figure 14, Panel B). Services prices were relatively stable in 1996 and most of 1997, but began to accelerate late that year, reflecting their higher labour content and the tight labour market, and have recently been rising at a year-on-year rate of nearly 4½ per cent. However, CPI inflation began to moderate in the autumn of 1998 due in part to the renewed appreciation of the effective exchange rate since the spring of the year, most notably after the decision to revalue the currency within the European Exchange Rate Mechanism in March 1998 in preparation for Economic and Monetary Union (EMU). Thus, year-on-year CPI increases have come down to 1.4 per cent since the summer 1998 peak. Of the 1.8 percentage point deceleration since August 1998, 0.9 point is attributable to lower mortgage interest payments resulting from EMU-based interest-rate convergence (see below) and a further 0.4 point to a combination of lower energy costs, resulting largely from lower world oil prices, and reduced indirect tax pressures on prices. Thus, the remaining slowdown was only 0.5 percentage point. It was held back by upward movements in food prices due to bad weather.

Despite moderate movements in consumer price inflation, the increase in the investment deflator rose from less than 4 per cent in 1995 and 1996 to over 7 per cent in 1997 and 1998, reflecting a significant rise in construction costs. The government consumption deflator also gradually accelerated to above 5 per cent in 1997 and 1998, keeping pace with the public-sector pay increase. These, together with a gain in terms of trade pushed up inflation as measured by the

Figure 14. **Consumer price inflation**

% change over four quarters



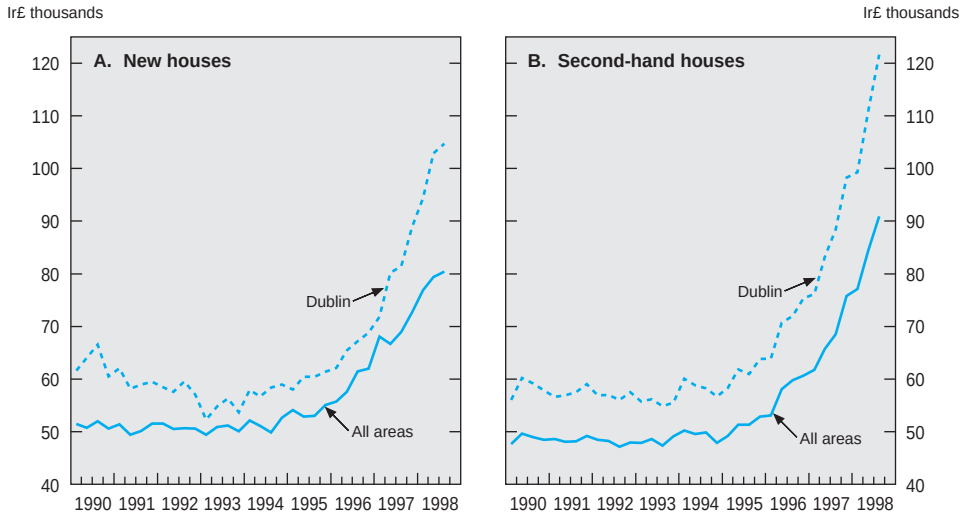
Source: Eurostat, OECD.

GDP deflator from 1.7 per cent in 1996 to 2.3 per cent in 1997 and an estimated 4 per cent in 1998, 2½ percentage points higher than the Euro-area average. These price developments indicate some signs of overheating.

Shortage of houses

Another symptom of excess demand in the economy is the boom in the housing sector. House prices have shown a marked rise since 1996. According to the official statistics published by the Department of the Environment and Local Government, new and second-house prices increased by about 16 per cent and 24 per cent per annum, respectively, in the 2½ years to the summer of 1998 in real terms (Figure 15).[73] Price hikes have been even faster in Dublin: a rise of about 23 per cent for new houses and 29 per cent for existing homes over the same period. However, some moderation was observed as from the middle of 1998, at least in the Dublin area. Nevertheless, according to official figures, prices still rose by 14.3 and 23.2 per cent at annual rates in the final two quarters of the year for new and second-hand houses, respectively. The modest slowing may have been the result of various measures taken following the policy recommendations by a government-appointed expert (Bacon, 1998).

Figure 15. **House prices**



Source: Department of the Environment.

According to the Bacon report, most of the price developments up to 1997 are explained by a standard demand-supply model (about 80 per cent of the variance for the whole economy and about 60 per cent for the Dublin area). There have been three economic factors boosting demand: strong growth in real income per capita, buoyant consumer sentiment backed by improvement in the labour market, and falling interest rates. The increased demand has also been bolstered by a social factor: rapid growth in the rate of household formation. This is the result of the large share of the population in the appropriate age group and a trend to declining household size, which remains still comparatively large in Ireland.[74] Furthermore, an increase in immigration has created additional demand. The gross rather than the net number of immigrants is important because immigrants are concentrated in the age group of 25 to 44, while most emigrants are 15 to 24 years old (see Table 5 in Chapter I). With regard to the supply side, the slope of the supply curve is usually steep in the short run and less so in the long run, and depends on institutional arrangements such as transactions costs, especially stamp duties in Ireland's case, as well as capital gains tax and infrastructure provision, for example. The faster increase in house prices in Dublin is due to higher income growth, a tendency for immigrants to settle there and a lower short-run price-elasticity of supply – 3.0 for the whole economy, but only 1.8 for Dublin, according to Bacon (1998). Although the tempo of economic expansion is likely to slow down from the current unsustainable pace, the deceleration may not be sufficient to stabilise house prices, as shown by a simulation contained in the Bacon report. Therefore, an important policy issue is how to raise the price elasticity of supply (see below).

Even though higher house prices have put pressure on the conventional lending criteria,[75] at present there is little concrete evidence that lending attitudes have been generally relaxed, apart from some of an anecdotal nature, although there have been calls for such a relaxation, despite the risks of such a change. Moreover, mortgage lending has not been the only driving force of the rapid increase in aggregate credit to households because its growth rate has been lower than that of non-mortgage credit since 1996.[76] Lower income earners have been unable to obtain a loan given conventional practices, and the financial sector has properly served to ration credit, rather than contributing to the boom. Nevertheless, the pressure was strengthened by competition among lending institutions. The division of business between banks and building societies has become blurred, and banks have penetrated the mortgage loan business. Efforts by building societies to fight back have resulted in an increase in their market share since 1995 (Bacon, 1998).

Fast growth of money and credit

The broad monetary aggregate (M3E) accelerated to a growth rate of above 20 per cent in 1997, reflecting buoyant economic activities and expected

declines in interest rates (Table 18). Domestic credit to the private sector continued to increase significantly. Disaggregated credit data show strong growth rates across all industries (Table 19). Although monetary and credit aggregates moved almost in line with nominal GDP until 1995, the more robust developments since then are worrying signs that easy monetary conditions and inflation might eventually generate an asset bubble. Sharp increases in share (Figure 16, Panel A) and house prices were observed in the two years to 1997, about 30 and 15 per cent per annum, on average, respectively. However, developments over the past year or so have given some comfort: the high monetary and credit growth rates have now stabilised albeit at very high levels; stock prices also peaked out a year ago, while new house price hikes have begun to moderate, even if the boom in existing houses is apparently unabated.

Table 18. **Money supply and its counterparts**

Year-on-year growth rate, per cent

	1995	1996	1997	1998
	December			
Liabilities				
Broad Money Stock (M3E)	12.4	15.9	22.1	17.3
of which:				
Currency	11.4	6.7	14.8	14.7
Non-government current account	15.6	22.0	28.8	30.8
Non-government deposit account	12.4	16.0	22.1	15.8
Others	4.3	6.4	2.9	1.7
Net external liabilities	65.1	–25.5	–4.1	122.0
Government deposits at Central Bank	34.4	9.1	1.1	40.6
Non-monetary liabilities	13.7	–5.9	58.9	9.5
Total	17.3	8.6	23.0	23.1
Assets				
Official external reserves	35.4	–9.4	–6.5	39.1
Domestic credit	14.9	11.5	26.8	21.6
of which:				
Credit to government	–1.9	–13.8	3.9	11.4
Credit to private sector	18.1	15.6	29.6	22.6
Memorandum items:				
Adjusted M3E [1]	11.6	15.7	19.1	18.1
Adjusted private-sector credit [1]	11.2	15.4	23.6	23.6

1. Adjusted for the effects of transactions between credit institutions and non-bank effects arising from exchange-rate movements.

Source: Central Bank of Ireland.

Table 19. **Domestic credit to private sector by industry**

Year-on-year growth rate, per cent

	1994	1995	1996[1]	1997	1998[2]
	November				
Agriculture, forestry and fishing	3.0	6.4	11.0	9.2	11.6
Manufacturing	–7.0	4.5	21.0	13.2	32.9
Building and construction	4.7	27.4	31.5	10.1	(87.4)
Financial	23.3	42.4	19.9	62.5	(25.1)
Real estate	23.8	–0.6	24.2	24.3	(46.0)
Personal	8.2	13.7	15.9	21.0	19.2
Mortgage finance	14.0	18.3	16.7	19.2	20.0
Others	10.1	3.6	17.8	15.1	(20.2)
Total	**10.4**	**16.5**	**17.7**	**27.6**	**23.7**

1. There is discontinuity between 1995 and 1996: the data up to 1995 are for all licensed institutions; and those from 1996 are for all credit institutions. Therefore, the latter has wider coverage, but slightly higher growth rate (0.4 percentage point) in 1996.
2. The industry classification was modified in February 1998, introducing discontinuity into the disaggregate data. Figures for modified categories are in parentheses.

Source: Central Bank of Ireland.

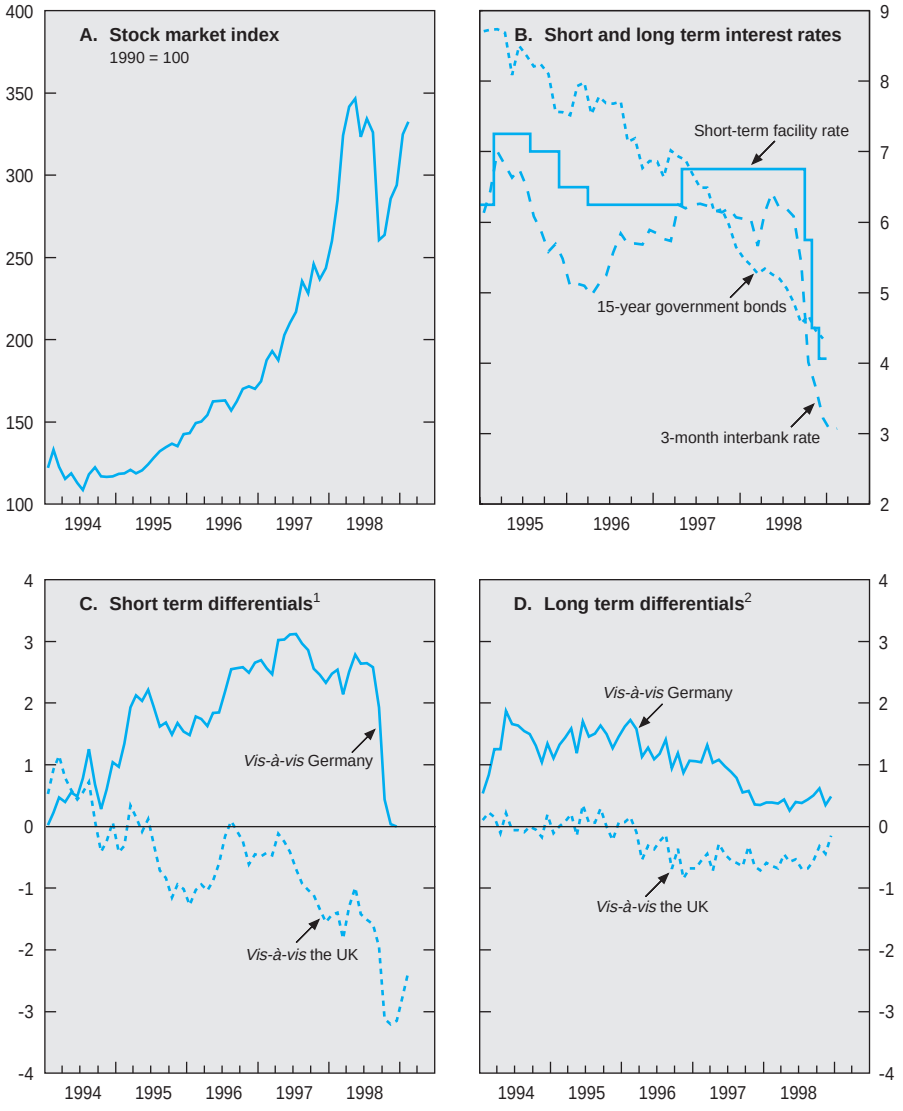
The setting of macroeconomic policies

In the past two years macroeconomic management has been severely restricted by the requirements of membership in Economic and Monetary Union from January 1999 and agreed terms in the Partnership 2000 Agreement covering 1997 to 1999. Although policy makers were concerned with overheating, they were not equipped with adequate tools to control demand. The largely predetermined convergence process in interest and exchange rates reduced monetary policy options, while fiscal policy has been tied to a large extent to implementation of measures promised in the centralised negotiations. Incomes policy has restrained wage growth, but, on the whole, the effects of monetary and fiscal policies together have been expansionary. Fiscal policy measures have been focused on improving the supply side, such as enhancing work incentives and attenuating infrastructure bottlenecks. Strong aggregate demand has put pressure on the labour market, thus testing the partnership approach to containing wage inflation.

Monetary policy

The monetary policy options available in 1997 and especially 1998 were limited by imminent membership in Economic and Monetary Union (EMU) from

Figure 16. **Financial market developments**



1. Three-month interbank rate for Ireland, three-month FIBOR for Germany and three-month interbank loan for the United Kingdom.
2. 15-year central government bonds for Ireland, 9-10-year public sector bonds for Germany and 10-year government bonds for the United Kingdom.

Source: OECD.

its outset, as the exchange rate and short-term interest rates were required to converge to the euro conversion rate and Euro area interest rates, respectively, by January 1999. Choices available to the Irish authorities were the level at which to set the new exchange rate for entry into EMU and the starting point and speed of the interest-rate convergence process.

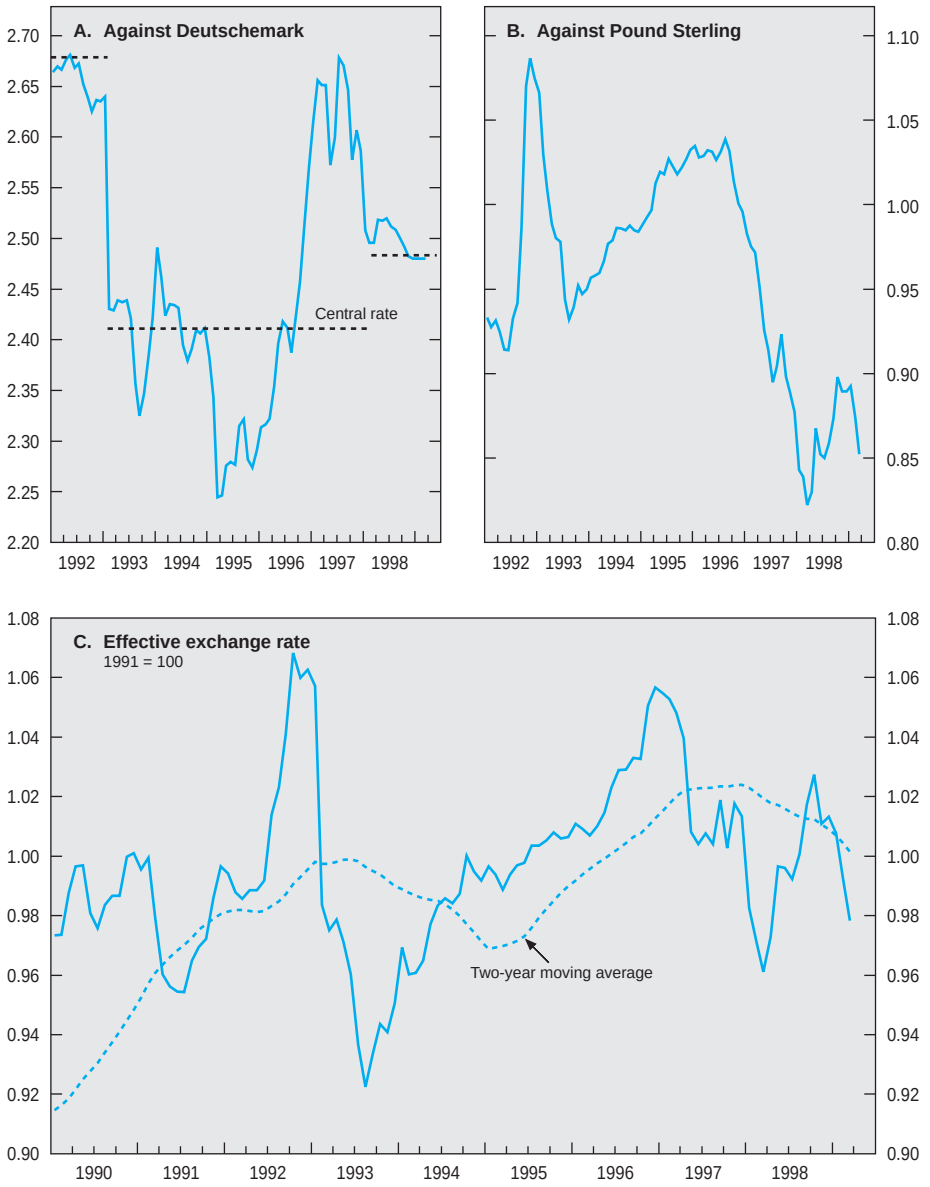
Exchange-rate convergence: limited appreciation

The Irish pound was the strongest currency in the European Exchange Rate Mechanism (ERM) during 1997, because of widened interest differentials carried over from the previous year (Figure 16 and Figure 17, Panel C). In fact, this strong currency position and expectations that Ireland would join EMU in January 1999, as well as excellent export performance, made market participants speculate about a possible revaluation: otherwise, a large depreciation toward the central rate would have made the potentially inflationary situation worse. The ebb and flow of this speculation was a source of volatility in the exchange rate in 1997 and early 1998.

The Irish pound continued to appreciate until the spring of 1997, when market participants reassessed the likelihood and possible magnitude of the revaluation of the Irish pound. This caused substantial capital outflows (Table 20) and thereby a sharp exchange rate depreciation. The Central Bank intervened heavily against the currency fall[77] and finally raised the short-term facility rate in May 1997 (Figure 16, Panel B). The exchange rate regained strength temporarily that summer, but started falling again in mid-September. On the other hand, it depreciated against Sterling from the summer of 1996 to the spring of 1998 by about 20 per cent,[78] reflecting higher interest rates in the United Kingdom (Figure 16, Panel C and D, and Figure 17, Panel B). Overall, the effective exchange rate continued to depreciate from the beginning of 1997. In March 1998, the ERM central rate of the Irish pound was finally revalued by 3 per cent, setting it close to prevailing market rates against other ERM currencies and effectively eliminating the possibility of further change. The new central rate was announced to be the irrevocable conversion rate against the Euro in May 1998; this was formally decided at the end of 1998. On an effective basis the currency appreciated sharply in the months following the ERM realignment. But since last autumn it has once again been on a downtrend, with very recent figures nearing the early-1998 trough.

Last year's revaluation by 3 per cent was meant to strike a balance between the need to retain competitiveness, especially in certain traditional sectors such as agriculture, and countering inflationary risks to the economy. It should be noted that the conversion rate was still 7½ per cent weaker than the old central rate up to February 1993. Moreover, thanks to large productivity gains, as well as structural change within the manufacturing sector, unit labour costs

Figure 17. **Exchange rate developments**



Source: OECD.

Table 20. **Capital account of the balance of payments**

Ir£ million

	1994	1995	1996	1997	1998
Current account					
Current account surplus	998	1 070	1 176	1 362	1 047
Market flows					
Irish government securities	–421	605	1 034	–1 122	–656
Transactions of credit institutions	140	1 798	–1 229	–303	4 350
Private capital flows and residual[1]	129	–1 700	–384	–258	–3 474
Non-market flows					
Capital transfers	251	511	489	578	661
Exchequer foreign borrowing	–416	–614	–986	–1 055	–697
Semi-state companies	–285	–260	–146	47	314
Other transactions[2]	–498	33	–10	–4	98
Official external reserves[3]	102	–1 443	55	754	–1 645
Counterpart to valuation changes	–135	–12	–459	430	168
Memorandum item:					
Level of official external reserves (end of year)	4 041	5 473	4 960	4 636	6 448

1. Private capital includes the net effect of reinvested earnings of direct investment enterprises.
2. Includes change in debtor position *vis-à-vis* EMCF (European Monetary Co-operation Fund).
3. Computed on a transaction basis, *i.e.* change in total reserves less valuation changes and allocations of SDRs.

Source: Central Statistics Office.

vis-à-vis the Euro area declined by over 6¼ per cent cumulatively between 1992 and 1998. From a demand management point of view, taking account of the substantial external surplus and observed bottlenecks to further growth, a larger revaluation would seem to have been preferable. It would have restrained net exports to some extent and alleviated more of the risk of overheating: exports from the modern sector would have been less affected than those from the traditional sector, where price competitiveness is likely to be a more important determinant of trade performance. From a supply-side point of view, which may be of much greater importance in such a small open economy, it would have encouraged labour supply through a damping effect on price hikes and hence higher real wages. Boosting real take-home pay in this way would have been a better measure than through personal income tax cuts (see below) in an overheating situation.

Interest rate convergence: the later, the better

At the beginning of 1997, interest rates in Ireland were higher than those in Germany as a result of precautionary measures taken in 1996 to prevent an excessively strong expansion of the economy. On the other hand, Ireland's

decision to participate in EMU from the outset made it necessary to eliminate (virtually all) the interest rate differential *vis-à-vis* German interest rates by the end of 1998. The strategy the authorities chose was to delay interest convergence as much as possible, given the buoyant economic situation. Long-term interest rates, however, tended to decline as from the spring of 1997, in anticipation of Ireland's participation in January 1999. The long-term interest differential *vis-à-vis* Germany fell to 40 basis points at the end of 1997. Further convergence has since been achieved: by mid-April it was down to 10 basis points, the lowest in the Euro area.

Because short-term interest rates are more under the influence of the authorities, their convergence process was more complex. In May 1997, the short-term facility rate was raised to deal with a fall in the Irish pound against other ERM currencies. The combination of subsequently stable short-term rates and a decline in long-term rates caused the yield curve to invert in mid-1997. Short-term market rates began to fall in March 1998 after the revaluation of the exchange rate made market participants expect that interest rate cuts were imminent. However, once the Central Bank's preference for delaying any cut for as long as possible was made clear, market rates bounced back. It was only in October that the Bank relented, and convergence was managed through a series of three rate reductions, with the last carried out in a co-ordinated fashion with other prospective Euro area countries. The cumulative decline in the repurchase rate was 3.2 percentage points – from 6.2 to 3.0 per cent.[79] The strategy of delaying the convergence is easily understandable, but how much difference it actually made remains unclear. Once the entry level of the pound into the Union was decided on, there was little further that monetary policy could do.

Fiscal and incomes policies

Fiscal policy has been seen mainly as an instrument for building consensus and strengthening the economy's supply side rather than to manage aggregate demand. The *ex ante* stance of fiscal policy remained slightly pro-cyclical as a consequence of the Partnership 2000 commitments. Nonetheless, rigorous economic expansion boosted tax revenues, partly offsetting the effects of successive tax cuts, and improved the fiscal balance further. Certain aspects of the tax cuts nevertheless frustrated some participants in the partnership process and further restrained government options. The partnership approach has been successful so far but will have to go through a challenging period in the near future.

Favourable out-turns of the 1997 and 1998 budgets

Both 1997 and 1998 out-turns were better than the government had expected, mainly a result of overshooting of tax revenues by about Ir£ 1 billion

each year (Table 21). Strong economic growth and asset price inflation boosted both tax and non-tax revenues.[80] As a result, the Exchequer deficit shrank more sharply than expected in 1997, and the balance turned to surplus in 1998 for the first time since the 1940s. The estimated central government account on a national accounts basis, which is relatively free from budgetary manipulations,[81] shows significant improvement in the government financial position (Table 22). Despite successive tax cuts worth an estimated Ir£ 0.9 billion cumulatively in 1997-98, tax revenues increased by around 14 per cent per year, notably more than 30 per cent per annum in the case of taxes on capital. Current and capital transfers, which are mainly provided by the EU, also increased, although they are expected to be scaled back as from 2000. As for expenditures, aggregate public service compensation grew at a fast pace, more than 9 per cent per annum on average in 1997-98, mainly because of wage adjustments as government

Table 21. **Budgetary developments**

Ir£ million

	1996 Outturn	1997		1998		1999 Budget
		Budget	Outturn	Budget	Outturn (estimate)	
Current budget						
Net expenditure[1]	12 662	13 391	14 015	14 388	14 412	15 401[2]
Social services	6 705	7 178	7 314	7 765	7 738	8 976
Service of national debt	2 360	2 560	2 755	2 625	2 543	2 542
Other expenditure	3 597	3 653	3 446	3 998	4 131	3 883
Revenue	12 954	13 584	14 619	15 497	16 503	17 736
Tax revenue	12 520	13 263	14 274	15 167	16 129	17 335
Non-tax revenue	434	321	345	330	374	401
Surplus		193	604	1 109	2 091	2 335
Capital budget						
Net expenditure	1 434	1 478	1 545	1 907	2 081[3]	2 247
Net voted expenditures	1 356	1 452	1 475	1 851	1 918	2 195
Other expenditure	78	26	70	56	163	52
Resources	705	648	706	709	737	836
Surplus (– Deficit)	–729	–830	–839	–1 198	–1 344	–1 410
Exchequer surplus	–437	–637	–235	–89	747	925
General government surplus	–120				1 410	1 057

1. In the budget, expenditure is shown net of receipts of Pay Related Social Insurance (PRSI) contributions and offsetting revenues attributable to expenditure activities.
2. This includes expenditures related to a newly established fund, the Local Government Fund, amounting to Ir£ 224 million.
3. This includes environment-related expenditure of Ir£ 224 million, which is not included in the 1999 Budget.

Source: Department of Finance.

Table 22. **Central government accounts**

Including extra-budgetary funds
Ir£ million

	1994	1995	1996	1997 Estimate	1998 Estimate
Current receipts					
Taxes on income and wealth	5 552	5 590	6 330	7 163	8 033
Taxes on expenditure	4 859	5 259	5 868	6 610	7 453
Social insurance contributions	1 883	1 959	2 004	2 174	2 422
Investment income	328	279	310	303	341
Transfers from the rest of the world (not elsewhere included)	353	296	388	449	523
Others	247	235	247	308	293
Total receipts – current	13 222	13 618	15 147	17 006	19 065
Capital receipts					
Taxes on capital	106	104	165	221	306
Loan repayments	301	401	183	141	276
Transfers from the rest of the world	222	509	404	482	498
Borrowing	3 712	4 186	6 721	6 608	1 407
Total receipts – capital	4 341	5 200	7 473	7 452	2 488
Total current and capital	17 563	18 817	22 619	24 457	21 553
Current expenditure					
Subsidies	390	440	527	641	674
National debt interest	2 028	1 988	2 091	2 039	1 816
Paid to residents	947	974	1 176	1 465	1 281
Paid to the rest of the world	1 081	1 015	915	574	535
Transfer payments	4 640	4 907	5 322	5 812	6 309
To residents	4 447	4 700	4 979	5 528	5 922
To the rest of the world	193	207	344	284	387
Expenditure on goods and services	2 951	3 072	3 250	3 530	4 043
Wages, salaries and pensions	2 221	2 287	2 404	2 595	2 947
Other	730	785	846	935	1 096
Grants to local government	3 022	3 161	3 323	3 586	3 737
Total expenditure – current	13 031	13 570	14 513	15 609	16 578
Capital expenditure					
Grants to enterprises	382	259	366	347	371
Other transfer payments	107	363	130	175	197
Redemption of securities	2 932	3 600	6 513	7 139	2 783
Loans and share capital	336	151	147	116	349
Gross physical capital formation	181	177	198	227	312
Grants to local government	581	691	744	838	958
Payments to the rest of the world	12	7	8	7	5
Total expenditure – capital	4 532	5 248	8 106	8 848	4 975

Source: Central Statistics Office and Department of Finance.

employment hardly changed. But interest payments on the debt, accounting for one-eighth of current expenditures, fell, holding total current spending increases to about 7¼ per cent per year. Thus, even though capital expenditure increased at more than 15 per cent annually, led by increases in capital outlays by central government in excess of 30 per cent per year, the general government balance turned to surplus in 1997. The surplus reached an estimated 2.4 per cent of GDP in 1998, implying a primary surplus (excluding interest payments) worth about 5 per cent of GDP.

This overall improvement in the general government fiscal situation, as measured by the estimated increase in net lending of 2¾ percentage points of GDP from 1996 to 1998, can be accounted for by two factors (Table 23). About 1¾ percentage points may have been due to above-trend growth, as well as that can be determined. The improvement was accelerated by reduced interest payments, further explaining about 1 percentage point. Thus, the cyclically-adjusted net primary surplus was slightly smaller in 1998 than in 1996: that is, the *ex ante* fiscal stance was probably modestly pro-cyclical and looks set to remain so in 1999.

Table 23. **Indicators of fiscal stance**

Per cent of GDP

	1994	1995	1996	1997	1998	1999	2000
General government balance							
Net lending	–1.6	–2.1	–0.3	1.0	2.4	2.3	2.1
Net primary balance	2.6	1.7	3.3	4.2	4.7	4.3	3.8
Cyclically-adjusted balances [1]							
Net lending	0.4	–1.9	0.0	0.5	1.0	0.9	1.0
Net primary balance	4.5	1.9	3.6	3.8	3.4	3.0	2.8
Gross general government debt	91.2	84.3	74.8	67.1	56.9	49.3	43.1

1. The structural balance was improved in 1994 by 0.8 per cent of GDP by a tax amnesty and worsened in 1995 by 0.7 per cent of GDP due to a one-off settlement of a legal award.

Source: OECD and Department of Finance.

The 1999 budget

The budgetary situation is expected to improve further this year, helped by robust albeit decelerating revenue growth (Table 21). Budgeted current expenditure, adjusted for the establishment of a new Local Government Fund, shows a

5.3 per cent increase from the previous year, that is, an average of 4 per cent per year growth compared to the 1997 out-turn – the government's self-imposed limit. While interest payments remain unchanged, expenditure for health, education and social welfare increased significantly. There are reasons to question the usefulness of the 4 per cent target for the increase in current expenditures. First, it should be noted that the targeted aggregate includes interest payments. Therefore, achieving the target in 1998-99 has been easier because of the reduction in interest rates associated with EMU entry. Furthermore, if Euro interest rates are cut in the future, a decline in interest payments on debt will make it possible to increase non-interest spending further without violating the target. Such a spending increase together with the decline in interest rates itself would give additional impetus to the economy. This destabilising aspect could be dealt with by redefining the target to remove interest payments. Second, changes in levy rates can keep the growth in net current expenditures within the limit. For example, the Employment and Training Levy rate was abolished while Health Levy rate was raised, resulting in net loss of revenue amounting to Ir£ 34 million in 1999. The budget, however, registers the two effects differently: the former is counted as a loss of tax revenues, while the latter is regarded as a decline of net current expenditures. Thus, while the government can claim that the spending growth was kept within the target, the growth in (gross) current expenditures remained higher. Capital expenditure growth, following a very strong rise in 1998, is expected to remain robust this year, with a rise of nearly 15 per cent. This slowdown is partly due to a new scheme for local governments to invest their own resources in roads. Overall, the budget projects a continuing large surplus for 1999, which will result in further debt reduction. The debt/GDP ratio should fall to around 49 per cent.

In the current budget, the government undertook significant tax changes (Table 24). The most noteworthy elements of the personal income tax package were the standard rating of the basic personal and PAYE (employee) allowances as the first step in a move to a full tax credits system. These standard rated allowances were increased significantly. The change to standard rating was accompanied by an increase in the standard rate band. Under the existing tax system, allowances have the effect of raising the threshold at which the higher rate of tax is payable. However, following the change to standard rating of the personal and PAYE allowances, these allowances will no longer have the effect of raising the threshold for the higher rate of tax. The standard rate band was therefore increased to ensure that there are no losers from the move to standard rating. These changes are of greater relative benefit to low income earners. This was in contrast to the changes in tax policy introduced in the 1998 budget: rate reductions and a rise in allowances benefited all tax payers but were of greater relative benefit to those at the high end of the income spectrum (Figure 18) because the increase in allowances was valued at higher tax rates for high-income earners.

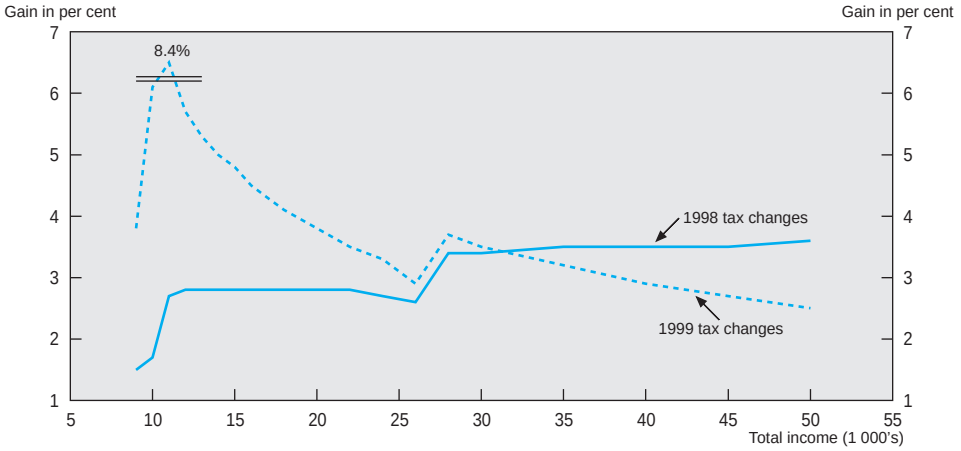
Table 24. **Outline of income and corporate tax changes in 1998 and 1999**

Item	Detail	Ir£ million Full year costs
Personal tax		
Changes in 1998		
Reduce tax rate (standard/top rate)	26/48 per cent to 24/46 per cent	315
Increase personal allowance (single/married)	Ir£ 2 900/5 800 to 3 150/6 300	125
Widen standard rate band (single/married)	Ir£ 9 000/18 000 to 10 000/20 000	15
Increase weekly PRSI-free allowance for full-rate employee contributions	Ir£ 80	38
Others		24
Total		517
Changes in 1999		
Standard rating personal and PAYE (employee) allowances		–733
Widen standard rate band (single/married)	Ir£ 10 000/20 000 to 14 000/28 000	778
Increase personal allowance (single/married)	Ir£ 3 150/6 300 to 4 200/8 400	400
Increase annual PAYE (employee) allowances	By Ir£ 200 to Ir£ 1 000	52
Others		84
Total		581
Corporate tax		
Changes in 1998		
Reduce corporate tax rate from 1999	Standard rate: 36 to 32 per cent, lower rate on the first Ir£ 50 000: 28 to 25 per cent	107
Others		–46
Total		61
Changes in 1999		
Reduce standard corporate tax rate from 1999	32 to 28 per cent	132
Increase the profit ceiling of the companies subject to lower 25 per cent tax rate	Ir£ 50 000 to 100 000	
	Memorandum: Future schedule of reduction from 2000: 24 per cent; from 2001: 20 per cent; from 2002: 16 per cent; from 2003: 12.5 per cent	
Others		–15
Total		117

Source: Department of Finance.

Figure 18. **Gains from tax changes by income class**

Per cent of net income



Note: The gain is calculated for the case of a married couple with one earner and no children, who are taxed under PAYE and full rate PRSI contributor.

Source: Department of Finance, *Summary of Budget Measures* (various years).

Box 2. The effect of the introduction of tax credits and the hybrid nature of the tax system

The differences in distributional consequences of tax cuts between the non-standardised rate system in 1998 and the standardised rate system in 1999 can be seen clearly in Figure 18. The gains (relative to income) from the 1998 changes are shown as an almost horizontal line with a jump at earnings over Ir£ 26 000, which is the sum of the married couples standard rate band (Ir£ 20 000), the married persons allowance (Ir£ 6 300) and the employee allowance (Ir£ 800). The change to standard rating in 1999 tilts the line in favour of the low-paid, even though the jump after Ir£ 26 000 remains. This is largely because of the increase in the married couples standard rate band to Ir£ 28 000. Therefore, the change in 1999 is one from a hybrid allowance system (where non-standardised allowances are dominant) to a hybrid credit system (where standardised allowances are dominant). The government intends to adopt a complete tax credit system in the near future.

Certain groups (including some representatives of low-paid workers) expressed dissatisfaction with this feature of the 1998 tax changes as well as with a cut in the capital gains tax rate, even though this measure was necessary to increase the supply of houses (see below). Therefore, the new tax package should go a long way towards easing workers' dissatisfaction and thereby put social dialogue back on the right track.[82]

Another aspect of the package is its response to labour shortages. Making the first 100 pounds of weekly wages free from taxation and levies is expected to remove an extra 80 000 people from the income tax net and thereby encourage labour supply by widening the gap between unemployment benefit and after-tax wages for low-paid workers. In fact, a large social welfare package was incorporated in the 1999 Budget, amounting to Ir£ 287 million (in a full year). The rise in social welfare benefit rates other than for old-age pensions) was 4 to 5 per cent depending on the scheme, well above the rate of inflation. Moreover, the government confirmed its intention to implement further cuts in both standard and higher income tax rates by 4 percentage points over the next three years, as set out in its policy programme.

As for corporate tax, the standard rate was reduced once again in 1999 following the previous year's cut, bringing the cumulative reduction to 8 percentage points. Following negotiations with the European Commission, a future reduction schedule was also announced, lowering the rate to 12.5 per cent in 2003, to which the preferential tax rate on profits of manufacturing and internationally traded services will be raised from the current 10 per cent at the same time.[83] The 28 per cent rate effective from 1999 is already the lowest in the EU (along with those of Finland and Sweden). Although there was no reduction in 1999 in the tax rate of 25 per cent for small businesses, the profit threshold for the application of this rate was doubled. The incentive for sole traders and partnerships to incorporate is likely to be reinforced by the significant gap between the top personal tax rate (46 per cent) and the small-business corporate tax rate. On the other hand, the contribution ceiling of Pay Related Social Insurance (PRSI) was raised for employers, even though the band for the lower rate was widened. Together these measures are expected to cost companies about Ir£ 24 million on a full-year basis. Although the cost itself is small compared to the gain from corporate tax cuts, employers were worried about its selective effect: according to IBEC, the cost will be borne primarily by the manufacturing sector.

Partnership 2000: a mid-term evaluation

The fourth partnership agreement concluded in December 1996 – Partnership 2000 for Inclusion, Employment, and Competitiveness ("Partnership 2000" hereafter) – covers the 39-month period to March 2000 (September 2000 for the public service). The total permitted wage increase was 9¼ per cent over

39 months, an average of 2¾ per cent per year. A further Ir£ 1 billion (over ½ per cent of GDP per year) was to be provided in the form of tax cuts, 90 per cent of which in the area of personal taxes. This was expected to raise after-tax incomes for those on the average industrial wage by up to 14 per cent (an average of 4.1 per cent per year). Despite much stronger economic performance than assumed in the agreement, according to a survey of the 778 private pay agreements concluded in the first 18 months of its applicability, 88.8 per cent followed the terms of the agreements. The survey may not, however, fully reflect the current situation, considering that a marked pick-up in wage growth was observed only as from early 1998. The unexpected buoyancy has probably put pressure on the Partnership agreement, as suggested by payment of the 2 per cent local bargaining element in advance of the centrally agreed date (bottom of Table 25). The early payment was observed in 37 per cent of the 206 local bargaining deals reviewed.

Several factors could explain the apparent divergence between the survey results and seemingly faster wage growth in the official statistics (see Table 17 in Chapter I). First, what is agreed centrally is the basic pay increase, while the official wage data include other payments such as bonuses, overtime, shift allowances, service pay, commissions, etc. Second, the official data may reflect compositional changes in the structure of employment. Although high measured wage growth in the official data is a sign of pressures on the terms of the agreement, it is not a convincing piece of evidence of overshooting. As for wage developments in the public sector, the official data seem to be consistent with the agreed terms, once account is taken of the fact that recent figures are still being boosted by major local bargaining settlements in the previous partnership agreement, the Programme for Competitiveness and Work.

A visible risk to Partnership 2000 was dissatisfaction with the 1998 Budget (see above) by some participants in the partnership process such as the CPSU, the largest civil service union, and the Community and Voluntary "pillar".[84] The CPSU went so far as to ballot its members during December 1998 on withdrawal from Partnership 2000; that suggestion was rejected by a 2 to 1 majority. The tax package in the 1999 Budget succeeded in preserving the partnership process. The cumulative cuts in personal and corporate tax from 1997 to 1999 amounted to some Ir£ 1 730 million, that is 73 per cent larger in total than the pledged amount of one billion pounds (Table 25).

Effectiveness of policy mix of fiscal and incomes policies

Since the start of EMU in January 1999, fiscal and incomes policies have become the only demand management tools available. One of the objectives of the Stability and Growth Pact (see Box 3) is to make budgetary room for manoeuvre. But the effectiveness of fiscal policy is perceived to be quite limited in the

Table 25. **Main features of Partnership 2000**

Assumption

GNP growth rate:	5½ per cent in 1997.
	4½ per cent in 1998 and 1999.
Inflation rate:	Around 2 per cent.
Employment growth:	38 000 per year.

Budget

An action programme for social inclusion and equity,
– additional expenditure of Ir£ 525 million to maintain real value of social welfare.

Tax reduction of Ir£ 1 billion on a full year basis (over 0.5 per cent of GDP each year),
– personal tax,
  • cumulative value over the three years of Ir£ 900 million on full year cost basis;
– business tax,
  • Ir£ 100 million on full year cost basis, including reduction in corporate tax.

Keep the annual growth of gross current supply service spending as close as possible to 2 per cent in real terms.
General government deficit should be not more than 1.5 per cent.
Debt/GDP of 70 per cent by 1999, 60 per cent in early 21 century.

Wage agreement

	Pay increase (per cent)	Applied period	
		Private sector	Public sector
Phase I	2½	01/97-12/97	Only for first Ir£ 200 of basic weekly pay in 07/97-03/98, for full basic pay from 04/98-06/99
Phase II	2¼	01/98-12/98	07/98-06/99
Phase III	1½	01/99-09/99	07/99-03/2000
Phase IV	1	10/99-03/2000	04/2000-09/2000
Local element	2	Not earlier than 07/98	From 07/99

Source: Partnership 2000 for Inclusion, Employment and Competitiveness.

first place because of the openness of the Irish economy. Although joining EMU may in theory strengthen the effectiveness of fiscal policy in the short run because of the disappearance of crowding-out effects as domestic interest and exchange rates become virtually exogenous, in practice the effectiveness of fiscal policy is likely to be rather small and will be eroded further by greater regional integration in the long run. Thus, the effect of fiscal policy on output seems limited; indeed, how sensible it is for a "regional" government to adopt active demand management is unclear. To the extent that there are imbalances between supply and demand, they will manifest themselves in the current account, which should become much less of a concern for policy makers for a small country in EMU.

Box 3. The Stability and Growth Pact

The Stability and Growth Pact clarifies the Maastricht Treaty's provision for dealing with "excessive deficits" and provides an institutional framework for its enforcement, in part through strengthened surveillance and co-ordination of economic policies *via* the annual review of national stability programmes. The Pact also calls on participants in the Economic and Monetary Union (EMU) and those countries with a derogation from initial participation in stage 3 of EMU – the actual introduction of a single monetary policy – to adopt budgetary balance (or even a surplus) as the medium-term objective.

The Pact considers a general government deficit above 3 per cent as excessive unless the European Union judges it to be temporary and there are special circumstances. Temporariness implies that, according to the projections of the Commission, the deficit would fall beneath the 3 per cent threshold in the following year. In the case of an excessive budget, the Council will, on the recommendation of the Commission, propose a course of action for the country which should be followed by effective measures within four months. The Council will monitor the measures and if they are found to be inappropriate it will make further more detailed proposals, which will be published. If corrective measures have not been implemented within ten months, sanctions will be imposed. These would initially take the form of non-remunerated deposits, with a fixed component equal to 0.2 per cent of GDP and a variable component rising in line with the size of the excessive deficit. Such deposits are limited to a maximum of 0.5 per cent of GDP per year, but would accumulate each year until the excessive deficit is eliminated. Provided the excessive deficit is corrected within two years the deposits are returned to the country, otherwise the deposits will become non-refundable. In the event that the measures taken by a country are ineffective, the whole process will be restarted, but sanctions will be imposed within three months.

When a country is judged to be in recession, which is defined as an annual fall in real output (GDP) of at least 0.75 per cent, the Pact will be enforced in a differentiated manner. If economic output in a member country declines by 2 per cent or more – and provided the deficit is temporary – exemption from the procedure is granted automatically. In the event GDP falls by between 0.75 per cent and 2 per cent, exemption can be granted in special circumstances by the Council. The country would need to convince the Council that the economic decline was "exceptional" in terms of its abruptness or in relation to past experience.

For Ireland, the risk of a deficit exceeding 3 per cent of GDP is rather remote in coming years, given that the general government enjoyed a surplus equal to around 2.4 per cent of GDP in 1998. Indeed, recent OECD work, along the lines of Dalsgaard and DeSerres (1999), shows that, based on the historical experience of the past 20 years or so, a zero deficit in cyclically-adjusted terms would suffice to ensure that the deficit ceiling is not violated over a five-year horizon with 90 per cent probability. OECD figures suggest that recent and prospective outcomes are on the order of a 1 per cent (of GDP) surplus. Thus, if the policy stance remains unchanged, the likelihood of a deficit greater than *2 per cent* of GDP in the next five years is only around 10 per cent.

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* Sanctions do not apply to those countries with an opt-out or derogation from participation in stage 3 of EMU.

It is not easy to assess the impact of a discretionary fiscal action on the economy when its supply-side effect is significant. Suppose, for example, that the announced future cuts in income tax rates by 4 percentage points (see above) were to be postponed as a way of restraining aggregate demand. If the postponement is perceived to be temporary, thereby leaving aggregate demand almost unchanged, inflation might even pick up as labour supply decreases (compared with the baseline of no postponement) in response to this announcement. The elasticity of labour supply in Ireland is likely to remain higher than those of other countries because another source of labour supply, migration, is available in addition to intertemporal substitution.[85] Hence, only if the postponement is perceived to be permanent will the effect on demand be important. However, because people understand that the partnership approach ensures low nominal pre-tax wage growth and competitiveness in exchange for tax cuts, the permanence of any postponement might not be credible. Thus, taking account of the effects on the supply side, the effectiveness of discretionary fiscal policy seems even more limited.

The authorities' inclination to use the budget as a policy instrument to improve the supply side, rather than to manage aggregate demand, makes sense. Given the current low tax rate compared to other countries, however, tax tools should be used selectively. In this sense, tax measures to reinforce work incentives incorporated in the 1999 Budget can be considered as a step in the right direction. As for spending policies, an increase in outlays for infrastructure seems necessary, taking account of existing bottlenecks and the forthcoming reduction in EU transfers, even though adoption of PPP (Public-Private Partnerships) and/or a service charge on the use of infrastructure[86] would help boost the efficiency of such outlays. The policy of embedding fiscal policy in the so-called "partnership approach" seems to have been successful, as their effectiveness has been mutually reinforcing. For example, it might have been more difficult to restrain the growth of public-sector pay without such an agreement. However, too much should not be expected from the partnership approach: the current situation will be a real test of its sustainability and usefulness in a period of increasingly scarce labour.

Short-term outlook and risks

Monthly indicators show that the vigorous economic expansion, especially in equipment investment and exports, during the first half of 1998 was followed by some moderation in the second half of the year (Table 26). The year-on-year growth rate in retail sales slowed in two consecutive quarters, although much of that was concentrated on the volatile garages and filling stations group. Despite a continuing boom in house building, overall activity in the

Table 26. **Recent economic indicators**

Percentage change over previous year

		1995	1996	1997	1998	1998 Q1	1998 Q2	1998 Q3	1998 Q4
Manufacturing									
Production	Index	20.1	8.2	16.6	16.7	18.2	16.1	16.5	16.2
Employment	Number	5.7	4.1	6.0		4.9	3.9	–	–
Construction									
Housing planning permissions	Number	15.6	27.9	6.3	27.9	20.8	3.7	63.9	28.9
Dwellings completed	Number	13.8	10.3	15.2		6.1	7.9	8.2	–
Employment	Index	4.5	8.2	9.3	8.8	9.6	9.0	8.5	8.2
Investment									
Industrial new construction	Value	32.1	25.9	42.3		107.2	25.5	–	–
Commercial new construction	Value	50.8	23.8	45.1		–9.7	6.4	–	–
Capital goods imports	Value	20.5	16.5	25.8		35.5	35.2	36.6	–
Goods vehicle registrations[1]	Number	8.3	20.7	11.3		21.1	19.2	14.3[2]	–
Capital acquisitions in industry	Value	n.a.	n.a.	6.6		25.7	27.8	11.8	–
Consumption									
Retail sales (total)	Volume	2.7	6.2	7.9	8.8	10.1	10.0	9.0	6.6
Retail sales (exc. garages)	Volume	2.9	4.7	7.1	8.7	9.0	9.5	7.7	9.1
Car licensings[1]	Number	6.8	23.5	8.8		10.0	12.8	12.1[2]	–
Exchequer spending	Value	7.9	6.3	10.5	5.5	3.5	9.9	1.6	7.1
Foreign trade									
Exports	Volume	20.1	9.9	14.3		24.1	22.4	23.5	17.4[3]
Imports	Volume	14.4	10.0	14.7		26.1	16.4	13.8	11.6[3]

1. New vehicles and newly registered second-hand vehicles.
2. July and August.
3. October and November.

Source: Central Statistics Office.

construction sector may be feeling the restraining effects of manpower shortages, judging by a deceleration of employment in that sector. Unfavourable outcomes in Ireland's major trading partners, Germany and United Kingdom in particular, have begun to affect business confidence somewhat, even though exports and production remain buoyant. Overall, the observed deceleration is quite modest.

The moderation of the economic expansion is expected to become a little more visible in 1999, mainly through more subdued export growth, brought about both by weakened market growth, particularly in Europe, and eroding competitiveness associated with some slowing in inward investment flows (as wages are bid up) (Table 27). However, the momentum of domestic demand may be maintained because of the stimulus from the substantial tax cuts in the 1999 budget and the lower interest rates that have resulted from the convergence process. The

Table 27. **Short-term projections**

	Estimates 1998	Projections 1999	Projections 2000
	Percentage change on previous year		
Demand and output			
Private consumption	8.0	8.0	6.6
Public consumption	5.0	4.0	3.1
Gross fixed investment	12.0	10.9	9.1
Business	12.0	10.7	9.5
Housing	13.7	12.4	9.0
Government	8.5	8.5	7.0
Final domestic demand	8.4	8.1	6.7
Stockbuilding [1]	–0.6	0.1	0.0
Total domestic demand	7.5	8.1	6.6
Exports of goods and services	20.9	11.4	10.2
Imports of goods and services	20.2	13.0	10.9
Foreign balance [1]	4.3	1.0	1.4
Gross domestic product [2]	10.4	7.5	6.7
Gross national product [2]	9.0	6.6	6.1
Inflation			
Private consumption deflator	2.6	2.9	2.8
Export deflator	3.0	2.0	1.6
Import deflator	2.2	2.0	2.1
GDP deflator	4.0	4.1	3.6
Labour market			
Total employment	8.4	5.2	3.8
Unemployment rate	7.7	6.4	5.9
	Per cent of GDP		
Current account balance	1.9	0.6	0.0
General government financial balance	2.4	2.3	2.1

1. Contribution to GDP.
2. Expenditure base estimates.
Source: OECD.

slowdown in import volumes is hence likely to be less marked than that of export volumes. As a result, output growth is projected to slow from 10 per cent in 1998 to 7½ per cent in 1999, entirely on account of the much smaller contribution of net exports. However, with the two expansionary effects mentioned above petering out, the domestic demand expansion is likely to ease as well by 2000 and push output growth down to some 6¾ per cent in 2000.

The milder pace of expansion is likely to cut employment growth. Labour force growth is also expected to slow only slightly because of a gradual depletion of would-be immigrants (see below). The unemployment rate is, thus, expected to

fall further, albeit less rapidly, to 6.4 per cent in 1999 and 5.9 per cent in 2000. This labour market tightness could raise total compensation per worker in the private sector by 6½ per cent each year, although this year the basic wage rate is assumed to rise in line with the terms of Partnership 2000. This is expected to push up services prices and investment deflators, thereby raising the GDP deflator by an average of nearly 4 per cent per year, as in 1998. Thus, overheating is likely to continue, although the projected slowdown in output growth may prevent further deterioration.

There are several uncertainties. The first concerns external developments. If Ireland's major trading partners go into a recession, inward investment as well as exports would be affected, although the buoyancy of the domestic economy might be able to absorb the deflationary shock to some extent, particularly if Euro-area interest rates are cut further in response. On the other hand, there is a non-negligible risk that wage inflation could become excessive, ultimately leading to a "hard landing" involving the simultaneous downsizings or shutdowns of several large employers. The risk of undue wage increases is greater in sheltered sectors such as construction and public services. Even if the agreed terms in Partnership 2000 have succeeded in damping wage expectations to this point, the next round of negotiations may be difficult to conclude: organised labour may try to recapture gains from the economic expansion, given recent declines in labour's income share.

The sustainability of the boom

As discussed above, the risks of serious overheating are not negligible in the short run, even if output growth slows down to a moderate extent. Ultimately, however, higher wages and prices will result in diminished competitiveness which should bring the economy back to a sustainable growth path. Given that the range of policy instruments to influence aggregate demand is now extremely limited and their effectiveness much reduced, the best policy makers can do is to focus on those measures which would make the supply side of the economy more flexible and adaptable. The following paragraphs assess the determinants of sustainable long-term growth rates and policies that could enhance those prospects.

Labour supply factors

Still strong demographic effects

The last official forecasts for population were undertaken in 1995 and are therefore somewhat out of date; for example, a lower bound for net emigration was set at 0.2 per cent of the population per year. Assuming zero net migration,

the Economic and Social Research Institute (Duffy *et al.*, 1997) based their 1997 medium-term projections on a population growth rate out to 2011 which averages between 0.4 and 0.5 per cent per year. However, the two intervening years have seen a natural increase in the population of between 0.5 and 0.6 per cent. As stated above, net immigration in the year to April 1998 was even greater than the natural increase in the population.

However, it is unclear whether this is sustainable for several reasons. First, more than half of all immigrants are returning Irish and the stock of such former emigrants who could conceivably return (those who have been gone less than ten years) is dwindling. Annual gross emigration peaked in 1993 at 1 per cent of the population and has since fallen by 40 per cent to 0.6 per cent in 1998. At the same time return migration has edged up from an average of 0.5 per cent of the population in the years 1993-96 to more than 0.6 per cent in 1998. But, even if it is shrinking, the stock remains large and will be so for several more years to come. Furthermore, migration is very sensitive to labour market conditions (Kearney, 1998), and with both unemployment and wages moving in a favourable direction, economic incentives to (re)locate in Ireland should strengthen. A second factor that has become dissuasive is the cost of housing. As discussed in Chapter I, housing prices have gone up sharply in recent years and look set to continue to outperform those in most other markets where prospective immigrants might be found, further boosting the cost of migrating to Ireland. Yet there is little evidence on how important a restraint on migration flows this is likely to be. Finally, there is ultimately some socio-political choice involved in migration. Only citizens of the EU have the automatic right to work in Ireland, although they form the large majority of all immigrants. Overall, it seems reasonable to say that an assumption of zero net migration is overly cautious for the next few years, even if in the very long run migration should tend to zero. Summing up then a conservative view of the population growth in the next few years would be for it to average around 0.6 per cent, with an alternative scenario with initially greater immigration, putting it at around 1 per cent (Table 28).

The working-age population will, however, continue to grow more quickly than the overall population in view of the continuing movement of the large cohorts born in the late 1970s and early 1980s into that age group. This effect will not be exhausted until around 2011. Ireland will be moving toward a dependency ratio which is more in line with those of other OECD countries from one that was unusually unfavourable. Initially, the inverse dependency effect might be expected to contribute around 0.8 percentage point per year to increasing labour supply, but in a few years that contribution will be down to around 0.5 point per year. Summing up then demographic change should add between 1.4 and 1.8 per cent per year to labour force growth, dropping to 1.1 to 1.5 a few years into the future.

Table 28. **Sustainability as viewed from the supply side**

Per cent per year

	Early years (2000-2002)		Medium term (2003-2005)	
	Base case	Optimistic scenario	Base case	Optimistic scenario
Contribution to growth from:				
Total population	0.6	1.0	0.6	0.6
Age structure	0.8	0.8	0.5	0.5
Participation rate	1.0	1.1	0.4	0.6
Reduced unemployment	0.0	0.0	0.0	0.0
Total labour input	2.4	2.9	1.5	1.7
Labour productivity	*2.3*	*2.8*	*2.0*	*2.5*
Real GNP	4.7	5.7	3.5	4.2
Factor income	*0.3*	*0.3*	*0.2*	*0.2*
Real GDP	5.0	6.0	3.7	4.4

Source: OECD.

Favourable participation effects as well

The final component of labour supply is the participation rate, defined here as a share of the population of working age. There seems little reason to expect any change in this activity rate for men as it has been relatively constant: the earlier slight downtrend, mainly reflecting those 55 and over, seems to have been reversed in the last year. Thus, it will be the evolution of the female rate that will determine the course of this component. As seen in Figure 5, its rise has been substantial in recent years, and the factors that have produced it (increased education, cohort-specific effects and greater job availability) look set to continue for some while yet, before they begin to wear off somewhat. Policy does have some role to play here. The recent tax and benefit changes should encourage greater participation. But above all, if the government wishes to encourage a continuing shift of prime-age women into the formal labour market, it will have to encourage employers to experiment with flexible forms of employment, reduce work disincentives in the area of child benefit payments and make more childcare available in the near future. This last requirement seems to have been accepted, but the commitment to act quickly has thus far been lacking. Nevertheless, the female participation rate should continue to rise through the medium term, although with a decelerating trend. And the contribution from this factor should initially be near the 1.0 percentage point it has managed in recent years, before tailing off to around half that rate.

But drawing upon the stock of unemployed is almost exhausted

Over the past few years employment has been augmented by drawing down the stock of the unemployed. With the standardised unemployment rate now at around 6¾ per cent, however, there is less scope for the rate to be pushed down further on a sustained basis. The question reduces to the assessment of what constitutes a "natural rate" of unemployment in Ireland. This is not necessarily a constant figure, as it will vary with such factors as demographics (it will be lower the higher the prime-age share of the labour force), structural change in labour demands (it will be higher the more mismatched are such demands by occupation and skill levels relative to supplies that are fixed in the short run) and policy (it will be influenced by a wide variety of choices related to benefit levels and duration, taxation, barriers to mobility, etc.). The *OECD Jobs Strategy* provided a detailed look at the subject. With a labour market that is very integrated with that of the United Kingdom, Ireland may not have a well-defined natural rate, however. While it is the case that the relationship between unemployment and either wage or price inflation broke down around 1987, most likely owing to reduced inflation expectations associated with both external changes as well as benefits from the centralised wage agreements, it is possible that this relationship is now beginning to reassert itself as labour is becoming scarce (see Chapter I). But lacking any aggregate wage figures since 1996, it is difficult to form strong judgements. The official target for the unemployment rate at end-2000 as set out in last year's *Employment Action Plan*, 7 per cent, was reached already in January. But even if the ultimate objective of the government of 5 per cent proves possible,[87] it looks likely that it will have to be approached rather gradually, and that it would be prudent not to count on any further net labour supply from the ranks of the unemployed after the projected decline to around 6 per cent by end-1999.[88] Summing up then, potential labour inputs can be expected to grow by around 2.4 per cent per year in the next few years and then to ease progressively to an average of about 1.6 per cent by the middle of the next decade.

What to expect from productivity growth

Labour productivity growth (here measured by real GNP per employee)[89] has averaged about 3.6 per cent over the last five years. In order to judge to what extent this rate can be sustained there are a number of relevant considerations. First, it is possible that productivity gains are in part cyclical and that when the boom comes to an end that portion of the increases will be lost. Indeed, going back to 1961 there does seem to have been a significant relationship between per capita output growth and productivity increases. Using the estimated elasticity between the two (in the range of 0.5 to 0.6), a first approximation is that around a percentage point of the increase might be taken to be cyclical, leaving an underlying annual trend of something closer to 2.6 per cent, but with a fairly wide confidence interval around that figure. By chance, that would appear similar to the

estimated outcome in 1998. Another possible influence on the rate of productivity increases would be the gap that remains between per capita output in Ireland and abroad. This would clearly point to some slowdown as well, since, as was pointed out above, Ireland is rapidly converging on the per capita outcomes of its trading partners, the EU in particular. A third important influence emanates from capital. However, there is virtually no relationship apparent between capital stock growth and labour productivity outcomes. For its part the contribution from human capital accumulation can be expected to continue to grow, even if at a somewhat slower rate than in the past, as average attainment levels for youth cohorts flatten out, more low-skilled people heretofore outside the labour force are encouraged to seek employment and the process of shifting away from earlier, low-attainment generations ebbs. According to Duffy *et al.* (1997), this factor is worth only about 0.1 percentage point per year, but others put the value higher. Finally, the benefits of EU membership, including the receipt of structural funds and other agricultural transfers and the effects of the SEM, will almost certainly decline radically in the coming years. The rate at which the EU transfers will be cut back is difficult to predict at this point, but the waning of the SEM-related gains in growth-rate terms might account for a slowdown of about 0.2 percentage point per year (Bradley, 1998).

Summarising then, we might expect annual labour input growth to be about 2.4 per cent as from next year and trend productivity growth of about the same order of magnitude, leaving real potential GNP growth of around 4 per cent per year. In a few years that figure might slow to around 3½ per cent per year. Optimists might choose figures about a percentage point higher in the early years and ¾ point in the more distant term. Another approach would be to bypass most of these details and to seek to pinpoint the rate at which the economy can grow with an unchanging unemployment rate, assuming that relationship (the Okun curve) is stable. Recently Walsh (1999*b*) has shown that there has indeed been such a stable relationship for any given unemployment rate in the United Kingdom. His results show that real GNP can only grow at a rate of 3½ to 4 per cent without leading to pressures in the labour market: for each percentage point by which growth exceeds that pace, the unemployment rate falls by around ⅓ percentage point. This is roughly consistent with the 1998 outcome when the unemployment rate fell by around 2½ points on average and real GNP looks to have risen by around 9 per cent. Thus, considering all the evidence, the range for reasonable potential growth estimates is from about 3½ per cent to 5 per cent in the near term and from 3½ per cent to 4¼ per cent in the medium term.[90]

What policy can do to prolong the high-growth era

It is inevitable that the low-technology traditional manufacturing sector will continue its trend decline as it loses out in competition for scarce capital and

labour inputs to more promising producers further up the value-added chain. But the speed with which the process takes place and the reasons for it (that is, whether it occurs in a context of continuing high utilisation of domestic resources) can be managed.[91] There are a number of policy areas that are essential to medium-term prospects. The key is to alleviate supply-side constraints so that serious overheating is avoided and competitiveness is as well maintained as possible. The first resource constraint is labour. While there is little that can be done to change demographics, the migration and participation questions are subject to policy influence. However, as has already been pointed out, both of these channels involve crucial non-economic choices.

Alleviating the housing constraint

Should it be seen as desirable to attract more immigrants then it will be necessary to house them, and housing availability and the related problems of commuting time and congestion are also serious constraints on drawing more workers into the market. The stock of housing units per capita is not much more than two-thirds of what it is in the rest of the EU. Even if that is to some extent justified by larger average household size, falling birth rates in the past 19 years will quickly cut that gap just as the demographics work to increase household formation in the near future.

In order to stabilise sharply rising house prices (see Chapter I), which have pushed up the number of households in the queue for public housing, the government announced a set of policy actions in April 1998, following recommendations made by the Bacon report (Bacon, 1998) (Table 29). They were officially presented as a three-pronged approach, aiming at increasing the supply of serviced land for an extra 100 000 units, reducing excessive investor demand and assisting lower-income house purchasers. The first objective was expected to be achieved by increasing infrastructure investment, encouraging local authorities to promote higher densities in suitable places, increasing the supply of second-hand houses by reducing transactions costs, and reducing the capital gains tax rate temporarily, followed by a later surcharge. The second was dealt with by reducing fiscal incentives to invest in residential properties and raising transaction costs for those other than owner occupiers. The third was handled by making the local authority shared ownership scheme more generous. Most of the measures announced were implemented by the summer of 1998, although some necessarily involve a lag. They may have been effective in moderating the rise in house prices because house prices showed a slower pace of increase in the second half of the year. However, their impact on the *overall* market situation may have been marginal, because the flow of new houses supplied per annum is small compared to the existing housing stock – about 3 per cent. Further changes were recommended in a follow-up Bacon report in March 1999 and drew an immediate policy

Table 29. **Summary of policy actions taken to stabilise house prices**

1. Serviced Land Initiative (SLI)
- Announced details of allocations under the Ir£ 15 million SLI to provide water and waste water services for some 44 500 additional housing sites.
- Provided an extra Ir£ 24 million to fund further releases under SLI.
- Provided a new provision of Ir£ 5 million to help local authorities to target (on the same basis as SLI) areas where roads infrastructure is a key constraint.

2. Planning
- Requested local planning authorities to promote increased densities in appropriate locations and advise them of the factors to be taken into consideration in moving towards higher residential densities.
- Will issue planning guidelines to local authorities in order to promote higher densities in new housing developments, particularly in major urban areas.

3. Local authority shared ownership scheme
- Extended the scope of the scheme significantly, by increasing the effective income limit (for example, a rise from Ir£ 15 000 p.a. to Ir£ 20 000 p.a. in the case of a single-income household).
- The shared ownership rent charged was reduced from 5 per cent to 4.5 per cent with effect from May 1998.

4. Stamp duty changes
- Rates were reduced in April 1998 (for example, the threshold for the 9 per cent rate was raised from Ir£ 170 000 to Ir£ 500 000).
- Duty is now charged on all new houses bought by non-owner occupiers, who had not previously been charged.
- Lower rates are applied to second-hand house buyers and new-house owner occupiers.

5. Tax changes
- Section 23 reliefs, which is tax relief available to investors purchasing new houses of a specified size and standard and offering them on the letting market for 10 years, were revised in August 1998 so that they would operate on a much more restrictive basis.
- Disallowed interest payments on borrowings undertaken for investment in residential property purchased on or after 23 April 1998, as an offset against rental income for personal tax purposes - subject to transitional provisions. The measures did not affect those who retained ownership of existing (as of 23 April 1998) rented accommodation.
- Reduced temporarily capital gains tax on disposals of land for residential development from 40 per cent to 20 per cent from April 1998 for 4 years and will increase the rate 60 per cent thereafter.

Source: Department of the Environment and Local Government.

response with a view to ensuring maximum supply of serviced land, reducing planning delays and providing early improvement in public transport. With regard to affordability, the government is also considering whether or not to legislate to force builders to provide more affordable housing either by setting a quota of houses at a specified cost as a precondition for planning permission or imposing a levy to subsidise a share of all new housing development.

There may be room for improvement in government policies. Strategic planning guidelines for the greater Dublin area have just been agreed. Their adoption together with the implementation of expedited planning procedures with a view to making even more serviced land available could go a long way in smoothing the current housing boom so that increases in activity in the construction sector persist and are not followed by a bust. Similarly, measures must be taken to broaden the rental market, at least in areas receiving substantial foreign investment. Ireland has one of the smallest private rental shares in the OECD (Table 30). Even if the measures taken to reduce the appeal of such investments made sense in order to ease demand-side pressures, serious thought has to be given as to how to renew such investments in the near future in order to see an expanded rental sector play its full role in an increasingly cosmopolitan society. Tax policy would seem to be an obvious possible route.

Table 30. **Housing tenure in OECD countries**[1]

Percentage shares

	Owner-occupied	Social rented	Private rented	Other[2]
European Union				
Ireland	**80**	**11**	**9**	**0**
Austria	41	23	22	14
Belgium	62	7	30	0
Denmark	53	20	21	6
Finland	72	14	11	6
France	54	17	21	8
Germany	38	26	36	0
Greece	70	0	26	4
Italy	67	6	8	10
Luxembourg	67	2	31	0
Netherlands	47	36	17	0
Portugal	65	4	28	3
Spain	76	2	16	6
Sweden	43	22	16	19
United Kingdom	66	24	10	0
Other OECD				
Australia	70			
Canada	63			
United States	64			
Japan	59			
Norway	62	4	21	14
Switzerland	31	3	60	6

1. 1995 except for Australia, Canada, United States and Japan which are for the early 1990s.
2. Basically municipally-owned and employer-provided apartments.

Source: OECD (1999*a*).

Policy changes could moderate some of the factors pushing up house prices. Housing is currently favoured as a form of investment since mortgage interest is partially deductible (at the standard rate) and there is no tax on the imputed rent from owner-occupied housing. While the value of mortgage interest deductibility has been curtailed considerably during the 1990s, either phasing out the deduction (as has been done in the United Kingdom) or bringing imputed rents into the tax base (as is done in Switzerland, for example) would remove the bias; the former would be preferred on the grounds of practicality. Advanced purchase may be a reasonable strategy for the prospective buyer because even an eventual deceleration of economic growth may not ensure moderation of house price increases, in which case it may make sense to buy a house first and keep it until it is actually utilised. A lack of holding costs[92] makes this strategy more attractive. Because it is possible to regard capital gains tax as deferred payment of property tax, appropriately structured capital gains tax rates, which become higher as properties are owned longer, would eliminate the distortion by effectively raising holding costs. Revisions in capital gains and acquisition tax and/or the introduction of property tax would help to moderate the uptrend in prices.

As argued in the Bacon report, it is clearly important to increase the price elasticity of supply of housing. Taking account of the lack of any sign of deceleration in second-hand house prices and of potentially available supply from increased density, it seems necessary to encourage more effective use of land, in and around Dublin in particular. Re-introducing a residential property tax and thereby increasing opportunity costs of inefficient land use would reinforce the incentives for landowners to use their land more effectively. Because property tax is a way to recapture the capitalised benefits on property created by infrastructure investment, its introduction would also be justified by distributional considerations. In addition, it would weaken to some extent the expansionary effects of concomitant increases in infrastructure investment, a desirable by-product in the current cyclical circumstances.

Filling the infrastructure gap

Other than housing there would seem to be shortages of a variety of other kinds of infrastructure[93] whose provision will increasingly have to be paid for out of domestic funds as EU sources are phased out.[94] These bottlenecks introduce inefficiencies in production and consumption and often entail unnecessarily high input costs hampering competitiveness.[95] They also attenuate the attractiveness of Ireland as a prospective investment location.[96] Transportation and sanitary services are the two most prominent, given that telecommunications, and to a lesser extent electricity, should be handled by market forces in the future[97] (see Chapter III). What data are available show that both rail (the lack of an electrified network and poor suburban rail services)[98] and road transport (the paucity of

motorways)[99] infrastructures are poorly developed (National Competitiveness Council, 1998; Fitzpatrick Associates, 1998) and that airports and sea ports will also require major upgrading and expansion in coming years. A recent private-sector estimate for the cost of filling the transport infrastructure gap is Ir£ 5.6 billion, over 10 per cent of GDP (Fitzpatrick Associates, 1998). The National Roads Authority recently estimated a need for road improvements costing Ir£ 6.1 billion over the next 20 years. Capital outlays for national roads in 1999 amounted to Ir£ 312 million, up 18 per cent on the 1998 total. The budgetary impact of further rises could be lessened by the introduction of pricing mechanisms (toll charges).

On the environmental front deficiencies are evident in a number of domains (see Chapter III). First, although the share of the population served by waste water treatment plants has risen from 45 per cent in 1993 to 68 per cent more recently, that is still less than in most other EU countries. Meeting the relevant EU directive will require substantial further capital outlays, even if other forms of treatment may be sufficient in low-density areas. Second, municipal solid waste treatment relies excessively on landfill, with almost no incineration and very little in the way of recycling. Existing landfill sites are nearing the end of their useful life and their replacement will be significantly more costly. Again several billion pounds of capital outlays may be required, especially if, in addition greenhouse gas emissions targets are to be met. It is clear that the public finance burden of meeting these needs will be enormous. However, it will be much facilitated if greater use is made of pricing mechanisms in their utilisation.[100] A combination of mixed public-private joint ventures[101] and charging for access to excludible infrastructure (especially urban road space) would go a long way to ensuring that provision is efficient and demand is controlled. Similarly, application of the polluter-pays principle requires charging for fresh water use[102] and for solid waste disposal.

The other area in which the government has an important role is in the production and dissemination of knowledge. Bottlenecks in certain skills areas are endemic in OECD economies, but it is not surprising that they are particularly acute in one that is growing as fast as Ireland's. The government has recognised its responsibilities in this area and in 1997 committed Ir£ 250 million to an Education Technology Investment Fund to be spent over the years to 2000. Ir£ 50 million of that will be spent equipping schools with 60 000 computers and training 20 000 teachers. The target is to have at least one personal computer for every five secondary students by the end of the programme. Substantial numbers of electronics and software engineers and technicians have been urgently needed for several years already,[103] and 1 750 post-secondary places have been created. A further 5 400 places will be created in the next few years. A two-year linguistic programme for the teleservices sector has also been expanded by 1 000 positions. But care should be taken not to over-endow the tertiary sector since demographic demands are expected to peak in the near future, and a sharp falloff

is expected thereafter, despite a high and apparently still rising (until 1994) return to tertiary education. More public research spending should be devoted to economic development. A National Seed Capital Fund was set up in mid-1998 with EU assistance to provide venture capital to early-stage small businesses. A similar strategy in Spain has not been entirely successful (OECD, 1998*d*, Chapter X). Links between the tertiary education sector and business need improvement, as there is no tradition of collaboration between the two. In response, a National Qualifications Framework for technical training in the tertiary, non-university sector is being finalised. It will deal with access, progression arrangements and standards setting.

Shifting the focus of industrial policy

Finally, the area that is possibly the most clearly in need of a shift in direction is that of industrial policy, at least as it concerns foreign investments. Although it was argued above that this policy had been highly successful in bringing about the boom, it is one that at a minimum needs to be rethought in the new era of scarce rather than excess labour. The justification for the selective project approach in terms of systemic market failures is elusive. Industrial policies should focus rather on how to generate more innovation, and indeed greater attention is now being paid to the role of innovation in the success of indigenous enterprise and the knowledge content of foreign investments. But even within such an approach, judging the success of the development agencies and their individual staff members by counting the number of jobs "created" or the cost per job "created" is inappropriate in a full-employment economy. While cost-benefit analysis is an appropriate methodology for distinguishing among the best potential projects, the parameters used must be re-evaluated in light of theoretical advances and economic realities. Until very recently the development agencies set the shadow price of labour at only 15 per cent of the wage paid based on a partial equilibrium approach to the sample profile of recruits. The authorities have recognised that this assumption was extreme and have boosted it to approximately 50 per cent. A survey of shadow wages used abroad showed that they varied from 26 to 113 per cent, with a mean of 67 per cent (Honohan, 1998). The shadow price should be higher the greater is the sensitivity of migration to the market wage and the greater are congestion costs, even if some external effects, for example, technological spin-offs (for foreign firms) and the benefits of providing jobs to returning migrants (for their families and friends), might go the other way. A still higher value than the 50 per cent recently adopted, probably close to 80 per cent (Honohan, 1998), should be implemented as soon as possible.

On the other hand the tax clawed back from grant-aided activities must be included, and the shadow price of public funds may have declined from the level of 2 or more that was appropriate in the late 1980s. A rule of thumb in this field is that the excess burden of taxation is proportional to the square of the tax

rate. Since widespread tax cuts have been made over the past decade or so, a value closer to 1.5 would seem to be more suitable. This is in line with values used in New Zealand (1.2) and Norway (1.7) and close to most experts' estimates for the United States (1.6 to 1.7), although these estimates exclude avoidance and evasion activities. In addition, account must be taken of the deadweight cost of grants, that is the share of all grants that will leak into infra-marginal projects that would have gone ahead even without a grant. The elasticity of investments with respect to the grant rate is difficult to estimate precisely, but it is likely that the deadweight losses will be high.[104] Taking such losses into account is equivalent to requiring approved projects to meet a very high hurdle benefit/cost ratio. Some account of this factor has been taken very recently. Finally, the job-quality and regional dimensions should be taken more explicitly into account. The agencies have just taken on board, on an experimental basis, the share of jobs which pay salaries in excess of Ir£ 25 000 per year, and something like this should be incorporated without delay into standard assessment procedures. Furthermore, with labour market tightness and congestion varying substantially from the east coast to the north west, for example, it seems only realistic to introduce a specific regional factor into the calculations (rather than just judgementally at the final stage of decision-making), even if industrial policy cannot be an entirely effective instrument of regional policy, since people will make location decisions including agglomeration effects and firms will tend to follow. Nevertheless, less infrastructure would probably also be needed if economic development were more balanced, and the employment costs of a national minimum wage would also be lessened.

Dealing with the EU: smoothing the phase-out of transfers

As was discussed above, 1999 will be a pivotal year for the EU finances and especially for Ireland, since it is one of the largest net recipients of EU transfers both absolutely and, especially, on a per capita basis. In recent years total receipts have averaged in excess of Ir£ 2.5 billion, about 4 per cent of the total EU budget and more than 4½ per cent of EU transfers. About 7½ per cent of total gross public expenditure has been co-financed by EU Structural and Cohesion Funds. While Ireland's payments to the EU budget rose sharply in 1998, they still did not reach Ir£ 0.8 billion, leaving a net receipt of around Ir£ 1.5 billion or 2 per cent of GDP.[105] GNP-based payments are likely to rise sharply in coming years, but the bigger economic shock will be on the receipts side of the ledger. Reforms to the Common Agricultural Policy will probably make the regime significantly less generous, and, with one of the EU's larger farming sectors, Ireland will be hit disproportionately. Furthermore, the European Commission's detailed Agenda 2000 proposals, tabled in March 1998, have other significant implications for Ireland. In particular, it recommended that Ireland continue to receive regional assistance as what is called an Objective I region in transition (that is, funding

would diminish linearly through the next round of financial arrangements until Objective II levels would be reached at the end) and that it should have continued access to the Cohesion Fund.

In November 1998 the government decided (principally for structural funds and state aids purposes and to facilitate a more targeted and focused regional development policy) to make an application to Eurostat for the reclassification of the existing single region of Ireland into two new regions. One of the proposed regions would have a per capita GDP below 75 per cent of the EU average for the relevant reference period[106] and would therefore qualify for so-called "Objective 1" status in terms of eligibility for structural funds. This would also allow it to continue to qualify for the highest level of state aid to industry.[107] In February 1999, the government announced details of the new regional authorities corresponding to the proposed new regions. These authorities will have responsibility for co-ordinating public service provision, advising on the regional aspects of the next National Development Plan (covering the period 2000-2006), managing the regional programmes under the EU Community Support Framework (CSF) and monitoring the impact of all CSF programmes. Adequate staff and resources will be provided by the government. However, whether the possible additional receipts of EU transfers over a probably limited span of years[108] is worth the extra layer of bureaucracy is unclear.

Agreement on Ireland's proposal was reached with Eurostat after the proposal was modified to take account of Eurostat's opposition to the inclusion of two counties with above-average income levels from the region with the lower level of per capita GDP. This agreement was then finally approved at the Berlin European Council. Thus, one of the two new regions will enjoy full Objective 1 status, while the other region will continue to receive substantial levels of assistance as a region in transition under Objective 1. Ireland will also continue to have access to the Cohesion Fund for the next four years. Altogether, about Ir£ 3 billion will be available to Ireland from the Structural and Cohesion Funds over the next seven years.

Ireland began to receive EU structural funds at a time when its per capita GDP was well below the EU average. It is clear from Chapter I that those funds made a significant, although not decisive contribution to the impressive economic growth of the last ten years. It is estimated that the level of potential output has already been boosted permanently by 1 to 2 per cent. But Ireland's success has now brought per capita GDP to more than 100 per cent of the EU average, implying less of a case for continued structural support. Any elimination would in any case be gradual, but it seems that Ireland's above-average growth does not depend on its continuation. Its prospects have more to do with labour supply and other aspects of the economy, including the structural reforms reviewed in Chapter III, than with the continued receipt of EU funds *per se*.

III. Other structural policy reforms

The authorities recognise the importance of well-functioning product and factor markets to the improvement of the economy's potential growth rate and to the full utilisation of labour and capital inputs. Competition is seen as vital in ensuring efficient resource allocation, longer-term innovation and dynamism, all of which are key factors determining national competitiveness and standards of living. In the new era of Economic and Monetary Union, enhancing flexibility will take on greater importance in view of the disappearance of national instruments of adjustment. This chapter focuses on those areas of structural reform that have not already been discussed elsewhere in the present *Survey*.

Environment: preserving natural advantages

Ireland faces a number of environmental challenges. Three of the most important are addressed here. The first is the long-standing problem of a lack of treatment of waste water (Table 31) and the associated dumping of sludge at sea with the resulting pollution of coastal waters. While a shrinking farm sector has helped, allowing such polluting actions to persist would be manifestly unsustainable.[109] Given finite supplies of fresh water, increased treatment is inevitable, and, fortunately, a sludge treatment facility will be imminently operational, ending the era of dumping at sea. The costs of treatment must be borne. Domestic water use has not been charged for since January 1997. Even though such mechanisms would be intensely unpopular, their return would both reduce the size of the problem and fairly allocate the costs of dealing with it. If funds from sources other than the EU Cohesion Fund[110] are to be spent on avoiding environmental degradation and preserving the commercial fishing and angling industries, then it would be better if financing were through user charges than tax raising: the polluter should not be spared. The government agrees that non-domestic users should pay the full operating costs of their water usage. The 1998 EU Court of Auditors report cited Ireland for failing to treat 85 per cent of total sludge produced, for delays in drainage construction projects and for neglecting to sanction firms over illegal discharges. However, some improvements have taken place in

Table 31. **Waste water treatment**[1]

European Union	
Austria	75.5
Belgium	75.4
France	81.0
Germany	92.2
Ireland	**68.0**
Luxembourg	87.5
Netherlands	98.0
United Kingdom	97.0
Other OECD countries	
Canada	91.0
Czech Republic	73.2
Finland	77.3
Hungary	43.0
Iceland	90.0
Mexico	64.6
Norway	73.0
Switzerland	94.0

1. Per cent of population connected to the sewage network, mid-1990's.
Source: OECD (1998*b*).

recent years.[111] Projects are also expected to get underway soon to deal with the dumping situation in order to conform to the EU Directive on Urban Waste Water Treatment.

The second challenge is with respect to solid waste. According to official estimates, the nation produces 328 000 tonnes of hazardous waste a year, of which 98 000 tonnes are unreported (including 28 000 tonnes of sheep dip), and projections are that, without preventive actions, quantities would rise by 47 per cent in the next eight years. In order to head off such a scenario, a National Hazardous Waste Management Plan has been in preparation for some time now and is expected to be published later this year. But for the moment much of this waste is exported, as there are few facilities available for its proper treatment. There is no national incinerator facility, and few landfills have the capability of accepting such waste. A more widespread problem results from municipal solid waste. While households produce less solid waste per capita than elsewhere in the EU,[112] they also recycle rather little of it (Table 32). Given that there is no incineration at all, it is not surprising that the landfill situation is deteriorating quickly and that in some areas it has assumed crisis proportions. Most existing landfill sites are nearing the end of their useful lives, and the "not-in-my-backyard" syndrome has hindered resolution of the problem in many cases. The 1996 Waste Management Act requires that all local and regional authorities develop waste-management plans entailing more recycling.

Table 32. **Disposal methods for municipal solid waste in OECD countries**

Share in per cent, mid-1990s

	Recycling and composting	Incineration	Landfill
European Union			
Austria	38	14	48
Belgium	14	31	55
Denmark	23	54	22
France	9	32	59
Germany	29	17	51
Greece	7	0	93
Ireland	**8**	**0**	**92**
Italy	0	6	94
Luxembourg	28	43	28
Netherlands	38	27	35
Portugal	12	0	88
Spain	12	4	83
Sweden	19	42	39
United Kingdom	7	9	83
Other OECD countries			
Canada	19	6	75
Finland	33	2	65
Iceland	14	17	69
Japan	4	69	27
Korea	24	4	72
Mexico	1	0	99
Norway	15	16	69
Switzerland	40	46	14
United States	27	16	57

Source: OECD (1998*b*).

Last, Ireland has now committed itself to the restraint of greenhouse gas (GHG) emissions as a result of the United Nations Framework Convention on Climate Change and an internal sharing agreement at the EU level. It has to hold the increase in its GHG emissions in 2008-12 to 13 per cent above their 1990 levels, while the overall EU level must be cut by 8 per cent. Government figures show that by 1996 the net increase over the 1990 level was around 4 per cent, and if real GDP growth follows the 1997 medium-term projections of the ESRI (see Chapter II above), then emissions would rise about 28 per cent above the 1990 levels, more than double the increase implied by the committed ceiling. A national abatement strategy is being developed. Implementation of a domestic emissions trading market is being considered. The government has mandated a national car-testing scheme, which will be implemented in January 2000. Already this year the Vehicle Road Tax system has been modified to favour smaller cars, and the Rural Environmental Protection Scheme (see below) will also have the

effect of reducing methane gas emissions through a shrinkage in animal herds. The biggest question, however, is whether the target can be achieved without a carbon tax. Employers are against any unilateral imposition of such a tax because of its feared effects on competitiveness, even though proponents claim that an offsetting cut in employer social insurance contributions could eliminate negative economic repercussions in the aggregate. There are also some administrative decisions that could be made that would have favourable effects, such as to eliminate the use of peat in electricity generation, to convert coal-fired plants to natural gas, to remove tax incentives to company cars and to make public transport relatively more attractive than private car-based urban commuting. Yet it is unclear how to encourage households and other private-sector users to switch fuels without resorting to some sort of tax/subsidy scheme.

Agriculture: struggling with lower product prices

The financial situation of Ireland's 148 000 farm households has deteriorated in the past two years (even if possibly less than elsewhere in the EU) under the weight of falling product prices (in part the result of the collapse of the Russian market) and poor weather conditions. According to Central Statistics Office preliminary figures, despite a rise in net subsidies of 10.4 per cent to over Ir£ 1 billion in 1998, aggregate farm income fell another 5.0 per cent after a decline of 6.2 per cent in 1997. This implies that, even with further shrinkage in the number of farms, average income per farm probably declined for a second straight year as well.[113] The government has responded by bringing forward payments from 1999 into 1998, providing extra funds for winter animal fodder, introducing a variety of tax concessions in the 1998 Budget and implementing a new assistance scheme for 13 400 low-income farmers. In addition, the Rural Environmental Protection Scheme (REPS), which was introduced in 1994, was extended to include Special Areas of Conservation to encourage up to 50 000 farmers to use less fertiliser, to reduce livestock levels and to preserve the countryside's natural features. REPS expenditures rose 15 per cent in 1998. A full evaluation of the Scheme is expected in 1999.

But the sector faces a challenging medium-term outlook, no matter what reforms are made to the EU Common Agricultural Policy (CAP). The CAP brought Irish farmers about 56 per cent of total farm income in 1998 – around Ir£ 1.3 billion in guarantee payments and a further Ir£ 186 million in "Guidance" payments,[114] up 30 per cent since 1995. The average disposable income per household in 1994/95 was Ir£ 16 644. About 30 per cent of farm households had incomes below Ir£ 9 000 per year, and, on average, only about half of farm household income now comes from farming activities. Around 70 per cent of all farms have less than 100 acres, and some experts believe that only around 30 000 farms are

economically viable in the medium term.[115] Further, Irish agriculture has an age structure in which only 12 per cent of farmers are under the age of 35 and 46 per cent are over 55 (of which nearly half over 65). Investment may be falling sharply – from Ir£ 450 million in 1997 to an estimated 270 million in 1998 and a projected 186 million this year – despite two consecutive years of major land price increases. Two important policy challenges are how to manage the sector's declining job prospects as well as increasingly important environmental externalities in the context of probably dwindling EU transfers and social and regional constraints. The REPS, for example, is a worthwhile attempt to deal with these problems. But the very recent CAP reform under Agenda 2000 also makes it desirable for the Irish food sector to reduce its reliance on world markets by modifying its product mix and developing the needed industry structures to increase its share of EU markets in order to reduce reliance on export subsidies.

Competition: only a few mostly semi-state markets with monopoly rents

As stated in the previous *Survey* of Ireland, the nation has a generally high degree of product market competition, thanks to its open trading system and a relatively low level of government regulation, even though market forces in the utilities sector have been in need of strengthening. Corporate behaviour may even have improved in recent years thanks to the more aggressive enforcement of the code by the Competition Authority. Its powers and resources were beefed up in 1996, when maximum penalties were boosted to Ir£ 3 million or 10 per cent of turnover along with two years imprisonment. Since then it has engaged in high-profile actions to modify business practices of some major corporations and has successfully used the courts to prevent price fixing agreements in the travel, veterinary services and road haulage industries. It has also instituted proceedings against the publicans and is investigating retail milk distribution. However, the Authority still lacks the power to investigate mergers without a referral from the Minister of Enterprise, Trade and Employment, something which happens only very occasionally. The Minister has a wide discretion to block mergers but only after making a referral to the Competition Authority. When a referral is made, the result is a published report to the Minister detailing the effects of the merger on competition as well as a number of subsidiary matters (*e.g.* effects on employment, shareholders, regional development, etc.). Proposals have been made to give the Competition Authority the power to review mergers without a referral while leaving the Minister final discretion to allow or disallow a merger. Adopting such a proposal would bring Ireland more into line with practices in most other OECD countries with merger review provisions in their competition statutes.

There have been two prominent industries where public attention has focused on officially enforced barriers to entry that have severely restricted supply, thereby raising prices above competitive levels.[116] The first is the Dublin taxi

industry, which was under State licensing control until 1997, but is now controlled by the local authorities. In late 1997 a paper written by some private economists claimed that consumers were paying Ir£ 12 million per year in excessive fees (other than queuing). Since then, 400 new licences have been issued at an average price of Ir£ 15 000 each, compared to only 150 in total over the previous 20 years. Currently there are 2 374 taxis and 3 500 hackney carriages. Despite the fact that the number of such vehicles in relation to the population is much higher than in other major European cities, there is an estimated demand for around 4 300 taxis, perhaps because of the lack of adequate public transport in the capital. In any case, licences are worth up to Ir£ 80 000 each, and an estimated Ir£ 30 million in scarcity rents is being earned every year. The Department of Enterprise, Trade and Employment favours the removal of entry restrictions, with regulation limited to quality, safety and insurance considerations. The Dublin authorities commissioned a report on the industry that was published in June 1998. It recommended issuance of 350 extra taxi licences per year for 10 years (at Ir£ 15 000 each),[117] followed by full deregulation, as well as free entry into the hackney services trade.[118] The relevant committee has yet to make a decision on the matter.

The other sector where legal entry barriers apply is public houses. A 1902 law severely circumscribes the conditions in which new licences can be issued, with the result that the number of licences has fallen 26 per cent since the law was passed, existing houses are getting larger in order to accommodate increased demand, and prices are higher in those parts of the country where supply is most severely restricted, principally Dublin.[119] In September 1998 the Competition Authority issued an interim report calling for the repeal of all entry restrictions other than controls over the suitability of the applicant and the premises, in particular provisions giving existing holders protection from new entry within one mile of their premises. One consequence of regulatory barriers to entry may have been to facilitate the formation of cartels. In any case, several cartel actions involving pubs have been brought by the Competition Authority and are now pending trial. The Department of Justice, Equality and Law Reform is currently examining the licensing laws.

Some progress has been made in deregulating and introducing competition into several utilities sectors, largely on the heels of EU requirements. The primary case is in telecoms, where the government decided in May 1998 to abjure its EU derogation and to liberalise voice telephony a year early (in December 1998) in light of signs that it would otherwise lose out on important investment projects. An independent telecoms regulator was set up, and twenty-one licenses have been issued. In preparation for that market opening, the State provider cut its internet access rates by one-third and its interconnection charges by up to 40 per cent. This was apparently enough to wipe out the difference in tariffs between Ireland and its main EU partners. In the electricity sector, the EU

Directive on the internal market calls for 28 per cent of Ireland's market to be opened to competition by February 2000. In December 1998 the relevant legislation was published and enactment is expected this spring. A new Commission for Electricity Regulation will be set up for licensing purposes.[120] The largest users comprising that share of the market will be allowed to choose their supplier, but final consumers will be subject to a levy to cover sunk costs for a five-year transitorial period. What has so far not been made clear is how this opening will affect the relative prices between residential and industrial users. Ireland has one of the largest price gaps between the two groups of any EU Member country. It is estimated that households benefit from tariffs that are at least 10 per cent below costs, while businesses pay at least that much more than the costs of supplying them with power, but industry representatives have gone on record promising even householders savings of up to 20 per cent of their electricity bills. In a related matter, in January 1998 the State for the first time awarded the contract to build, own and operate a peat-fired power station to a Finnish firm that outbid a variety of others, including the domestic Electricity Supply Board.[121] Finally, in February 1999 some 4 per cent of the market for postal services was liberalised, again following an EU directive. All items weighing more than 350 grams or where the price exceeds Ir£ 1.50 are now open to competition from private-sector couriers. A much larger liberalisation (25 per cent) is scheduled for 2003.

An inter-departmental debate is currently taking place concerning the wisdom of leaving the Competition Authority as the sole enforcer of competition law in the deregulated sectors. The telecoms regulator in particular is seeking powers in this area, and it is unclear whether the yet-to-be-set-up electricity regulator will have a mandate to enforce competition law. This is an important debate, because effective enforcement could be critical to ensuring that hoped-for benefits from liberalisation are not delayed or denied by anti-competitive conduct, especially by strong incumbent firms. While there may be clear synergies in combining regulation and enforcement of competition law, there are other factors weighing against such a combination, at least within sector-specific regulators. Prominent among such factors are: sector-specific regulators appear to be in greater danger of regulatory capture than are economy-wide agencies such as the Competition Authority; and different enforcers could make for important differences in enforcement policy resulting in distorted competition (for example, between gas and electricity providers).

Public sector efficiency issues: partnership and privatisation

Competition will also benefit from several planned privatisations. The government does not have a single ideal model for all state-owned enterprises; rather, it has shown a willingness to consider various options to raise needed

additional capital, including privatisation, depending on their cost effectiveness. For example, at least 20 and up to 35 per cent of the shares in Telecom Eireann will be sold in an initial public offering this summer. Foreign partners KPN and Telia hold 20 per cent and are expected to exercise an option to buy a further 15 per cent. The State has offered 14.9 per cent to the firm's employees, 5 for changes in work practises and the rest financed by a combination of loans and as a contribution to an Employee Share Ownership Trust.[122] The government has also decided to dispose of ICC, the state-owned bank specialised in industrial lending, most likely this year. In addition, the other state-owned bank, ACC, will be merged with the private TSB Bank and then floated on the Stock Exchange. It is also likely that Irish Fertiliser Industries Limited (jointly owned with Imperial Chemical Industries of the United Kingdom) and the cable television firm Cablelink will be sold later this year. The State airports authority is reviewing its future development options, and Air Lingus is examining the possibility of joining a strategic alliance, a move that might involve selling an equity stake. But while air travel within the EU has been opened up to some extent, Ireland seems to be losing out to the United Kingdom in terms of transatlantic air travel because its market is not fully open to competition.

The government is also examining to what extent it can make use of the private sector to help deliver public services. It is exploring the range and nature of activities in which it might involve so-called Public-Private Partnership schemes. The Minister for Finance will shortly announce the projects that will be included in the initial pilot phase. These are expected to be mainly in the area of economic infrastructure (such as roads).[123] But the authorities are aware that involving the private sector is worthwhile only if the long-term value for money offered exceeds that resulting from conventional public procurement. That will be the case only if the gains from the transfer of risk are worth the higher capital cost the private partner faces.

Financial sector developments: reacting to rising supervisory concerns

Besides the recent announcement that two of the nation's largest financial institutions, Irish Life and Irish Permanent, would merge to form a new third force in the sector, and periodic public concerns over bank charges and lending rates,[124] the most important developments in this sector have been related to supervision. In the past much of the supervision has been self-imposed, but a number of well-publicised incidents has caused growing concern that the system is inadequate. In July 1998 a joint committee of the Parliament recommended that a new independent financial services authority be set up. In October 1998 the government agreed in principle to the establishment of a single financial

regulatory authority and set up a working group to work out the necessary details. Its report is expected shortly. Also last summer the government implemented the EU Investor Compensation Directive through an Act that provides clients of investment companies, insurance intermediaries and stock brokerages with the right to compensation of up to the lesser of ECU 20 000 or 90 per cent of losses in the event of default. Earlier in the year the Central Bank had been given the job of regulating investment intermediaries, and there have been some in the private sector who have called for insurance brokers, credit unions and mortgage intermediaries also to come under its supervision. Finally, in the 1999 Finance Bill the government allocated new powers to the Revenue Commissioners to inspect accounts at financial institutions in order to combat tax evasion.

Government debt management has also been the subject of some significant changes. First, the debt was restructured last year in order to eliminate exposure to any non-euro currencies other than Sterling, in which only some 6 per cent of the debt is now denominated. Second, the budget surplus and the consequential lower debt issuance has resulted in the shortening of the averaged debt maturity. Finally, a bond exchange programme is being finalised in order to comply with all of the conventions agreed for the Euro area and in order to offer bonds which would have interest coupons close to market yields. With such an exchange annual debt service costs would fall (in line with the lower interest rate on euro-denominated paper), but that would be offset by a higher nominal outstanding debt (to keep the same present value of the debt plus the stream of interest payments) and, therefore, an increased debt/GDP ratio (so that the present value of the stock would be unchanged). To maintain liquidity in individual issues in order to minimise funding costs in the new era, Ireland and other small EMU Members have an incentive to confine their issuance to a small number of maturities.

Copyright law reform: complying with international standards

Following a US complaint that Ireland's law was not in conformity with the WTO Trade Related Aspects of Intellectual Property (TRIPS) agreement,[125] the authorities passed legislation in July 1998, which shifted the burden of proof in civil actions for breach of copyright in favour of plaintiffs and boosted the penalties allowable for such criminal offences. Further legislation has just been published that will overhaul the whole regime. Besides transposing into law all outstanding EU Directives in this field and enacting all the nation's obligations under international copyright and neighbouring rights law, the Bill is intended to provide comprehensive protection for copyright and related rights, including performers' rights.

Pensions: reform now before ageing gets underway

Currently pension arrangements in Ireland combine a relatively low level flat-rate benefit (first pillar) for all individuals and occupational pension schemes (second pillar). However, less than half the working population is currently covered by these schemes. The flat rate benefit is financed by a combination of the Exchequer for means-tested pensions, and of social insurance payments for contributory pensions. Currently the official retirement age is 66, but the effective age of retirement (the transition of older workers into retirement) is much lower – 63.4 years for men and 60.1 years for women in 1995.

The ageing of the population is not expected to accelerate before another decade. However, thereafter the old-age dependency ratio is expected to rise very quickly (from 0.20 in 2006 to 0.53 in 2056). Official projections published in September 1997 show an increase in the over 65s from today's 420 000 to more than a million in less than fifty years, alongside a decline in the adult population expected to set in around 2026. In a report published in May 1998, the Pensions Board (1998) recommended a major reform to both the public and private components of the nation's old-age pension system. First, it suggested that the contributory social welfare pension, considered to be the first pillar of retirement income,[126] be raised from the June 1999 level of 28 per cent of average industrial earnings[127] to a target level of 34 per cent over the next 5 to 10 years. This was estimated to cost only 0.7 percentage points of GNP in 2003,[128] but could be higher after the population ages. It also endorsed the longer-term indexation of pensions to wages rather than prices, although it advised against any such *guaranteed* indexation, something which would cost as much as an extra 5 percentage points of GNP by 2056.[129] Second, it recommended that the State set up a reserve fund financed by a budget contribution of Ir£ 250 million per year from 1999 to 2003, 500 million for the following five years and then half of all social security contributions. Assuming the fund was able to earn a real return of 5 per cent per year, the Board estimated that this would cumulate to as much as Ir£ 30.3 billion (26 per cent of GNP) in 2031. This would allow the State to contain the increase in its contribution (currently 1.7 per cent of GNP), to a maximum of 3.8 per cent of GNP, rather than 6.8 per cent without a fund, assuming private-sector contribution rates remain unchanged. While funding is desirable, significant rises in tax and social-insurance contribution rates in the next few years may be required, unless part of this State contribution were found from privatisation proceeds.[130] There is also no assurance that the assumed real rate of return of as much as 5 per cent on such substantial investments will be achieved. While average market returns have surpassed 5 per cent in real terms in recent years, there have been several occasions in history with prolonged periods of much lower earnings, and the increase in saving could put downward pressure on rates of return, at least to the extent that investments comprise domestic assets. Finally, the management of

such a fund and the choice of assets could pose serious corporate and public governance questions. Government-managed retirement plans around the world have tended to underperform privately-managed plans in both the 1980s and the early 1990s, because they invest too conservatively and/or have used non-economic investment criteria – for example, investments aimed at bolstering the local economy (World Bank, 1994; Nofsinger, 1998).

With regard to the second pillar of the pension system, occupational pensions,[131] the Board recommended that the government introduce Personal Retirement Savings Accounts (PRSAs), similar to 401k plans in the United States.[132] These would allow all adults from 20 to 65 to contribute to an individual account of their own and benefit from tax relief. The amount contributed could be up to 10 per cent of earned income at the outset, with the ceiling rising by a half percentage point per year to reach a maximum of 30 per cent at age 60. These funds would be professionally managed, with at least four investment options and withdrawals allowed at any time from age 55 to 70. The advantages of such a scheme are several.

- First, it would allow improved coverage under the second pillar. The latest estimates (for 1995) show that only 46 per cent of all workers have occupational pension cover (83 per cent in the public sector, 38 per cent in the private sector and 27 per cent among the agricultural and other self-employed), and that this proportion is static or falling. The ultimate target set by the Board would be to boost the share of those over 30 years of age with second-pillar coverage to 70 per cent. If retaining the voluntary nature of the system does not appear to be sufficient to reach intermediate targets of 53 per cent in 2003 and 57 per cent in 2008, then the Board argued that mandatory measures should be considered.
- Second, pension portability would be significantly enhanced, especially if mandatory vesting of employer PRSA contributions was shortened from the current five years to two.[133] Implementation of such a scheme would improve labour market flexibility, as workers would be able to switch employers without fearing a loss of pension rights.

The government accepted in principle the Board's recommendations except for two key proposals: the target increase in the social welfare pension and the funding proposal. Both of these suggestions will be the subject of further study.

In addition, the government will soon be reviewing the pension system for public servants, following a final report from the Commission on Public Service Pensions expected in the near future. One of the key issues will be that of the retirement age, currently 65. Another will be advance funding in view of the projected average annual increase of 4½ per cent of such pension costs in real terms in the next 30 years. Full funding of future liabilities on a private-sector

actuarial basis would require total contributions of some 16 per cent of the *pensionable* pay. However, not all of the pay bill is pensionable, and there is no risk of the employer going out of business. On the basis of an employer contribution rate of 7 per cent of *total* pay (equivalent to a rate of 9.2 per cent of pensionable pay) in addition to the existing employee contribution rate of 5 per cent of pensionable pay and a hypothetical rate of return of 1 percentage point more than the yield on national debt, a funded scheme would reach a peak of some 25 per cent of GDP by 2040. Full funding has obvious transparency benefits. But once again the crucial nature of the assumption regarding returns and the questions concerning governance cannot be over-emphasised.

Health care: adjusting a system that works well

The last *Survey* of Ireland devoted a chapter to the health care system and concluded that it works well. It made some specific recommendations, and the authorities have acted on several of them, although the idea that drugs costs should be included in capitation payments for those eligible for state-provided care was not adopted. In response to the suggestion that budgetary authority within hospitals be devolved to a greater extent, a pilot project was set up in 1998 to try out different models such as management teams ("clinical directorates"). To reduce incentives for patients to make excessive use of public hospital casualty departments rather than private general practitioners, whose charges were lower, the fee for Accident and Emergency departments was raised from Ir£ 12 to 20 at the beginning of 1998. Finally, following on the *Survey*'s concern regarding the unusual length of medical manpower training, a Forum on Medical Manpower has been established in order to seek a solution to this and other career problems.

Hospital waiting lists were another problem identified in the previous *Survey*. Unfortunately, despite spending about Ir£ 60 million over the past five years on more than 100 000 additional procedures, little headway has been made in reducing the numbers: some 34 000 people are still waiting for non-emergency treatment. While recognising that there will always be excess demand in a system that does not charge, the government decided to raise the annual allocation of funds devoted to their shortening to Ir£ 20 million and to implement the measures proposed by an Expert Review Group, such as the creation of "step-down facilities" to free up acute-care beds. Another dilemma is over equity considerations, as those who can afford it may seek private hospital or specialist treatment. Private insurance coverage has risen from the 37 per cent figure cited in the *Survey* to nearly 42 per cent most recently. In order to ensure that those aged 70 or over receive adequate primary care, the government announced in the 1999 Budget that eligibility for free care for that age group will be significantly extended over the next three years through a loosening of the means-testing criterion: over the next three years the income limit to qualify for a medical card is to double.

Human capital development: continuing to upgrade

Ireland's education and training system is still maturing and is continuing to be upgraded and modernised, as it has for the past generation (OECD, 1995). For example, a new Education Bill was passed in December 1998. For the first time education has been placed on a statutory footing, rather than governed by regulation. It also established a National Council for Curriculum Assessment and explicitly prohibited "league tables". But the earlier idea of Regional Education Boards was abandoned. Also, in January 1999 the State announced that henceforth, subject to certain conditions, it would assume significantly greater financial responsibility for school sites and buildings, thus providing greater relief for parents and local communities in respect of this burden.

Nevertheless, there remain outstanding weaknesses at every level. Preschool education remains underdeveloped, with only 19 per cent of all children aged 2 to 4 enrolled in 1996, compared to an OECD average of 41 per cent. Calls have therefore been made for equal access to all those aged 3 and over either through free provision or tax concessions. The government is in the process of preparing a White Paper on early childhood education following a report by a group of interested parties that it convened last year. At the primary level the big issue is related to demographics: the authorities did not anticipate the rebound in births and the rising number of immigrants under 14 in recent years, with the result that class sizes will soon be bigger than intended. The problem is compounded by a reduction in the number of graduating teachers and increased retirements. The government has responded by boosting the number of teacher training places to prepare for the expected rebound in primary enrolments as from 2004, even though there is no guarantee that they will be taken up. It is also reviewing the primary curriculum with a view to introducing science. The school leaving age is probably the key issue at the secondary level. The government has promised to raise it from the relatively low level of 15 to 16 in the near future, but in December 1998 the main employers' group called for it to be lifted to 18, especially if wage levels attract teens into the labour market once minimum wage legislation is implemented (see below).[134] The "demographic dividend" at this level (the opportunity for reduction in costs associated with falling student numbers) will persist for some time and will continue to be used to increase quality. For example, in December 1998 a two-year Ir£ 57 million package of measures to deal with special needs education was announced.

Another issue at the upper-secondary level may be the emphasis on general academic programmes. In 1996, only 20 per cent of all upper secondary education was provided in vocational and technical programmes, the second lowest in the OECD and 33 percentage points behind the country mean (OECD, 1998*b*). This seems to be in the process of changing: Vocational and Applied Programmes at the Leaving Certificate level are in the process of gearing up substantially, with the number of pupils enrolled rising from 3 053 in 1997 to 9 460

in 1998 and an expected 50 000 in 2000 (National Competitiveness Council, 1998). Also underway is a transformation of the apprenticeship system, which in 1993 began a shift from one based on time served to a standards-based programme with better-qualified entrants.[135] The old system will disappear by the year 2000. There are now more than 18 000 youth apprentices and pre-apprentice trainees, the highest figure since 1985 and 2 times the 1996 figure. Another question at the secondary level is whether the time spent on foreign languages and mathematics and science is adequate. There is a need for greater linguistic proficiency in light of the increasing role of foreign trade described above and the expanding telemarketing sector in particular. Foreign languages are now being introduced into some primary schools, but this needs to be generalised as quickly as possible. Similarly, a better grounding in mathematics and natural science would facilitate meeting the urgent skills requirements in engineering and information technology (see below).

The number of students likely to enrol at the tertiary level is also still the subject of some debate. One factor that may have boosted numbers in recent years is the abolition of fees, especially as the United Kingdom has since instituted tuition charges. The result has been a marked fall in the number of students applying to study in the United Kingdom, something of the order of 3 000 per year. The modest level of the maintenance grant in Ireland is also a continuing subject of controversy. Consideration should be given to restoring fees and instituting an adequate programme of income-contingent student loans, especially since a sizeable share of the returns to education at the tertiary level are private (OECD, 1997*b*). In April 1998 the government announced that it would fund 14 000 more post-secondary places in the next two years, mainly at the Institutes of Technology. In June of that year the Expert Group on Future Skills Needs pointed to a significant gap between supplies of and demand for professionals and technicians in the engineering and computer software fields. It suggested that the deficiency be met by the creation of more than 4 000 new degree and technicians' full-time places, development of one-year conversion courses and enhancement of part-time channels of study. The government has just agreed to implement these recommendations over the next few years. Another possible worry is that the strength of the labour market has meant a steady reduction in the share of first degree-holders going on to further academic study or training.[136] More encouraging, however, was the 1997 announcement of an Ir£ 250 million Education Technology Investment Fund.[137] Following up on that, in November 1998 the State announced a new three-year programme to boost science and research at the tertiary level,[138] with fixed transparent procedures to ensure that funds are allocated in an efficient fashion.

Finally, human capital development is restricted by the lack of adult education and training. Job-related training was undertaken by fewer than one in four 25 to 64 year-olds in employment in 1994-95 and fewer than one in 14 of the

unemployed of that age. In the leading countries those fractions are closer to one in two and one in four, respectively. The lack of suitable course offerings, especially in part-time options, and above all the lack of on-the-job training[139] would seem to be the key barriers in this area. The importance of this deficiency was highlighted by an international literacy study undertaken in 1995 (Morgan *et al.*, 1997) that showed Ireland in a rather poor light.[140] Accordingly, the government issued a Green Paper on Adult Education in November 1998. This paper called for the development of a National Adult Literacy Programme, the phased development of a Back to Education Initiative (for those who have not finished secondary school), improved support services such as guidance and childcare and the implementation of a national qualifications framework (in part because accreditation of non-formal learning is weak). Adult literacy programmes currently have around 5 000 participants per year, but the Green Paper calls for tripling that allocation. A six-month consultation period is now underway, with legislation to follow.

Labour market policies: steps forward and backward

With the unemployment rate declining so quickly, it is obvious that the nature of many labour market problems has changed over time. Now that it is obvious that there is no scarcity of jobs, the challenge is to deal with the scarcity of workers, in large part by improving work incentives, since that is probably the single factor that has most restrained labour supply.[141] The government has recognised the importance of the interaction between taxes and benefits (by, for example, making unemployment and disability benefit liable to income taxation) and has responded to some of the suggestions made in the chapter on "Implementing the OECD Jobs Strategy" in the previous *Survey* (Table 33). Marginal tax rates facing low-paid workers have been lowered by removing all those earning less than Ir£ 100 per week from the tax net (some 80 000 in all) and by cutting those on the 40 per cent "marginal relief" rate from 11.6 per cent of all taxpayers in 1997/98 to 2.2 per cent (mostly aged 65 and over) in 1999/2000. Furthermore, the benefit system has been adjusted in several ways, as will be described below. The overall result has been to reduce net replacement ratios further since 1995[142] (Table 34), even if they remain disturbingly high for married people with children.

Even if joblessness has dwindled, there remains a sizeable stock of unemployed who could conceivably fill existing labour-market vacancies but possibly only with additional training or other employability measures.[143] The main policy change that has been implemented since the last *Survey* is the adoption of the Employment Action Plan in April 1998 (Department of Enterprise, Trade and Employment, 1998).[144] The most significant reform in the package resulted from EU guidelines that called for the introduction of a "preventative" approach to the

Table 33. **Implementing the OECD Jobs Study – an overview of progress**

Jobs Study proposal	Action taken	Assessment/recommendations
I. The tax system		
– Reduce the high marginal tax rate on low-paid workers.	All those earning less than Ir£ 100 per week no longer pay tax. "Marginal relief" rate applies to fewer taxpayers.	A promising start.
II. The benefit and wage formation systems		
– Change the rules of the unemployment benefit system to strengthen work incentives.	Conditionality has been further strengthened in 1998.	Maintaining existing level of case reviews. Reduce replacement rates further by indexing benefits to prices. Either shorten benefit duration or cut them through the spell.
– Reform the child benefits system, which favours the unemployed.	Child Benefits have been raised. Child Dependent Allowances have been frozen. Family Income Supplement reformed (see text).	No effect on work incentives. A step forward. Previous poverty trap now likely of minor importance.
– Improve the system of housing-related benefits.	Under review.	Allow benefits to continue for a time after beginning work. Weaken link between income and rents in local authority housing.
– None.	Planned introduction of national statutory minimum wage.	Level is too high. Either exempt all those under 18 or raise school leaving age.
III. Active labour market policies		
– Rationalise the schemes currently in operation.	No action.	Reducing complexity should yield cost savings.
– Integrate the activities of the government agencies assisting the unemployed.	Training and Employment Agency's (Fas') mission has been focused.	Still not quite "one-stop shops". All unemployed should have to register with Fas, and more training should be offered to them.
– Focus on targeted employment subsidy programmes.	Community Employment scheme is being shrunk.	Unclear if alternative "social economy" use will be better.
– Emphasise specific skills training programmes.	See above.	See above.
IV. Increase labour force skills and competencies		
– Intensify efforts to cut the share of students who leave school without qualifications.	Goal adopted in Employment Action Plan.	School leaving age not raised. Age 16 is probably not high enough.
– Increase further the resources devoted to schools in disadvantaged areas.	Two-year scheme to spend Ir£ 57 million.	Monitor the results of this well-endowed programme.
V. Enhance the diffusion of technology		
– Encourage the diffusion of technology to domestic firms.	Consolidated several agencies to form Enterprise Ireland.	A logical reorganisation, so long as different corporate cultures can be merged.
VI. Enhance product market competition		
– Reduce the role of state-owned firms.	Telecoms fully deregulated. Partial liberalisation of electricity and postal services. Some privatisations planned.	All desirable moves, but many undertaken only with EU compulsion. State does not need to have monopoly in these areas.

Source: OECD.

Table 34. **Replacement ratios, 1995-99**[1]

Per cent

In Ir£	Single person		Married with 2 children	
	1995	1999	1995	1999
8 000	59	56	79	78
9 000	53	51	78	75
10 000	49	46	81	73
11 000	45	44	83	71
12 000	42	41	79	70
13 000	39	38	75	68
14 000	38	35	70	66
15 000	36	34	66	64

1. These provide, in general terms, an indication of the relationship between net of tax incomes at different income levels and out-of-work benefits. Median annual earnings for all full-time employees was around Ir£ 13 000 in 1995 and will be possibly around Ir£ 15 000 in 1999.

Source: Department of Finance (1998).

treatment of long-term unemployment. As from September 1998 all those under 25 who have been unemployed for six months have been called into an interview with the public employment and training service (FAS) in order to assess whether they are apt for an existing vacancy or in need of training. Very preliminary indications are that a surprisingly high share of these cases can be dealt with in this fashion: nearly half either failed to attend the interview or refused the intervention, and 28 per cent were struck off the rolls either because of a voluntary departure or because they had accepted a job or training place. This year the authorities are moving on to focus on the stock of under 25s who have been out of work for at least 18 months and those 25 to 34 year-olds who have been unemployed for over a year. Thus, while unemployment benefits have not been made digressive over time, as was suggested in the *Jobs Strategy*, conditionality has been further strengthened, after initial moves in this direction in October 1996.[145] Previously the availability and search criteria were applied in a normative fashion, but as from mid-1998 specific job-search actions and a commitment to availability have been imposed. These various developments have led to an increase in the number of claimants removed from the rolls either because they were not available for work (4 700 in 1997, over 6 000 in 1998) or not genuinely seeking work (900 in 1997, over 1 800 in 1998).[146]

The *Jobs Strategy* was also concerned with other aspects of the benefit system which dull work incentives, thereby resulting in unemployment or poverty traps. The child support system was specifically cited. Since then, budget changes have frozen Child Dependent Allowances but have raised Child Benefit. The latter, available to workers and non-workers alike, should have no impact, but the

former should improve incentives, since this transfer payment is only available to the unemployed. Also cited by the *Jobs Strategy* was the system of housing-related benefits.[147] The withdrawal rate of rent supplement for those beginning work is under review. But entitlement to the Family Income Supplement has been changed from a gross to a net income basis, and the income threshold has been raised in order to strengthen the employment incentive of this in-work benefit.

In the area of active labour market policies (ALMPs) there has certainly been no rationalisation of the plethora of schemes (Table 35), nor has the payment of benefits and the other aspects of programmes for the unemployed been integrated, both policy actions recommended by the *Jobs Strategy*. There have been few major changes in fact. In 1998 a new programme called Job Assist was introduced. It is a special tax allowance for people unemployed over a year. The employee receives a tax break of Ir£ 3 000 plus 1 000 for each dependent child; this is then phased out over a three-year period. The employer also gets to deduct double the actual wages from its corporate income tax for the same

Table 35. **Expenditure on active labour market programmes**

Ir£ million

	1998	1999
Public employment service	**16**	**18**
Initial education and training	**306**	**359**
Apprenticeship	59	75
Other (mainly for the disadvantaged)	97	120
Mid-level technician	138	160
Advanced technical skills	12	4
Continuing training for the unemployed	**62**	**66**
Employment subsidies	**95**	**109**
Social exclusion	**398**	**434**
Community employment	316	329
Vocational training opportunities scheme	26	26
Training for people with disabilities	41	43
Other (including community training)	15	36
Adaptation to industrial change	**22**	**26**
Training support scheme (FAS)	9	11
Other	13	14
Other	**11**	**18**
Certification	3	3
Promotion of equal opportunities	1	2
Capital	7	14
Total	**900**	**1 030**

Source: Department of Finance.

three-year period. There is thus a strong incentive for the long-term unemployed to take up employment and for employers to hire such people as opposed to others in the labour market. Overall employment support and incentive schemes cost around Ir£ 800 million in 1998 (1½ per cent of GDP) (Table 35). These programmes involved something like 150 000 participants. But although ALMPs are well endowed financially, one disquieting feature is how few of the unemployed receive training: according to OECD (1998*a*), only 9 per cent of all jobless over 25 receive any education or training, compared with a simple average of nearly 30 per cent in 10 other OECD countries. This is especially critical for the long-term unemployed, since they are marked by an especially low level of educational attainment (Department of Enterprise and Employment, 1996). It is unclear why all the unemployed are not required to register with the public employment and training service, a change which would admittedly necessitate improved staffing for that body.

By far the largest ALMP is the Community Employment (CE) programme, which focuses on re-integrating the long-term unemployed into the labour market and had 43 000 participants in 1998. Participants are paid an average of Ir£ 7 500 per year to work in the public and community/voluntary sectors. There have been two recent evaluations of the CE scheme. A firm of private consultants (Deloitte and Touche, 1998) concluded that, although CE was largely meeting its assigned objectives, it was the least effective of all FAS programmes. Thus, it recommended that, because of improved labour market conditions, CE should be shrunk by 6 000 to 8 000 places and that the minimum age be raised from 21 to 25, since it is both too generous and too part-time for younger workers. Furthermore, participants should be obliged to get more training with certification outside of working hours and should not stay on the programme indefinitely, as some have done in the past.[148] The European Social Fund Evaluation Unit raised a concern that it was the programme that has the worst record in terms of progression to employment[149] and training certification and yet it has the highest pay, secondary benefits and flexibility. It clearly suffers from having the dual goals of improving the employment outcomes of its participants and of supporting the not-for-profit sector; normal public services should probably not be provided by such labour-market schemes. For 1999 there was a reduction of 2 500 places on the part-time element of CE, with an increase of 875 places on the full-time option. The savings have been used to fund a range of measures designed to increase participation of the long-term unemployed on training and to fund additional training places for them. There will also be a gradual reduction of 5 000 places on the programme with the resources being used to fund a new "social economy" programme. However, it is unclear to what extent the outcomes will truly differ, as this programme is described as providing services to disadvantaged communities and jobs for the long-term unemployed. It is to be hoped that at least the training component will be higher after the shift.

The *Jobs Strategy* also made recommendations in the labour force skills, technology diffusion and product market competition areas. A variety of actions on skills have been undertaken (see above), but the number of children leaving school without qualifications remains significant and the school leaving age has not been raised. As mentioned above, schools in disadvantaged areas are to get extra funding in 1999-2000. The role of state-owned firms in product markets is set to decline in the coming year (see above).

There are two other important labour market issues that are still outstanding. One is the sensitive question of labour union recognition. This came to a head following a bitter dispute involving an airline in 1998. The subject is on the agenda for discussions among the social partners in the run-up to the expiry of Partnership 2000. A breakthrough agreement was reached this spring. It is believed to retain the core principle of voluntarism. Procedures to be followed where a negotiating forum does not exist have been agreed. The nation's Labour Court will be charged with making an initially non-binding recommendation, followed, if necessary, by one that will be binding for one year and ultimately by binding arbitration.

But more important have been the developments in the minimum wage domain. The current government committed itself in its Action Programme for the Millenium to the introduction of a national minimum hourly wage in order to address the issues of exclusion, marginalisation and poverty.[150] In July 1997 it appointed a National Minimum Wage Commission with a view to advising on how best to implement the commitment; it reported in April 1998 (National Minimum Wage Commission, 1998). The Commission recommended a minimum wage of two-thirds of median wages of full-time workers be introduced by April 2000. At 1997 rates, that would amount to Ir£ 4.40 per hour. It would be applicable to all those over 18, with no sectoral or regional exceptions. Those under 18 would be subject to a minimum of 70 per cent of the adult rate. Trainees, apprentices and job entrants would be subject to 75, 80 and 90 per cent of the regular rate in the first, second and third years. Some 23 per cent of all employees were earning less than the proposed minimum wage in 1997[151] (Table 36), with the total cost estimated at just under 4 per cent of the national wage bill, Ir£ 813 million in 1997 (1 per cent of GDP). The incidence of the proposed minimum would be especially significant[152] on part-time workers, young people,[153] women, those without tertiary education, employees of small firms[154] and workers in the primary sectors, retail and personal services. It will also be much more significant elsewhere than in the Dublin region, where starting wages in the retail sector are already over Ir£ 4 per hour.[155]

The government welcomed the Commission's Report and is currently drafting the required legislation. At the same time an Inter-Departmental Group charged with examining the impact of its implementation has commissioned a

Table 36. **The probable incidence of the proposed national minimum wage**

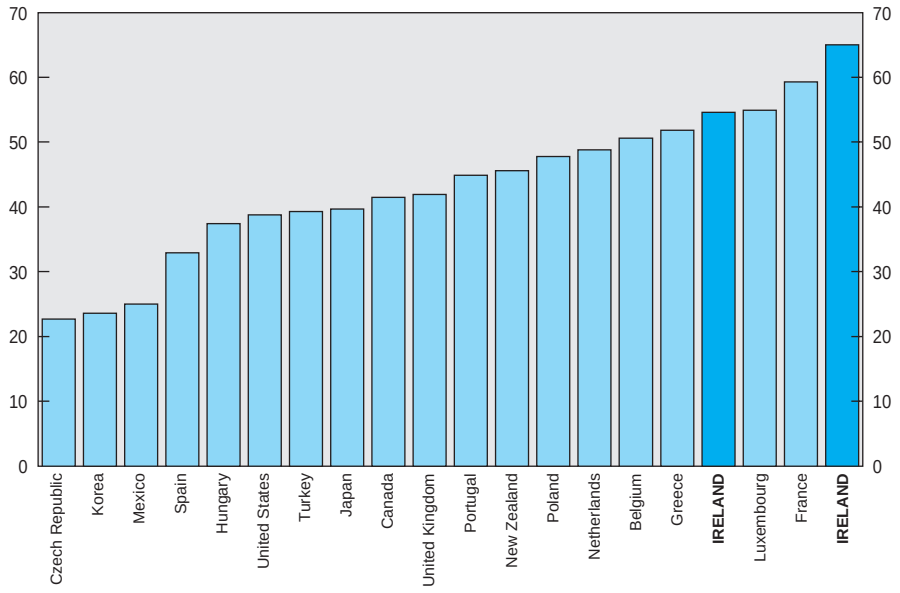
	Per cent below proposed level	Per cent of all those below proposed level
I. Full-time versus part-time employees		
More than 30 hours per week	20.5	75.2
Less than 30 hours per week	36.4	24.8
Less than 18 hours per week	49.0	14.8
All	23.0	100.0
II. Age		
Under 18	100.0 [1]	3.1
18 to 20	76.2	19.6
21 to 24	46.1	27.0
25 to 34	15.0	19.2
35 to 44	13.6	14.9
45 to 54	14.7	11.2
55 and over	14.7	4.8
III. Gender		
Male	17.8	45.9
Female	30.5	54.1
IV. Education		
Primary Certificate	26.5	11.0
Group Certificate	28.6	18.1
Inter/Junior Certificate	31.6	22.3
Leaving Certificate	24.2	34.7
Diploma	22.0	10.8
Degree	4.6	3.1
V. Occupation		
Agricultural workers	63.3	6.4
Producers, makers and repairers	21.1	19.6
Labourers and unskilled workers	35.1	6.5
Transport and communications workers	20.7	6.3
Clerical workers	11.6	7.9
Commerce, insurance and finance workers	51.0	21.9
Service workers	44.7	25.6
Professional and technical workers	5.6	4.7
Others	4.0	1.3
VI. Industry		
Agriculture, forestry and fishing	63.4	7.2
Building and construction	24.8	5.8
Other production industries	19.0	20.2
Wholesale	23.0	5.1
Retail	56.1	17.6
Insurance, finance and business services	9.0	2.2
Transport, communication and storage	11.2	3.9
Professional services	38.2	2.0
Teaching	3.0	1.0
Health	13.0	5.3
Public administration and defence	4.4	1.8
Personal services	67.7	21.8
Other	21.9	6.2

1. The National Minimum Wage Commission proposal is for a special rate for those under 18 of 70 per cent of the full rate. The source does not provide estimates for this variant, but if it had been 75 per cent the resulting figure would be 90.2.

Source: Nolan (1998).

further study on the dynamic effects on employment and competitiveness at the macroeconomic and sectoral levels. Ireland's main trading partner, the United Kingdom, has adopted a lower level for its minimum wage, the equivalent of around Ir£ 4.30 at current rates of exchange. Indeed, the Irish minimum could be the highest in relation to median earnings of any statutory minimum in the OECD, should it be uprated to keep it at the two-thirds figure recommended by the Commission, as is the claim of the union umbrella organisation (Figure 19). Even if the absolute figure of Ir£ 4.40 is retained (rather than the figure which would correspond to two-thirds of estimated median earnings at the point of implementation), it would still be higher than all others except those in France and

Figure 19. **Minimum wage ratios, 1997**[1]
Per cent of median earnings of full-time workers



1. Except the United Kingdom,which is estimated for April 1999 (the implementation date), assuming median earnings have risen 5 per cent over the past year, and Ireland, which is shown at the recommended I£4.40 rate, assuming average annual median earnings increases of 6 per cent from 1997 to 2000, and at the alternative two-thirds level. Both Irish figures have been adjusted, since the raw data are in relation to all employees rather than full-time workers (the difference is some 2½ per cent).

Source: OECD (1998*c*, Table 2.3) and OECD estimates for Ireland and the United Kingdom.

possibly Luxembourg. The Commission did recognise the risks involved in adjusting the level automatically in the future and, therefore, specifically recommended that the National Competitiveness Council should make submissions to future reviews of the level of the minimum wage. The Commission was also concerned by the possible knock -on effects on the wages of other workers earning above the proposed statutory rate. It, therefore, endorsed a union proposal that there be a centralised labour-management agreement "that any claims on the foot of the introduction of the minimum wage would not be recognised or entertained by the Labour Court". It is, however, doubtful if any such agreement could prevent relativities from re-establishing themselves in a free labour market over any term other than the shortest.

The main argument cited by proponents of the minimum wage in Ireland, including the Commission, is that it will reduce poverty. However, even the Commission admitted that a minimum wage is a weak anti-poverty device because most of the poor are unemployed. Indeed, only 1.7 per cent of all those over 16 years of age and living in poor households are low-paid employees. Nevertheless, the Commission argued that this strategy could be effective, even if not efficient, for aiding the working poor, whose low pay, it claimed, is not being addressed by existing systems. It recognised that it is unemployment that is the principal cause of poverty[156] and social exclusion, and it wished therefore not to set the minimum wage so high as to boost unemployment. But it seems doubtful that employers who will be faced with an average increase for such employees of at least 35 per cent (assuming that the minimum implemented is not uprated) will be able to retain all such employees or, in many cases, remain in business.[157] Furthermore, many may be tempted to substitute capital for such labour. In addition, there are reasons to believe that imposing a minimum wage is a particularly inefficient strategy against low pay in Ireland's case:

> "Ireland stands out as having a particularly weak connection between low pay and low household income, with more than 90 per cent of low-wage workers living in households with moderate-to-high incomes. This result stems in large part from the fact that the incidence rate for low-paid employment for all individuals in moderate-to-high income families is more than four times the comparable rate for low-income families. The data suggest that the labour force participation rate in Ireland's moderate-to-high income families relative to that in other families must be high in comparison to other countries" (OECD, 1998*c*, p. 52).

In fact, only 3.3 per cent of all low-paid workers in 1993 lived in households with adjusted income below conventional poverty levels (half of median adjusted income), compared to a simple mean of 11.1 per cent in 12 other OECD countries. Clearly, imposing wage increases on over 20 per cent of the work force in order to combat poverty that afflicts only 3 per cent of them is extraordinarily inefficient.[158]

Another argument cited by supporters is that a minimum wage will exert a favourable substitution effect on the supply of labour and at the same time reduce welfare dependency. But much of the supply effect may result from teenagers quitting school prematurely, and inadequate work incentives for those on welfare are surely best dealt with by appropriately modifying the tax/benefit system. A further claim in favour of a minimum wage is that it will impel employers to take a high value-added strategy instead of trying to compete on costs and prices. But there are already plenty of pressures on low-wage sectors of the economy, especially indigenous firms. As was seen in Chapter I, their pay levels are in general much lower than foreign-owned firms. While it may sound reasonable to encourage employers to upgrade their employees' skills in order to justify the higher wage, such incentives might be better undertaken through the tax system. Indeed, the entire policy concern – poverty, exclusion and marginalisation – would probably best be undertaken by a combination of tax and spending policies whose costs could be closely monitored in budget documents, rather than imposed on the private sector and hidden from view.

Income distribution: falling absolute poverty but possibly increasing dispersion

In earlier times the problem of poverty in Ireland was always strongly associated with that of unemployment, its long-term component in particular. Despite the large fall in unemployment in recent years, around 200 000 people are still registered as unemployed, equivalent to nearly one-eighth of the labour force. Putting these people back in full-time, year-round employment would certainly go a long way to reducing poverty. With such a large decline in unemployment, the residual poverty problem is now attributable to a greater extent to other factors. The income distribution may have become slightly more dispersed in recent years,[159] probably because the increases in real transfers have been less than in other incomes, and the number of multiple-earner households has risen sharply in the current boom, leading to a polarisation between those households and others that lack any earners at all.[160] Some 88 per cent of all poor households include no earners (Nolan, 1998). But the severity of poverty, as proxied by the degree to which those in poverty fell below the threshold, has declined (Callan and Nolan, 1999). Furthermore, absolute poverty is clearly less severe than it had been prior to the boom: in 1994 only 8 per cent of the population had real incomes below the 1987 poverty line (half that year's median disposable income), compared with 20 per cent in 1987 itself (Callan *et al.*, 1996).

The government launched its National Anti-Poverty Strategy in April 1997. Over a ten-year horizon the plan aims to cut persistent poverty by at least one-third and to bring the overall and long-term unemployment rates down to 6 and

3.5 per cent, respectively. In keeping with that, all significant proposals coming before the government have to be evaluated for their impact on poverty, and a Cabinet Committee on Social Inclusion has been set up. In the last two years social welfare rates have been raised in real terms and, in some cases, relative to average earnings as well. Much of the remaining problem is concentrated on housing estates on the fringes of urban areas.[161] Here the issue is complex and heavily interwoven with other social problems. Several of the major foreign multinationals have instituted voluntary training programmes for the residents of such estates that look to have had some success.

Notes

1. Annex Table A1 shows that even the most reputable series of medium-term forecasts – those undertaken at the Economic and Social Research Institute – have been fairly consistently overly cautious during the past decade. Real GNP has averaged about a percentage point more than they have been predicting. But for the years beginning in 1994 that error has averaged more than 2½ percentage points. Its April 1997 *Medium-Term Review* predicted a fall in the unemployment rate to 9.1 per cent in 1998 and ultimately to 7 per cent in 2005, whereas the rate reached 7 per cent already in January 1999.
2. OECD *National Accounts* figures show an increase of some 15 percentage points in the operating surplus as a share of value added in industry plus transportation and communication from 1986 to 1995. Over the same period the rise in its share of GDP was around 7 percentage points (to 34 per cent).
3. The difference between GDP and GNP comprises all earnings of foreign-owned factors of production less returns to indigenous factors of production earned abroad. Ireland has traditionally had a small surplus on employee remuneration along with a sizeable deficit on investment income. Part of the investment income flows are of a portfolio nature, and this component has moved progressively in Ireland's favour in recent years as a result of the persistent current surpluses; in 1998 it withdrew more than 1 per cent from GDP. But this has been easily swamped by developments on so-called direct investment income account – dividends, distributions of branch profits, inter-affiliate interest payments and re-invested earnings. Up until 1995, only those profits that were repatriated were counted.
4. This adjustment, which yields gross national disposable income (GNDI), amounted to over 5 per cent of GDP in the early 1990s, but it has been of steadily declining importance in recent years, because of the stabilisation in transfers from the European Union in nominal terms (see below), and it is now less than 2 per cent. However, when the effects of inflation are removed (to assess changes in "real" terms), then an additional adjustment must be made to GNDI to account for changes in the terms of trade since the base year (1990 in Ireland's case). This has been negative (because of the trend deterioration in the terms of trade resulting from the product composition of Ireland's main manufactured exports) and has grown to over 4 per cent of real GDP in recent years.
5. It is not known what share of convergence is attributable to within-sector catch-up and what to inter-sectoral shifts, but there has clearly been both. Catch-up within agriculture, construction and manufacturing has been important (Barry, 1999), even if the last is possibly exaggerated by transfer pricing – which cannot account for more than about a quarter of its value added – but gaps remain for utilities, transportation and communications (Birnie, 1994).

6. One widely cited set of statistics pertaining to the year 1995 (from the European Observatory for SMEs, *Fourth Annual Report 1996*, Table 11.1, quoted in National Competitiveness Council (1998) shows that for firms with less than 10 employees Ireland's labour productivity was only 65 per cent of that of the United Kingdom; it rises to 87 per cent for those with 10 to 49 employees and 117 for the 50 to 249 employee range. This data set shows that in 1995 Irish gross output per employee in 12 major manufacturing sectors varied from 38 to 152 per cent of corresponding UK levels, with an unweighted average shortfall relative to the United Kingdom of 16 per cent. Only the food, beverages and tobacco, non-metallic minerals and miscellaneous manufacturing industries had higher labour productivity than their UK counterparts. Firm size does not seem to be the major factor: the simple mean across the 12 sectors of employment per enterprise in Ireland was 103 per cent of the UK figure.
7. Participation rates are here defined over the working-age population (15-64), whereas they are normally defined in relation to the entire population 15 and over.
8. Fitz Gerald (1999) speculates that this pickup resulted from European integration and that firms needed to become convinced that the changes were irreversible before undertaking the capital formation.
9. The total fertility rate was around 1.9 in 1995 and had been expected to decline further to 1.6 in 2006.
10. In fact, by 1998 there was even a migration surplus of Irish nationals with respect to the United Kingdom (that is, there were more returning Irish than emigrating Irish with respect to that nation). Around half of all immigrants (53 per cent in 1998) are of Irish nationality. Irish nationals (that is returning emigrants) represent more than half of all immigrants in each origin-country grouping except the non-UK European Union (no further disaggregation is available).
11. The country is fortunate today in that a relatively large outflow of fairly high-skilled people took place in the 1980s (a "brain drain"). By the end of the decade over one-third of all emigrants had third-level education, up from less than one-fifth in the 1960s. Some 25 per cent of male emigrants with third-level education have been shown to return eventually, compared to only about 10 per cent for less-educated male emigrants (Fitz Gerald, 1997; Fahey *et al.*, 1998), and both returning migrants and non-Irish immigrants are better educated than the resident non-migrant population (Barrett and Trace, 1998). Return migrants are also more heavily concentrated in higher-skilled occupations than non-migrants. Looking at the 15-29 age group over time there has been a steady drop in the share of the least educated who emigrate and a steady rise among those with upper secondary or tertiary education (Fitz Gerald and Kearney, 1998). Ireland is attractive to high-income earners, according to the Borjas relative earnings inequality theory of immigration. That theory holds that such earners have an incentive to seek out destinations where earnings dispersion is greatest because in such places the cost of labour-intensive services is relatively low. Ireland's dispersion, as measured by the income share of the top decile divided by that of the bottom decile, is high and got higher from 1987 to 1994 (Barrett and Trace, 1998), as is discussed in Chapter III.
12. The initial exposition and testing of this hypothesis is in Honohan (1992). He found that in equilibrium the Irish unemployment rate would be around 4 percentage points above that in the United Kingdom. More recently Walsh (1999) has estimated a model which gives a varying, albeit smaller, equilibrium gap between the two rates. The latest estimate for the actual gap is less than 1 percentage point.

13. However, with the much higher costs of housing, especially in the Dublin area, the actual costs of migrating to Ireland may in fact have risen.

14. The generosity of the social welfare system – as proxied by the replacement rate (see Chapter III) – jumped from 1979 to 1984 and again from 1987 to 1991. Since then it has fallen gradually by a cumulative 10 per cent or so, but it remains fairly high in real terms by historical standards, especially for single people.

15. As elsewhere, participation increases with educational attainment. Over the past decade rates have risen especially strongly for those with Leaving Certificates (upper secondary education). Overall female rates have been rising by an average of about 1 percentage point per year, while male rates have been essentially unchanged.

16. Replacement rates are higher for married women than for single women and for those with children than those without (Callan and Nolan, 1996). OECD figures show that Ireland's gross and net unemployment replacement rates are the second lowest in the EU after the United Kingdom and amongst the lowest in the OECD (OECD, 1999*a*). The importance of the replacement rate for welfare transfers in labour force participation decisions in Ireland has been demonstrated by Newell and Symons (1990).

17. The difference in the number of years of employment expected for someone with tertiary education and someone with less than upper secondary education in Ireland in 1996 was 7.4 years for men and 15.5 years for women. The male figure is in line with outcomes in other OECD countries, but the female differential is much higher than the OECD mean of 11.0 years. Thus, growth should benefit more in Ireland than elsewhere from this trend increase in female schooling duration.

18. Already in 1995 the National Economic and Social Forum (NESF, 1995) had flagged the need for affordable and accessible childcare in order to boost women's labour force participation. A very recent newspaper survey found that there is a chronic shortage of childcare places in the Dublin area and spiralling costs throughout the country, due to wage increases for qualified staff and stricter health board regulations. Certain steps were taken in the 1999 Budget to encourage the supply of childcare places. At that time the government also pointed out some of the complex issues raised by possible childcare tax relief, such as equitable treatment of parents who stay at home to care for their children and those who have to meet additional childcare expenses when they go out to work, and between taxpayers and those relying on social welfare benefits. At the same time it recognised the need to find the right balance between measures to lower childcare costs and those to enhance supply. In January 1999 came a report from the Expert Working Group on Childcare set up under the aegis of the Partnership 2000 process. It called for a seven-year strategy including personal tax credits, subsidies to non-taxpayers, tax support to providers and employers investing in childcare and a national framework of qualifications for providers. The government has now established an inter-departmental committee to evaluate, cost and prioritise the recommendations made by the various working groups that have recently reported. This committee is to present its report to the government within six months.

19. In 1997 part-time employment for women was 27 per cent of total female employment, the same as in the OECD, but less than the 30 per cent EU average. However, by 1998 that gap was probably reversed, as the part-time share for women jumped sharply.

20. Census figures show that the average distance travelled to work rose 22 per cent from 1991 to 1996, according to the *Irish Independent* of 22 January 1999.
21. The EU average agricultural employment share is below 4 per cent, so it is possible that the recent decennial rate of shrinkage could continue for another five years or so.
22. Variations in human capital formation are important in all theoretical approaches to the empirics of economic growth – convergence/catch-up, growth accounting and endogenous growth models.
23. As pointed out by Bradley (1998), the assumptions required to construct such an index imply that the contribution of human capital enrichment will be understated to the extent that the returns to such investments are rising over time.
24. Nearly three-quarters of all men and two-thirds of all women born before 1930 had only primary schooling (Fitz Gerald, 1997).
25. Such longer-term estimates may turn out to be too conservative: they neglect the fact that parental educational attainment in Ireland is a more important factor in determining attainment than in almost any other country for which such figures exist. Thus, there is likely to be a knock-on effect of lengthened schooling on future generations.
26. By 1993 the share of science and technology graduates in the labour force 25 to 34 years old was, at 2.75 per cent, the highest of 25 OECD countries.
27. A study undertaken for the Irish Business and Employers Confederation (IBEC) (Fitzpatrick Associates, 1998) argued that there are enormous capital spending backlogs at all levels of the education system, independent of the expected increase in the number of third-level students in coming years. Comparative data from the OECD for 1994 show that Ireland did indeed spend relatively little on capital outlays per student at the primary and secondary levels (USD 123 compared with a country mean of USD 282, third lowest out of 19 countries). The gap at the tertiary level was proportionally much smaller (USD 631 compared with a country mean of USD 843, with Ireland ranking fifteenth out of 22 countries).
28. The Youthreach programme to give early leavers a second chance at some kind of education, training and work experience is being expanded.
29. Already by 1995 a number of OECD Member countries had compulsory education beyond the age of 16: in Germany, the Netherlands and Belgium it was 18, and in the United States and Austria it was 17.
30. The number of students at third level rose from some 18 500 in 1965 to 55 000 in 1985 and then to 103 000 in 1998. The expansion took place both among universities (full-time university students tripled from 18 600 in 1970 to 56 700 in 1996) and in the technology sector. The latter was largely non-existent prior to the inception of regional technical colleges in 1969. That sector had over 38 000 students in 1996.
31. More than half of all CSF spending has been allocated to human resource development and physical infrastructure spending. The remainder has gone to aids to the private sector for marketing and research and development, for example.
32. This might appear meagre to some, but with total structural fund spending in Ireland (excluding "guidance" outlays in agriculture) of around Ir£ 6 billion (as spent) over the decade and a level of GNP in 2000 of around Ir£ 57 billion, the internal rate of return implied would not be unreasonable – something in the range of 6 to 7 per cent per year.

33. X-inefficiencies are those production-related inefficiencies which are unrelated to scale and persist due to a lack of effective competition.

34. About one-third of the increase in real after-tax income per worker during the decade to 1997 resulted from tax reductions.

35. There does remain scepticism in some quarters as to how successful social partnership has actually been in achieving wage moderation. For example, Fitz Gerald (1999) argues that, even if the partnership approach has had an effect on expectations formation, it is the evolution of the unemployment rate differential with the United Kingdom that has been primarily responsible for trend changes in real wage gains. If there has been a structural change in wage formation, it probably occurred before the first accord in 1987, and, he argues (page 162), "the partnership approach served more to validate the results, which market forces had made inevitable". Walsh (1999) too attributes wage moderation over the past decade or so to initially slack labour market conditions and a highly elastic supply of labour.

36. There were some periods when the consensus view that this policy was advantageous came under attack. For example, the government's so-called Telesis report of 1982 concluded that a sustainable competitive position cannot be built on foreign direct investment. The usual arguments for FDI as a catalyst for domestic growth are largely based on technology upgrading (besides the additional competition, which is of little importance in Ireland's case because most of the FDI has been in sectors where there is little domestic activity). However, recent empirical work on the growth, capital and productivity impact of FDI inflows in a broad sample of countries excluding Ireland (de Mello, 1999) shows, somewhat surprisingly, how tenuous the relationships are. There may be a relationship of substitutability between the capital stocks through, for example, a higher rate of obsolescence of the old-technology stock in the presence of FDI (de Mello, 1999). Much seems to depend on the absorptive capacity of the host country, which is determined by a variety of institutional and scale factors that would seem to place Ireland in very good stead.

37. The low-tax regime was originally designed to promote exports, but the export relief system was replaced by a flat 10 per cent corporate profits tax on all manufacturers (as well as internationally traded services operations, that is those that are fully export-based or for which exports are more than half of revenues and which plan further export expansion) in 1981. It has now been agreed with the EU Commission that this rate will be raised to 12½ per cent by 2002 in order to merge it with a declining rate in all other sectors. The preferential manufacturing tax rate has led to attempts to have some operations classified as manufacturing in order to benefit from the lower rate: banana ripening, for example. It has probably also caused manufacturers to avoid outsourcing and internalise more of their related activities.

38. Forfas is the policy advisory and co-ordination board for industrial development in Ireland. The State has vested in it legal powers for industrial promotion and technological development.

39. Indigenous firms with less than 10 employees are dealt with by county enterprise boards.

40. Since the nation's main problem has long been that of unemployment, a perennial debate has been whether support should be redirected more towards subsidising labour rather than capital at the margin (*e.g.* Barry, 1989). But that debate has now largely ended, as capital grants form a smaller part of the support budget than in the past and as the economy has moved towards full employment.

41. There is relatively little information available regarding even the first-round employment impact of industrial support. Forfas did a special analysis of the employment status of all new hires by 111 grant-aided firms (who may not have been representative of all such firms) in 1994-95. The results showed that 37 per cent had formerly worked in another firm, 33 per cent had been receiving unemployment benefits, 17 per cent had been in full-time education in Ireland, 5 per cent had immigrated and the rest had been either on government employment schemes, were spouses previously inactive or in other miscellaneous categories. Interestingly, there were significant differences between the foreign-owned, IDA-assisted firms and the other Irish-owned companies. The former hired more people previously employed in another company (44 versus 28 per cent) or resident abroad (7 versus 2 per cent) and fewer unemployed (26 versus 44 per cent). A broader survey on recent new hires would obviously be most useful.

42. See Audretsch (1998) for an exposition of the latest thinking along these lines.

43. Data just released by the European Commission for the period 1995-97 show that state aid to the manufacturing sector rose to 1 454 euros compared with an average of 1 261 euros for the fifteen Members of the EU. As a per cent of value added, however, the Irish figure was still below the EU average (2.2 *versus* 2.8).

44. The disparity between the two trends is presumably the result of the lag in the second figure. Costs per job have been reduced by the shift away from manufacturing projects, where investment costs are higher.

45. US investment in Ireland tripled from 1991 to 1993, just at the time the SEM programme was being implemented. At that point foreign investment represented more than one-quarter of entire gross national investment. The subsequent fall-off was sharp, so that even with the more recent rebound the share was only back to one-sixth in 1997.

46. As Krugman (1997) has pointed out, the stock of US direct investment per capita in Ireland is over 50 per cent greater than that in the United Kingdom and six times greater than in Germany or France.

47. For a discussion of the role of clusters in industrial development, see Chapter VI of OECD (1998*d*).

48. One of the other clear differences between the sectors dominated by foreign firms and those by indigenous businesses is that the former ones tend to exhibit increasing returns to scale. Using an independent identification of such sectors, Barry (1996) shows that for the 20-year period ending in 1993 foreign firms nearly doubled their employment share in these sectors while indigenous firms showed a slight decline. He later refined his view to argue (Barry *et al.*, 1999) that the scale economies involved are at the firm rather than at the plant level.

49. Some, but by no means all, of the productivity difference between indigenous and foreign-owned manufacturers is attributable to the manipulation of inter-affiliate transfer prices. As Barry (1999) has shown, foreign-firm output per worker is reduced from 3¾ to 2½ times the indigenous-firm average if one removes all the profits earned by the former, surely an upper bound to the impact of such manipulation.

50. Labour compensation represents a much smaller share of total inputs in the leading sectors dominated by foreign-owned producers than in manufacturing overall. This low labour share may be the result of the kind of firm attracted by the tax regime (Honohan *et al.*, 1998).

51. Comparing Tables 10 and 11, there might seem to be an inconsistency between the "profit" earned by US-owned manufacturing plants and the returns on US-manufacturing FDI in Ireland. However, the definitions differ substantially: in particular, net output in the Irish Census figures includes purchased services and depreciation, for example.

52. Such comparisons are also likely to be heavily influenced by the manipulation of transfer prices on the trade in goods and services between affiliated companies.

53. Annual figures show that there may well have been some favourable symbiotic effects up to 1993, but once the level of around 20 per cent of such inputs was purchased locally, it became more difficult to make further progress. See Forfas (1998*a*).

54. The share of Irish exports destined for the United Kingdom has fallen from around 75 per cent in 1960 to around 22 per cent today.

55. Irish-owned firms lifted their R&D to gross output ratio from 0.47 per cent in 1988 to 1.04 per cent in 1995, while the corresponding increase for foreign-owned firms was from 0.56 to 0.91 per cent (O'Malley, 1998). In any case, given the specialisation in high-technology products (nearly one-third), the overall R&D intensity in manufacturing is unusually low by international standards (OECD, 1998*f*, Figure 46).

56. Recent examples of such shifts include the Seagate closure and the Apple Computer downsizing.

57. The prediction error from a cross-country regression of one on the other was some three percentage points (0.6 standard errors of estimate).

58. The error in the regression equation ballooned to over 12½ percentage points, the largest of the 24 countries in the sample and a statistically significant 1.9 standard errors of estimate. The unusually small size of the market services sector has been known for a long time and has even been the subject of recent official study (National Economic and Social Forum, 1995). To some extent, of course, it results from classification bias, as noted above in footnote 37. A further explanation is that manufacturers have probably avoided outsourcing in order to qualify as much of their gross output as possible for the favourable tax rate.

59. Employment in internationally traded services more than tripled to nearly 30 000 in 1997. Nearly 20 per cent of that was at the International Financial Services Centre in Dublin (Forfas, 1998*b*), which was launched in 1987 and which now has employment of around 5 000. Operations located there are mainly back-office operations in the funds management sector, corporate treasury, international lending, life insurance and captive insurance and re-insurance (managing the funds firms appropriate when they self-insure). These operations are subject to a rigorous licensing and approval process as well as Basle and EU directives.

60. Furthermore, even the construction sector has not been spared; its output share is fifth smallest among Member countries.

61. An unweighted average of 25 OECD countries' output shares of this sector was around 48 per cent in the mid-1990s.

62. The overlap between traditional and Irish-owned manufacturing has always been imperfect and looks like it is growing more so. Of course it is also a function of one's definition of what constitutes the modern sector and what is therefore residually left to the traditional sector. In 1990 34 per cent of the output of traditional sectors was already being produced by foreign firms (O'Malley, 1998). By 1996, with a heavy dose

of inward investment in other traditional sectors, that figure had risen to 47½ per cent. Similarly, some of the output of the modern sector comes from Irish-owned firms: 8 per cent in 1990 but only 6 per cent in 1996.

63. Another advantage of the modern sector is that in general it faces a more income-elastic demand (Barry *et al.*, 1999) and therefore superior medium-term growth prospects.

64. A further 2 percentage point increase in the share of the index which is shrinking is expected in the near future with the recent series of announcements of shutdowns in the clothing sector.

65. These transactions are recorded on accrual basis. That is, all the profits multinational companies earn are recorded as factor income payments on the current account, retained profits of which are recorded as direct investment inflow on the capital account.

66. There is a break in the data after April 1997: the annual Labour Force Survey (LFS) was discontinued at that point and replaced by the Quarterly National Household Survey (QNHS). The QNHS was launched in the autumn of 1997 in order to fulfil an EU requirement. Therefore the QNHS for the three months covering March to May 1998 is the first QNHS providing comparable data on a year-on-year basis. However, there are methodological differences, the most notable of which are as follows: a change in survey reference period from point-in-time to continuous survey during a quarter; and changes in the questionnaire made possible by the use of laptop computers for data collection. According to the Central Statistics Office, an estimated 20 000 of the 115 000 increase in the level of employment is due to increased coverage of part-time employment, which raises the unadjusted growth rate of employment by 1.4 percentage points.

67. Almost all these part-time workers are satisfied with the number of hours they currently work. The remainder, officially labelled "underemployed", are only 4 per cent of total part-time employment, down from 15 per cent in April 1996.

68. All figures in this paragraph are calculated after adjustment for the methodological changes described in footnote 66.

69. The numbers on the Live Register fell by almost exactly the same amount in the two years to the spring of 1998, but the percentage decline was much smaller (18 rather than 29).

70. However, it should be recalled that, as recently as 1994, Ireland's long-term unemployment share of total unemployment, 64.3 per cent, was the highest in the OECD.

71. Until 1997 the typical survey showed only 2 per cent reporting skilled labour constraints and none unskilled labour constraints.

72. Despite this significant wage hike, it is noteworthy that overall house building costs have still grown at only a 3 to 4 per cent rate, helped by subdued price movements of construction materials.

73. The official data are average prices of houses for which loans were approved, based on responses from lending institutions. Therefore, the index suffers from compositional effects, which may account for the higher price level of second-hand houses than of new ones; it is also thought to have a downward bias for second-hand house prices. This is because the very top end of the second-hand houses are unlikely to be financed through main lending institutions (Bacon, 1998) and transactions involving them are unlikely to register in the official index. On the other hand, the Bacon

report argues "by comparing the data with those collected by the tax authority" that the house price index published by a leading private real estate agent tends to overestimate the inflation trend.

74. The average size of households in Ireland was 3.28 in 1991 and 3.04 in 1998, well above EU figures (2.63 in 1992) and UK figures (2.5 in 1989). This downsizing trend is confirmed in the high growth rate in the number of one- and two-person households in Labour Force Survey data.

75. According to the Bacon report, the conventional criteria in providing a mortgage are: *a)* income multiple of no more than 2.5 (3.5) in the case of single (double) earner(s); *b)* loan-to-value ratio of 80 to 90 per cent; and *c)* a requirement to undertake indemnity insurance arrangements.

76. Credit institutions have kept their balance sheets fairly healthy: capital adequacy ratios were around 12 to 14 per cent in 1997.

77. The authorities sold assets worth about 1.2 billion pounds in April 1997, 19.3 per cent of their external reserves at the end of the previous month.

78. The exchange rate of the Irish pound *vis-à-vis* Sterling in April 1998, 0.83, was the weakest since 1989.

79. The European Central Band decided to cut its official repurchase rate a further ½ percentage point in April 1999, bringing it to 2½ per cent.

80. Non-tax revenue includes the Central Bank's surplus income, National Lottery profits, interest on loans and dividends.

81. For example, it was widely claimed in the press that more expenditure that might alternatively have occurred in 1998 was booked in 1997. That is, spending was boosted in 1997 by an increase in the small saving reserve fund, allocations relating to post and telecom pension funds and expenditure under the education vote. However, the Minister for Finance rejected criticism that the 1997 outturn figures were manipulated in any way. He stated publicly that all the aforementioned payments were fully matured and unavoidable liabilities and that the government decided that it was prudent to use the opportunity created by the exceptionally strong Exchequer position in 1997 to meet these inescapable liabilities.

82. The extension of tax relief available to employees who save for a fixed period to purchase shares in their employers on favourable terms, SAYE (Save As You Earn) scheme, is also expected to reinforce the partnership approach by encouraging profit-sharing arrangements.

83. There are some existing companies who have acquired the right to the 10 per cent rate until the end of 2005 and 2010.

84. There are five groups (often called pillars) in the partnership process: Farmers; Trade Unions; Community and Voluntary; Employers; and Government.

85. If migration involved mainly workers in the tradable-goods sector, the effect on the supply side might be overwhelmed by that on demand side. This is because the price of tradable goods is exogenous, and, therefore, the decline in labour supply is more directly translated into a fall in income (Walsh, 1998).

86. Introduction of property tax can be seen as a variant of collecting a service charge on the use of infrastructure (see Chapter III).

87. This may be ambitious since frictional unemployment alone is thought to be around 4 per cent of the labour force.

88. This is not to say that the number of people on the Live Register will have been reduced to an unavoidable minimum, since there remains a large gap between the two measures (6.4 per cent of the labour force in the spring of 1998), due, for example, to the eligibility of some part-time, seasonal and casual workers to receive Unemployment Assistance or Benefit. Providing such workers with full-time year-round jobs would therefore have no effect on the unemployment rate but would reduce the numbers on the Live Register.

89. Here it might be more logical to use real *GDP* per employee, since most of the distortion is on the profits rather than the wages side. However, since most Irish work in this area limits itself to real GNP, the same approach is followed here. Once full employment has been reached, there should be a far smaller trend difference between the two than has been the case in the 1990s, since foreign investment will be able to make use only of already utilised labour through upgrading, rather than under-utilised labour.

90. Some in the financial community are decidedly more optimistic. For example, NCB Stockbrokers (1998) project that both labour force and productivity growth will be maintained at an undiminished pace for another eight years. Thus, potential growth of real GNP would remain at 6 per cent per year. However, most analysts expect that a significant slowdown in potential will get underway in the not too distant future, mostly emanating from reduced labour input. For example, Durkan *et al.* (1999) project a halving of labour force growth in the period 2001-2006 compared with the years 1996-2001.

91. Furthermore, it behoves the authorities to begin to think about the trade-off between the level of output and its variability as the economy becomes increasingly concentrated on a small number of industries.

92. There is no property tax for residential use in Ireland, only one on commercial owners.

93. Ireland's overall competitiveness ranking according to the World Competitiveness Yearbook has improved sharply in recent years. But the category in which it ranks the lowest (22nd out of 48 in 1997) is infrastructure.

94. The government is presently putting together a National Development Plan for the years 2000 to 2006. To that end it commissioned a major report from a group of private experts (Fitz Gerald *et al.*, 1999) which has just been published. It calls for average annual increases in capital spending on public physical infrastructure, human resources, R&D and commercial infrastructure of 6.7 per cent in real terms. This would imply Ir£ 13.6 billion in extra spending (in 1999 prices) over the seven-year horizon. This includes sharp cuts in support to industry and tourism. Others have recently placed the bar even higher at some Ir£ 20 billion. The EU is expected to cover only about Ir£ 3 billion of such spending.

95. The costs of post, transport, employers' and public liability insurance and electricity (at least for smaller firms) are often cited as being higher than those of major trading partners. Higher financing costs (an exaggerated fear of devaluation meant that there was an excess return on Irish assets of some 2½ per cent compared with German yields, after realised exchange rate depreciation was taken into account) than EMU partners have now been largely eliminated with the realisation of monetary union.

96. That infrastructure may be used as a policy instrument to compete for FDI has been hypothesised by Taylor (1992) and demonstrated by Coughlin *et al.* (1991).

97. The sector has now been fully deregulated and the main public provider is expected to begin the privatisation process in the near future. Nonetheless the government is keeping its attention focused on this sector. For example, the National Competitiveness Council (1998*b*) recently issued policy recommendations for the sector. It said that Ireland needs to make electronic commerce eligible for development agency support and to move quickly to invest in broadband communications. The authorities have decided to provide public funding to assist in the latter.
98. Given that the inter-city rail network is underdeveloped, care should be taken in assessing whether it is appropriate for it to be developed or if alternative forms of transport should be the focus for upgrading. A long-standing problem is how to improve the traffic situation in and around Dublin. The authorities have recently taken the controversial decision to build a light-rail system whose initial development will cost at least Ir£ 400 million.
99. In mid-1996 there were only 70 kilometres of motorway in the nation. While road density is 14 per cent below the EU average, once population density is accounted for the road network is denser than anywhere else in the EU. However, dependence on road transport in Ireland is greater than in most other EU countries: 97 per cent of passenger travel and 86 per cent of freight is accounted for by roads.
100. Where any pricing of infrastructure takes place at present (for example, in the electricity sector), it tends to be based on average variable cost, that is, it omits depreciation. Furthermore, when the cost curve is rising, this level will fall short of marginal cost.
101. The government has agreed in principle to implement a system of public-private partnerships but has yet to clarify the details or the range of activities the scheme might cover.
102. Neither domestic nor agricultural users pay for fresh water, despite the need to improve treatment, just discussed, and the advisability of reducing leakage rates, said to be about 40 to 45 per cent, double EU recommended levels (Fitzpatrick Associates, 1998).
103. Two-thirds of surveyed firms in the software industry already reported skilled labour shortages in the summer of 1996.
104. Honohan (1998) uses a figure of 80 per cent for illustrative purposes.
105. This share of GDP is already well down on earlier years. The peak was 6.6 per cent in 1991, and even the 1997 figure was 4.1 per cent.
106. However, there are some problems with the use of these figures. They are actually based on gross value added (GVA), which suffers from the same deficiencies for the regions as does GDP for the nation as a whole; indeed the distortions are greater the smaller the economic unit. Some regions have low GVA per capita merely because their population includes a substantial number of people who work in another region. Given that the guide should really be income rather than output per capita, GVA is a rather poor proxy. For example, two of the component regional authority areas of the western region with the lowest GVA per capita have amongst the highest average household incomes (O'Leary, 1998), and in one (the West) the unemployment rate has fallen around 5 percentage points in the most recent two years to only 6.6 per cent, the lowest in the nation.
107. According to EU rules which took effect this year, projects in the least-developed regions are eligible for 40-65 per cent of start-up costs in assistance, as opposed to only 10-40 per cent for other regions (small and medium sized firms are eligible for an extra 15 points in the first and 10 in the second).

108. Press reports claim that the additional receipts could total a cumulative Ir£ 80 to 120 million.

109. Already in 1996 12 per cent of drinking water samples were found to have excessive amounts of coliforms. Water supplies also suffer from too much phosphorus emanating from fertiliser use.

110. Some Ir£ 563 million of EU Cohesion funds will have been allocated to water and sewerage projects in the period 1995-99. Overall capital expenditure on water and sewerage rose by 50 per cent to Ir£ 185 million from 1996 to 1998.

111. Eurostat figures for the early 1990s showed that only 45 per cent of Ireland's population was served by waste water treatment plants, the lowest of any EU members other than Greece, Portugal and Spain. But the more recent OECD figures covering the mid-1990s in Table 31 show a figure of 68 per cent.

112. In Ireland 260 kilograms per capita of such waste were generated annually in the early 1990s, the same as in Finland and substantially less than the unweighted EU average of 360 kilograms (ESRI, 1995).

113. However, farm incomes have risen by 24 per cent in real terms since 1990, double the EU average.

114. The Guidance section of the European Agricultural Guarantee and Guidance Fund, which is one of the EU's four Structural Funds, co-finances measures to assist structural change in the agricultural sector and to promote rural development.

115. Dr. Jim Phelan, Head of the Agribusiness Department at University College Dublin cited in the Irish Times, January 1999.

116. Large-scale supermarkets have also been recently ruled out on at least a provisional basis: in June 1998 a temporary cap on store size was imposed by the Minister for the Environment. A period of consultation is now underway, before permanent planning guidelines are adopted.

117. It is strange to suggest that the price should be as high at the end of the period as at the beginning.

118. There is presently a moratorium on such entry.

119. Prices in the capital are said to be around 6 per cent higher than elsewhere, despite lower off-licence prices. The total value of all pub licences is estimated to be around Ir£ 500 million.

120. An independent regulator for the gas industry is also expected to be established in due course.

121. The expected price per kilowatt-hour is 3.5 pence, 40 per cent more than gas-fired production. This is only part of the price paid for using domestic fuel; the rest is the higher production of greenhouse gases.

122. Under the terms of these Trusts, employers can give tax-free compensation in the form of shares, so long as they are held for a fixed period. With steadily rising estimates for what this IPO will bring in the way of a per share price, the value of this award to the employees has unofficially gone up to around Ir£ 50 000 per worker.

123. The last budget also included a provision for a State position in a venture that will include Ireland in a high capacity fibre optic international telecoms scheme designed to exploit electronic commerce.

124. Bank charges came under the supervision of the Office of the Director of Consumer Affairs in 1995. While the government maintains that bank interest margins have

come down steadily in recent years (Department of Finance, 1998), lending rates on riskier loans (such as those to farmers and small and medium-sized firms) have risen in relative terms.

125. It has been estimated in the press, for example, that around 70 per cent of all software used in Ireland is pirated. This is not the highest figure in the EU but well above the average.

126. The first pillar also includes a means-tested non-contributory social assistance pension of Ir£ 72.50 per week for those under 80 and 77.50 for those over that age.

127. After the June 1999 increase from Ir£ 83 to 89 per week, the government is committed to raising the public pension further to Ir£ 100 per week by 2002. That implies an annual average rise of 4 per cent, much more than projected price inflation indeed but likely less than wage gains. To reach the 34 per cent figure already that year would require a rate of Ir£ 125 per week, assuming wage gains in the industrial sector continue at the latest year-on-year rate of 5.1 per cent. Most other OECD countries with similar real income per capita levels do have higher first-pillar replacement ratios than 28 per cent, although the United States and Italy are notable exceptions (OECD, 1998*e*, Figure IV.3).

128. This could be funded by a rise in all social insurance contribution rates of 1⅓ percentage points. The Pension Board also examined the effects of slower growth, raising the retirement age and unexpected net immigration.

129. The current GNP share of social welfare pensions is around 4.8 per cent. With price indexation alone following the promised increase to Ir£ 100 per week (see below), that might fall to 2.8 per cent in 2056, but wage indexation would boost it to 7.9 per cent (Pensions Board, 1998). The Minister for Finance's representative on the Board advised against indexation to wages. In the longer term it might also make a big difference whether indexation was to gross or net wages.

130. The initial contribution to the fund would represent around 10 per cent of all social insurance contributions. Adding in all personal income tax revenues would lower the share to 2 per cent.

131. At end-1997 there were some 55 000 private-sector occupational pension schemes with 313 000 members. Some 96 per cent of them were of a defined-contribution type, but their average size is much smaller: they cover less than a third of all members. But that fraction has been rising quickly. It was only a fifth at end-1992.

132. See OECD (1999*b*) for more details on these plans.

133. Related to this are proposals to change the relevant legislation to redefine workers' rights to access pension schemes, their preservation and revaluation and the integration of those occupational schemes that are of a defined-benefit nature with the social welfare pension.

134. The importance of educational attainment for youths' labour-market success is attested to by citing the unemployment rate for 15-24 year-olds with no secondary qualifications: 46 per cent in 1996.

135. More than 55 per cent of 1999 entrants had completed upper-secondary education.

136. This share has fallen from 48.2 per cent for 1992 graduates to 37.5 per cent for those in 1997. At the same time more than one-quarter of all third-level students do not complete their courses, leading to calls for increased provision of guidance and counselling. The Government has responded by promising a minimum of eight hours a week at all secondary schools.

137. This entailed spending of Ir£ 100 million in 1998 and a total of Ir£ 150 million the following two years in order to modernise tertiary-level infrastructure, especially in the technology area, to develop new areas of study and to promote innovation. Ir£ 80 million is being allocated to Regional Technical Colleges and Institutes of Technology; Ir£ 60 million to providing software places; Ir£ 30 million for equipment at tertiary institutions and 25 million at primary and secondary schools; and Ir£ 20 million each for the hotel and tourism sector and apprenticeship courses.

138. This programme comprises Ir£ 50 million in state capital outlays (only 30 million of which is incremental), Ir£ 75 million to be raised by the institutions themselves, Ir£ 25 million in tax reliefs for corporate investment and Ir£ 30 million on current expenditures (three-quarters of which is from the State and the rest from the institutions). This dwarfs current spending levels on such research of about Ir£ 5 million per year.

139. It appears that the respondents to this question in the OECD (1998*a*) source (p. 220) seem to have answered this question differently, biasing this comparison. However, the gap is so large that the qualitative point made in the text is likely still to be valid. Not unrelated to this is the fact that only 42 per cent of all courses received any employer funding, the lowest share of any of the ten OECD countries in the survey. An unweighted mean of the other nine was 56 per cent. The training levy that was introduced in 1981 does not seem to have had much effect and was abolished in the most recent budget.

140. Ireland's rating was clearly next to last among a dozen OECD Member countries in terms of both the paucity of those at the highest levels and the abundance of those at the lowest level. One-quarter of all adults were found in the lowest literacy group (the second lowest in the country sample), and even one-sixth of those in the 16-25 age group lacked basic literacy skills. One explanation might be the lateness at which free secondary education was introduced. But even controlling for the level of educational attainment Ireland's scores were near the bottom. Dispersion was also relatively high. Thus, the problem is not merely a matter of less schooling but of less satisfactory outcomes at any given level of schooling. This is somewhat surprising since Ireland had always scored at least near the average in other standardised international tests administered to youths. For example, Ireland placed sixth in mathematics and eighth in science out of 17 countries in the Third International Mathematics and Science Study for nine year-olds, the results of which were published in 1997.

141. The impact of the excessive generosity of the benefit system was to generate an "unemployment trap", which pushed up unemployment sharply (Scarpetta, 1996; and Burda, 1997). Note that 54 per cent of all households received some social welfare benefit in 1994, and for 36 per cent these amounted to half or more of total income.

142. These had already fallen since the early 1990s according to Figure 20 of the previous *Survey* (OECD, 1997*a*).

143. Annex Table A2 shows how important are the labour market costs of low educational attainment in Ireland. Nearly two-thirds of the unemployed have not completed upper-secondary education, compared to less than two-fifths of the employed. The unemployment penalty of low attainment is especially severe for those under 40.

144. It contained recommendations under four headings: improving employability, developing entrepreneurship, encouraging adaptability and promoting equal opportunity.

It also set three targets for the turn of the century: of 7 per cent for the unemployment rate; of 5 per cent of payroll for employer-provided training; and of 90 per cent of each age group for the share that finishes upper-secondary school. The latter was 81 per cent in 1994/95 and has been fairly stable.

145. At that point the Youth Progression Programme was introduced. It required all 18 and 19 year-olds hitting the six-month threshold in unemployment duration to register with FAS for job placement or employability-enhancement measures with the threat of forfeiture of benefits for failure to comply.

146. These removals are still fairly modest, given that the number of case reviews in 1997 reached 169 000, about two-thirds of the average stock.

147. Murphy and Walsh (1997) have shown the contribution of residence in local-authority housing to the risk of unemployment.

148. Currently participants need to take a six-month break after completing a session, but the consultants recommended that it be raised to a year. Single parents have a very strong financial incentive to enter and stay on the programme, since they are allowed to retain their single-parent benefits. Their share of the stock of participants rose from 4 per cent in 1994 to 22 per cent in 1997.

149. Another survey of the experience of participants in labour market programmes was recently undertaken. It covered those on FAS schemes in the period April to July 1997 and enquired into their situation 12 months later. An average of 61 per cent were in employment, much higher than the outcomes of a similar survey of participants in 1992. Once again the CE programme had below-average outcomes. More generally, there is mounting evidence that ALMPs with strong labour-market linkages, such as skills training and employment subsidies, are the most effective, while direct job-creation schemes are among the least (O'Connell and McGinnity, 1997).

150. Currently Ireland has a system of selective minimum wage protection which applied to 23 per cent of employees in 1997 and which operates under the auspices of the Labour Court through 17 sectoral Joint Labour Committees, similar to Wages Councils in the United Kingdom. There are also similar Registered Employment Agreements, mainly in the construction sector. Pay minima under these systems vary from Ir£ 2.33 to 6.53 per hour.

151. OECD figures (OECD, 1998*c*) show the shares of employees affected by minimum wages in seven other countries. In Mexico the share was 17.6 per cent and in France 11 per cent, but in all other cases it was only 3.7 to 5.1 per cent. Of course, with probable knock-on effects on some of the 23 per cent recently below the proposed minimum wage level in Ireland, the ultimate share earning that rate might well be lower.

152. The incidence on the public sector might appear minimal at first sight. However, there are a variety of people who are paid for jobs such as sleeping in the homes of the elderly for a nominal rate of about one pound an hour. In any case the health boards would probably face a large increase in their costs.

153. Even with a reduced rate for those under 18 it is likely that the overwhelming majority of such workers were earning less than the minimum in 1997. Indeed, the 70 per cent figure is very high for under 18s, both relative to existing wages and to practices abroad. For example, the UK minimum will not apply at all to under 18s and is about Ir£ 3.30 for those aged 18 to 21. In New Zealand the teenage minimum is 60 per cent of the adult rate. Thus, with a school leaving age that is likely to be 16 when implementation occurs, there will be an incentive for youths to enter the labour market prematurely.

154. Fifty-one per cent of all those earning below Ir£ 4.40 per hour in 1997 were employed in organisations with fewer than 20 employees, compared with an employment share of only one-quarter for such small organisations.

155. For example, in 1997 the index of pay rates compiled by the Irish Business and Employers Confederation was over 30 per cent higher in Dublin than in the north-west of the country.

156. In fact, Callan and Nolan (1999) point out that there has been a shift in the share of households in poverty from those headed by retirees, widow(er)s and lone parents to those headed by someone who is unemployed.

157. Employers are also facing another mandated cost increase resulting from the implementation of the Parental Leave Act. The Chairman of the Small Firms Association claimed that would cost small business alone Ir£ 144 million per year, even though the leave is unpaid.

158. The comparison is actually slightly unfair. The 3 per cent figure refers to full-time workers only. The corresponding figure for the incidence of the minimum wage on full-time workers (from Table 36) is 20.5 per cent. However, as shown in OECD (1998*c*, Table 2.10) for a few other OECD countries, including part-timers may well increase the overlap between low pay and poverty.

159. The latest data available are still only from 1994, and they show that 2.2 percentage points more households were below the 50 per cent of mean equivalent income threshold often taken to define the poverty category (Nolan and Hughes, 1997). More than all the increase is attributable to households headed by women, both single person units and families headed by single mothers. Such households represented over 31 per cent of all poor households in 1994, up from less than 10 per cent in 1987. (O'Connell, 1998, Table 3.8). However, using more sophisticated distribution measures, such as the Gini coefficient, there does not seem to have been any increase in *inequality* over this period (Callan and Nolan, 1999).

160. However, this polarisation is believed to be not as severe as in the United Kingdom. In any case care should be taken to distinguish the distribution of earnings from that of overall income.

161. The 1996 Census of Population showed 110 unemployment "hot spots" (with mean unemployment rates of 37.6 per cent), 47 of which were in the Dublin region.

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Annex I

Supplementary tables

Table A1. **Medium-term growth outcomes – actual versus forecast**[1]

Percentage change in real GNP, average of output and expenditure measures

Date of forecast	Average annual rate (actual rate)	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
1987	2.6 (3.8)	–0.4	3.0	3.3	3.7	3.6								
1989	4.9 (4.5)		4.0	7.1	5.6	4.6	4.8	3.2						
1991	3.4 (4.8)				2.0	3.7	4.3	3.3	3.6	3.7				
1994	4.7 (6.3)						2.4	4.3	6.9	5.7	4.6	4.8	4.7	4.5
1997	5.8 (6.2)									6.4	5.7	5.9	5.3	4.5
Latest CSO estimates and ESRI forecasts		2.3	5.3	7.2	1.9	2.3	2.7	7.5	8.8	6.0	7.7	7¼[2]	5¾[2]	4.5[2]

1. Forecasts by the Economic and Social Research Institute.
2. Forecasts and estimates made in December 1998 (for 2000) and February 1999 (for 1998 and 1999).

Source: Economic and Social Research Institute and OECD.

Table A2. **The labour market costs of low educational attainment, 1997**

Per cent

	Employed	Unemployed	Labour force
A. Activity status by education level and age group			
Primary	14.6	28.3	16.0
Lower secondary	24.3	35.0	25.4
Upper secondary	32.1	25.4	31.4
Third level	28.7	10.9	26.9
Other	0.3	0.4	0.3
Total	100.0	100.0	100.0

	Primary	Lower secondary	Primary	Lower secondary	Primary	Lower secondary
15-24	4.9	16.9	14.9	32.0	6.7	19.0
25-39	25.9	40.7	33.3	43.1	19.1	41.0
40-54	44.8	33.4	41.6	21.9	44.2	31.8
55-64	24.0	7.7	9.1	2.9	21.3	7.0
Over 65	10.4	1.3	1.1	0.2	8.7	1.1
Total	100.0	100.0	100.0	100.0	100.0	100.0

B. Unemployment rates

By education level		By age group for the low attainment classes	Primary	Lower secondary
Primary	18.3	15-24	40.6	23.9
Lower secondary	14.3	25-39	31.8	15.0
Upper secondary	8.4	40-54	17.1	9.8
Third level	4.2	55-64	7.8	5.8
Other	14.6	Over 65	2.3	2.3
Total	10.3	Total	18.2	14.2

Source: OECD, based on Labour Force Survey figures quoted in Department of Education and Science (1998, pp. 71 and 78).

Annex II

Chronology of main economic events and policy measures

1997

January

The 1997 Budget is presented. It includes reductions in personal tax and social security contributions worth Ir£ 393 million in a full year. The 38 per cent rate of corporate tax is reduced to 36 per cent, while the 30 per cent rate on the first Ir£ 50 000 of profits is cut to 28 per cent.

March

The post-Budget Estimates of public expenditure are published, providing for gross current expenditure of Ir£ 12 949 million.

The Central Bank Act 1997 is approved by Parliament. It deals with regulation of payments systems, updating of supervisory powers and supervision of the Bureaux de Change.

April

The government launches the National Anti-Poverty Strategy, aiming to cut persistent relative poverty by at least one-third and to bring the overall and long-term unemployment rates down to 6 and 3.5 per cent, respectively, in the next ten years.

May

The Finance Bill 1998 is passed by Parliament. It provides for substantial reductions in personal and corporation tax as well as introducing important tax reliefs in the areas of education and training.

The Minister for Finance publishes a National Plan for the Changeover to the euro, providing a framework within which financial institutions, companies and government departments can plan for the introduction of the new currency.

July

A double taxation agreement between the United States and Ireland is signed, replacing the previous agreement, which had become outdated in relation to the OECD Model Convention and the tax policies of both countries.

November

Parliament approves the Taxes Consolidation Bill 1997, which brings together income tax, corporation tax and capital gains tax legislation since 1967.

December

Publication of the 1998 Budget, which includes reductions amounting to Ir£ 517 million in personal tax, thereby reducing the standard rate of corporate tax from 36 to 32 per cent and the rate of tax on the first Ir£ 50 000 of profits from 28 to 25 per cent. Capital gains tax is also lowered.

1998

January

End-December Exchequer Returns are published, showing current and capital Ex-chequer spending for the year at Ir£ 43 million less than budgeted. There is an improvement of Ir£ 402 million on the 1997 Budget target of an Exchequer Borrowing Requirement of Ir£ 637 million.

February

The Department of Finance determines that Ireland achieved a general government budgetary surplus of 0.9 per cent of GDP in 1997 compared with an original 1997 Budget projection of a deficit of 1.5 per cent of GDP. The ratio of general government debt to GDP fell to 66.3 per cent at end-1997 from 72.7 per cent a year earlier.

March

The Irish pound's central rate in the Exchange Rate Mechanism is revalued by 3 per cent from DM 2.41105 to DM 2.48338 with effect from 16 March.

The European Commission determines that eleven Member States, including Ireland, fulfill the necessary conditions for membership of Economic and Monetary Union.

April

The government introduces a package of measures to restore balance to the housing market and to provide for the orderly growth of housing supply in coming years. The main measures include government investment to overcome infrastructural constraints, removal of incentives for investment in residential property and reductions in Stamp Duty rates for residential property.

The Employment Action Programme is released. It sets three targets for the turn of the century: 7 per cent for the unemployment rate, 5 per cent of payroll for employer-provided training and 90 per cent of each age group for the share that finishes upper-secondary school.

The National Minimum Wage Commission recommends that a minimum wage of two-thirds of the median earnings of full-time workers be introduced by April 2000 (equal to Ir£ 4.40 per hour at the average 1997 outcome).

Parliament approves the Central Bank Bill 1998, which provides for the organisational integration of the Bank into the European System of Central Banks and into the European Central Bank.

May

The Minister of Finance launches the Euro Changeover Board of Ireland. This is a follow-up to the confirmation by the EU that Ireland would be among the Member States joining the final phase of EMU on 1 January 1999.

The government decides to liberalise voice telephony in December 1998, and an independent telecommunications regulator is set up.

The Pensions Board publishes a major report recommending significant changes to the pension system. The basic contributory pension should be more generous, an element of funding should be introduced and Personal Retirement Savings Accounts should be allowed.

July

The government reaches agreement with the EU Commission on the arrangement for phasing in the 12.5 per cent general rate of corporation tax for trading activities by 1 January 2003.

Legislation is passed to make copyright protection in Ireland consistent with the WTO Trade Related Aspects of Intellectual Property (TRIPS) agreement.

August

The government agrees to a proposal from the Minister of Finance to endorse a Public Private Partnership approach, on a pilot basis, to meet infrastructure needs.

October

The Central Bank lowers the repurchase rate from 6.19 to 4.94 per cent. The short-term facility (STF) rate is also cut from 6.75 to 5.75 per cent.

November

The repurchase and STF rates are reduced to 3.69 and 4.50 per cent, respectively.

December

The repurchase and STF rates are reduced to 3 and 4 per cent, respectively, in a co-ordinated fashion with other Economic and Monetary Union member countries.

Legislation is published to partially deregulate the electricity supply industry by February 2000 in conformity with EU requirements.

1999

January

The Economic and Monetary Union is launched on 1 January 1999 with the fixed conversion rate between the euro and the Irish pound at 0.787564 per euro.

February

The European Commission turns down an application by the government to sub-divide the country into two regions to minimise future reductions in financial aid from the EU structural funds.

The Finance Bill 1999 passes the Parliament, providing for tax cuts and allocating new powers to the Revenue Commissioners to inspect accounts at financial institutions.

OECD PUBLICATIONS, 2, rue André-Pascal, 75775 PARIS CEDEX 16
PRINTED IN FRANCE
(10 1999 18 1 P) ISBN 92-64-16985-7 – No. 50671 1999
ISSN 0376-6438